

Technical Appendix to Accompany
“On the Design of Price Caps as Sanctions”

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Part A of this Technical Appendix provides detailed proofs of the formal conclusions in the paper.¹ Part B analyzes the benchmark setting in which R is a monopoly supplier. Part C considers the setting where R 's profit replaces R 's revenue in the welfare function. Part D analyzes a benchmark setting with exogenous prices. Part E explores another benchmark setting in which R has a different cost structure. Part F examines how equilibrium outcomes change as parameter values change in the modified baseline setting, where iso-elastic demand prevails. Part G presents two supplemental figures.

Equations from the Text

$$C^R(q_A, q_N) = c_A q_A + \frac{k_A}{2} [q_A]^2 + c_N q_N + \frac{k_N}{2} [q_N]^2 + \frac{k^R}{2} [q_A + q_N]^2. \quad (1)$$

$$D \equiv [2b + k] [k_N (k_A + k^R) + k_A k^R] + b k_A [3b + 2k] - b^2 [b + k] > 0. \quad (2)$$

$$k_A [(a - c_N)(2b + k) - b(a - c)] > [c_N - c_A] [3b^2 + 2b(k + k^R) + k k^R]. \quad (3)$$

R 's problem is:

$$\begin{aligned} & \underset{q_A \geq 0, q_N \geq 0}{\text{Maximize}} \quad P_A(q_A + q_N + q) q_A + [a - b(q_A + q_N + q)] q_N - C^R(q_A, q_N) \\ & \text{where } P_A(Q) = \begin{cases} \bar{p} & \text{if } P(Q) \geq \bar{p} \\ P(Q) & \text{if } P(Q) < \bar{p}. \end{cases} \end{aligned} \quad (4)$$

The rival's problem is:

$$\underset{q \geq 0}{\text{Maximize}} \quad [a - b(q_A + q_N + q)] q - C(q). \quad (5)$$

The inequality in (2) holds because:

$$\begin{aligned} D &= [2b + k] [k_N (k_A + k^R) + k_A k^R] + b k_A [3b + 2k] - b^2 [b + k] \\ &= b [k_N (k_A + k^R) + k_A k^R] + [b + k] [k_N (k_A + k^R) + k_A k^R] \\ &\quad + 2b k_A [b + k] + b^2 k_A - b^2 [b + k] \\ &= b [k_N (k_A + k^R) + k_A k^R] \\ &\quad + [b + k] [k_A k_N + k_N k^R + k_A k^R + 2b k_A - b^2] + b^2 k_A \\ &> [b + k] [k_A k_N + k_N k^R + k_A k^R + 2b k_A - b^2] \end{aligned}$$

¹Some of the formal conclusions below generalize their counterparts in the paper.

$$= [b + k] \{ k_A [2b + k_N] + k^R [k_A + k_N] - b^2 \} > 0.$$

The final inequality here holds because $k_A [2b + k_N] + k^R [k_A + k_N] > b^2$, by assumption.

A. Proofs of Formal Conclusions in the Paper.

Proposition 1. *There exist values of the price cap, $0 < \bar{p}_0 < \bar{p}_d < \bar{p}_b$, such that, in equilibrium, $q_A = 0$ if and only if $\bar{p} \leq \bar{p}_0$. Furthermore: (i) $\bar{p} < P(Q)$ if $\bar{p} \leq \bar{p}_d$; (ii) $\bar{p} = P(Q)$ if $\bar{p} \in (\bar{p}_d, \bar{p}_b]$; and (iii) $\bar{p} > P(Q)$ if $\bar{p} > \bar{p}_b$.*

Proof. The proof follows directly from Lemmas A1 – A6 (below), which refer to the following definitions.

$$\bar{p}_0 \equiv c_A + \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2} [b + k^R]. \quad (6)$$

$$\begin{aligned} \bar{p}_d \equiv \frac{1}{D_2} \{ & [a(b + k) + bc] [(b + k^R)(k_N + k_A) + k_N k_A - b k_N] \\ & + b[b + k][k_A - b]c_N + b[k_N + b][b + k]c_A \} \end{aligned}$$

$$\begin{aligned} \text{where } D_2 \equiv & b[b + k][k_N + k_A] + k_N[k_A - b][2b + k] \\ & + [k_N + k_A][2b + k][b + k^R]. \end{aligned} \quad (7)$$

$$\begin{aligned} \bar{p}_b \equiv \frac{1}{D_3} \{ & [a(b + k) + bc] [(b + k^R)(k_N + k_A) + k_N k_A] \\ & + b c_N [b + k]k_A + b k_N [b + k]c_A \} \end{aligned}$$

$$\begin{aligned} \text{where } D_3 \equiv & b[b + k][k_N + k_A] + k_N k_A [2b + k] \\ & + [k_N + k_A][2b + k][b + k^R] = D_2 + b k_N [2b + k]. \end{aligned} \quad (8)$$

Lemma A1. Suppose $\bar{p} \leq \bar{p}_0$. Then in equilibrium:

$$\begin{aligned} q_A &= 0, \quad q_N = \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2}, \\ q &= \frac{[a - c][2b + k_N + k^R] - b[a - c_N]}{[2b + k_N + k^R][2b + k] - b^2}, \text{ and} \\ Q &= q_A + q_N + q = \frac{[a - c][b + k_N + k^R] + [a - c_N][b + k]}{[2b + k_N + k^R][2b + k] - b^2}. \end{aligned} \quad (9)$$

Proof. (4) implies that R 's problem when $q_A = 0$ is:

$$\text{Maximize}_{q_N \geq 0} [a - b(q_N + q) - c_N]q_N - \frac{k_N}{2}(q_N)^2 - \frac{k^R}{2}(q_N)^2. \quad (10)$$

(10) implies that R 's profit-maximizing choice of $q_N > 0$ is determined by:

$$a - 2bq_N - bq - c_N - k_N q_N - k^R q_N = 0 \Rightarrow q_N = \frac{a - c_N - bq}{2b + k_N + k^R}. \quad (11)$$

(5) implies that the necessary condition for an interior solution to the rival's problem is:

$$\begin{aligned} a - b[q_A + q_N + q] - c - bq - kq &= 0 \Leftrightarrow [2b + k]q = a - b[q_A + q_N] - c \\ \Leftrightarrow q &= \frac{a - c}{2b + k} - \frac{b}{2b + k} [q_A + q_N]. \end{aligned} \quad (12)$$

(11) and (12) imply that when $q_A = 0$:

$$\begin{aligned} q_N &= \frac{a - c_N}{2b + k_N + k^R} - \frac{b}{2b + k_N + k^R} \left[\frac{a - c - bq_N}{2b + k} \right] \\ &= \frac{[a - c_N][2b + k] - b[a - bq_N - c]}{[2b + k_N + k^R][2b + k]} \\ \Rightarrow q_N \left[1 - \frac{b^2}{[2b + k_N + k^R][2b + k]} \right] &= \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k]} \\ \Rightarrow q_N [(2b + k_N)(2b + k) - b^2] &= [a - c_N][2b + k] - b[a - c] \\ \Rightarrow q_N &= \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2}. \end{aligned} \quad (13)$$

(12) and (13) imply:

$$\begin{aligned} q &= \frac{a - c}{2b + k} - \left[\frac{b}{2b + k} \right] \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2} \\ &= \frac{[a - c] [(2b + k_N + k^R)(2b + k) - b^2] - b[a - c_N][2b + k] + b^2[a - c]}{[2b + k][2b + k_N + k^R][2b + k] - b^2} \\ &= \frac{[a - c][2b + k_N + k^R][2b + k] - b[a - c_N][2b + k]}{[2b + k][2b + k_N + k^R][2b + k] - b^2} \\ &= \frac{[a - c][2b + k_N + k^R] - b[a - c_N]}{[2b + k_N + k^R][2b + k] - b^2}. \end{aligned} \quad (14)$$

(13) and (14) imply:

$$Q = q + q_N = \frac{[a - c][b + k_N + k^R] + [a - c_N][b + k]}{[2b + k_N + k^R][2b + k] - b^2}. \quad (15)$$

From (6):

$$\bar{p}_0 = \frac{1}{[2b + k_N + k^R][2b + k] - b^2}$$

$$\begin{aligned}
& \cdot \{ [a - c_N] [2b + k] [b + k^R] - b [a - c] [b + k^R] \\
& + c_A [(2b + k_N + k^R) (2b + k) - b^2] \}.
\end{aligned} \tag{16}$$

(15) implies:

$$\begin{aligned}
P(Q) &= a - b \frac{[a - c] [b + k_N + k^R] + [a - c_N] [b + k]}{[2b + k_N + k^R] [2b + k] - b^2} \\
&= \frac{a [(2b + k_N + k^R) (2b + k) - b^2] - b [a - c] [b + k_N + k^R] - b [a - c_N] [b + k]}{[2b + k_N + k^R] [2b + k] - b^2}.
\end{aligned} \tag{17}$$

Observe that:

$$[2b + k_N + k^R] [2b + k] > 4b^2 > b^2.$$

Therefore, (16) and (17) imply:

$$\begin{aligned}
\bar{p}_0 < P(Q) &\Leftrightarrow [a - c_N] [2b + k] [b + k^R] - b [a - c] [b + k^R] \\
&+ c_A [(2b + k_N + k^R) (2b + k) - b^2] \\
&< a [(2b + k_N + k^R) (2b + k) - b^2] \\
&- b [a - c] [b + k_N + k^R] - b [a - c_N] [b + k] \\
&\Leftrightarrow [a - c_N] [2b + k] [b + k^R] - b [a - c] [b + k^R] \\
&+ c_A [(2b + k_N + k^R) (2b + k) - b^2] \\
&< a [(2b + k_N + k^R) (2b + k) - b^2] \\
&- b [a - c] [b + k_N + k^R] - b [a - c_N] [b + k] \\
&\Leftrightarrow 0 < [a - c_A] [(2b + k_N + k^R) (2b + k) - b^2] - b [a - c] k_N \\
&- [a - c_N] [(2b + k) (b + k^R) + b (b + k)] \\
&\Leftrightarrow 0 < [a - c_A] [2bk + 2bk^R + kk^R + 3b^2 + 2bk_N + kk_N] \\
&- b [a - c] k_N - [a - c_N] [2bk + 2bk^R + kk^R + 3b^2] \\
&\Leftrightarrow [c_N - c_A] [2bk + 2bk^R + kk^R + 3b^2] \\
&+ k_N [(a - c_A) (2b + k) - b(a - c)] > 0.
\end{aligned} \tag{18}$$

The last inequality in (18) reflects (3). Therefore, $\bar{p} < P(Q)$ when $\bar{p} \leq \bar{p}_0$.

It remains to show that $q_A = 0$ when $\bar{p} \leq \bar{p}_0$. Because $\bar{p} < P(Q)$ when $\bar{p} \leq \bar{p}_0$, $q_A = 0$

when:

$$\begin{aligned}
& \frac{\partial}{\partial q_A} \left\{ [\bar{p} - c_A] q_A + [a - b(q_A + q_N + q) - c_N] q_N \right. \\
& \quad \left. - \frac{k_A}{2} [q_A]^2 - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_N + q_A]^2 \right\} \Big|_{q_A=0} \leq 0 \\
& \Leftrightarrow \bar{p} - c_A - b q_N - k^R q_N \leq 0 \\
& \Leftrightarrow \bar{p} \leq c_A + \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2} [b + k^R] = \bar{p}_0. \tag{19}
\end{aligned}$$

The equality in (19) reflects (13). $\square$

Lemma A2. Suppose $\bar{p} \in (\bar{p}_0, \bar{p}_d]$. Then in equilibrium:

$$\begin{aligned}
q_A = \frac{1}{D} \{ [3b^2 + 2b(k + k_N + k^R) + k(k_N + k^R)] [\bar{p} - c_A] \\
+ b[b + k^R][a - c] - [2b + k][b + k^R][a - c_N] \}; \tag{20}
\end{aligned}$$

$$\begin{aligned}
q_N = \frac{1}{D} \{ [2b + k][k_A + k^R][a - c_N] - b[k_A + k^R][a - c] \\
- [b(b + 2k^R) + k(b + k^R)] [\bar{p} - c_A] \}; \tag{21}
\end{aligned}$$

$$\begin{aligned}
Q^R \equiv q_A + q_N = \frac{1}{D} \{ [2b + k][b + k_N][\bar{p} - c_A] + [2b + k][k_A - b][a - c_N] \\
- b[k_A - b][a - c] \}; \tag{22}
\end{aligned}$$

$$\begin{aligned}
q = \frac{1}{D} \{ [k_N(k_A + k^R) + k_A k^R + 2b k_A - b^2][a - c] \\
- b[k_A - b][a - c_N] - b[b + k_N][\bar{p} - c_A] \}; \text{ and} \tag{23}
\end{aligned}$$

$$\begin{aligned}
Q = q + q_A + q_N = \frac{1}{D} \{ [b + k][b + k_N][\bar{p} - c_A] + [b + k][k_A - b][a - c_N] \\
+ [k^R(k_A + k_N) + k_A(b + k_N)][a - c] \}. \tag{24}
\end{aligned}$$

Proof. (4) implies that if $q_A > 0$ and $\bar{p} < P(Q)$, R 's problem, [P-R], is:

$$\begin{aligned}
& \underset{q_A, q_N}{\text{Maximize}} \quad \bar{p} q_A + [a - b(q_A + q_N + q)] q_N - c_A q_A - \frac{k_A}{2} [q_A]^2 \\
& \quad - c_N q_N - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_A + q_N]^2.
\end{aligned}$$

The necessary conditions for a solution to [P-R] in this case are:²

$$q_A : \quad \bar{p} - b q_N - c_A - k_A q_A - k^R [q_A + q_N] = 0; \quad (25)$$

$$q_N : \quad a - b [q_A + q_N + q] - b q_N - c_N - k_N q_N - k^R [q_A + q_N] = 0. \quad (26)$$

(25) implies:

$$\bar{p} - b q_N - c_A - k^R q_N = [k_A + k^R] q_A \Rightarrow q_A = \frac{\bar{p} - c_A}{k_A + k^R} - \left[\frac{b + k^R}{k_A + k^R} \right] q_N. \quad (27)$$

(26) implies:

$$\begin{aligned} a - b [q_A + q] - c_N - k^R q_A &= [2b + k_N + k^R] q_N \\ \Rightarrow q_N &= \frac{a - c_N}{2b + k_N + k^R} - \frac{[b + k^R] q_A + b q}{2b + k_N + k^R}. \end{aligned} \quad (28)$$

(25) also implies:

$$\bar{p} - c_A - k_A q_A - k^R q_A = [b + k^R] q_N \Rightarrow q_N = \frac{\bar{p} - c_A}{b + k^R} - \left[\frac{k_A + k^R}{b + k^R} \right] q_A. \quad (29)$$

(28) and (29) imply:

$$\begin{aligned} \frac{a - c_N}{2b + k_N + k^R} - \frac{[b + k^R] q_A + b q}{2b + k_N + k^R} &= \frac{\bar{p} - c_A}{b + k^R} - \left[\frac{k_A + k^R}{b + k^R} \right] q_A \\ \Rightarrow \left[\frac{b + k^R}{2b + k_N + k^R} - \frac{k_A + k^R}{b + k^R} \right] q_A &= \frac{a - c_N}{2b + k_N + k^R} - \frac{\bar{p} - c_A}{b + k^R} - \frac{b q}{2b + k_N + k^R} \\ \Rightarrow \left\{ [b + k^R]^2 - [k_A + k^R] [2b + k_N + k^R] \right\} q_A &= [b + k^R] [a - c_N] - [2b + k_N + k^R] [\bar{p} - c_A] - b [b + k^R] q \\ \Rightarrow q_A &= \frac{[b + k^R] [a - c_N] - [2b + k_N + k^R] [\bar{p} - c_A] - b [b + k^R] q}{[b + k^R]^2 - [k_A + k^R] [2b + k_N + k^R]}. \end{aligned} \quad (30)$$

(5) implies that the rival's problem in this setting, [P], is:

$$\text{Maximize}_q \quad [a - b(q_A + q_N + q) - c] q - \frac{k}{2} (q)^2. \quad (31)$$

The necessary condition for an interior solution to [P] is:

$$a - b [q_A + q_N + q] - c - b q - k q = 0 \Leftrightarrow [2b + k] q = a - b [q_A + q_N] - c$$

²It is readily verified that the determinant of the Hessian associated with [P-R] in this setting is $[k_A + k^R] [2b + k_N + k^R] - [b + k^R]^2$, which is strictly positive if $k_A \geq \frac{b}{2}$.

$$\Leftrightarrow q = \frac{a-c}{2b+k} - \frac{b}{2b+k} [q_A + q_N]. \quad (32)$$

(29) and (32) imply:

$$\begin{aligned} q &= \frac{a-c}{2b+k} - \frac{b}{2b+k} \left[q_A + \frac{\bar{p}-c_A}{b+k^R} - \left(\frac{k_A+k^R}{b+k^R} \right) q_A \right] \\ &= \frac{a-c}{2b+k} - \frac{b}{2b+k} \left[\frac{\bar{p}-c_A}{b+k^R} \right] - \frac{b}{2b+k} \left[1 - \frac{k_A+k^R}{b+k^R} \right] q_A \\ &= \frac{a-c}{2b+k} - \frac{b}{2b+k} \left[\frac{\bar{p}-c_A}{b+k^R} \right] - \frac{b}{2b+k} \left[\frac{b-k_A}{b+k^R} \right] q_A. \end{aligned} \quad (33)$$

(30) and (33) imply:

$$\begin{aligned} q_A &= \frac{[b+k^R][a-c_N] - [2b+k_N+k^R][\bar{p}-c_A]}{[b+k^R]^2 - [k_A+k^R][2b+k_N+k^R]} \\ &\quad - \frac{b[b+k^R]}{[b+k^R]^2 - [k_A+k^R][2b+k_N+k^R]} \\ &\quad \cdot \left\{ \frac{a-c}{2b+k} - \frac{b}{2b+k} \left[\frac{\bar{p}-c_A}{b+k^R} \right] - \frac{b}{2b+k} \left[\frac{b-k_A}{b+k^R} \right] q_A \right\} \\ \Rightarrow q_A &\left[1 - \left(\frac{b[b+k^R]}{[b+k^R]^2 - [k_A+k^R][2b+k_N+k^R]} \right) \left(\frac{b}{2b+k} \right) \left(\frac{b-k_A}{b+k^R} \right) \right] \\ &= \frac{[b+k^R][a-c_N] - [2b+k_N+k^R][\bar{p}-c_A]}{[b+k^R]^2 - [k_A+k^R][2b+k_N+k^R]} \\ &\quad - \frac{b[b+k^R]}{[b+k^R]^2 - [k_A+k^R][2b+k_N+k^R]} \left[\frac{a-c}{2b+k} - \frac{b}{2b+k} \left(\frac{\bar{p}-c_A}{b+k^R} \right) \right] \\ \Rightarrow q_A &\left[1 - \frac{b^2[b-k_A]}{[2b+k] \{ [b+k^R]^2 - [k_A+k^R][2b+k_N+k^R] \}} \right] \\ &= \frac{[2b+k] \{ [b+k^R][a-c_N] - [2b+k_N+k^R][\bar{p}-c_A] \}}{[2b+k] \{ [b+k^R]^2 - [k_A+k^R][2b+k_N+k^R] \}} \\ &\quad - \frac{b[b+k^R][a-c-b(\frac{\bar{p}-c_A}{b+k^R})]}{[2b+k] \{ [b+k^R]^2 - [k_A+k^R][2b+k_N+k^R] \}} \\ \Rightarrow q_A &\left\{ [2b+k] \left([b+k^R]^2 - [k_A+k^R][2b+k_N+k^R] \right) - b^2[b-k_A] \right\} \end{aligned}$$

$$\begin{aligned}
&= [2b + k] \{ [b + k^R] [a - c_N] - [2b + k_N + k^R] [\bar{p} - c_A] \} \\
&\quad - b [(a - c) (b + k^R) - b (\bar{p} - c_A)].
\end{aligned} \tag{34}$$

Observe that:

$$\begin{aligned}
&[2b + k] \{ [b + k^R]^2 - [k_A + k^R] [2b + k_N + k^R] \} - b^2 [b - k_A] \\
&= [2b + k] \{ b^2 + 2bk^R + (k^R)^2 - 2bk_A - 2bk^R - k_A k_N - k^R k_N - k_A k^R - (k^R)^2 \} \\
&\quad - b^3 + b^2 k_A \\
&= [2b + k] [b^2 - 2bk_A - k_A k_N - k^R k_N - k_A k^R] - b^3 + b^2 k_A \\
&= 2b^3 - 4b^2 k_A - 2bk_A k_N - 2bk^R k_N - 2bk_A k^R \\
&\quad + b^2 k - 2bk k_A - k k_A k_N - k k^R k_N - k k_A k^R - b^3 + b^2 k_A \\
&= b^3 - 3b^2 k_A - 2bk_A k_N - 2bk^R k_N - 2bk_A k^R \\
&\quad + b^2 k - 2bk k_A - k k_A k_N - k k^R k_N - k k_A k^R \\
&= b^2 [b + k] - b k_A [3b + 2k] - [2b + k] [k_N (k_A + k^R) + k_A k^R].
\end{aligned} \tag{35}$$

Further observe that:

$$\begin{aligned}
&[2b + k] [2b + k_N + k^R] - b^2 = 2b [2b + k_N + k^R] + k [2b + k_N + k^R] - b^2 \\
&= 3b^2 + 2b [k_N + k^R] + k [2b + k_N + k^R] \\
&= 3b^2 + 2b [k + k_N + k^R] + k [k_N + k^R].
\end{aligned} \tag{36}$$

(2) and (34) – (36) imply:

$$\begin{aligned}
q_A &= \frac{1}{D} \{ [3b^2 + 2b (k + k_N + k^R) + k (k_N + k^R)] [\bar{p} - c_A] \\
&\quad + b [b + k^R] [a - c] - [2b + k] [b + k^R] [a - c_N] \}.
\end{aligned} \tag{37}$$

(2), (29), and (37) imply:

$$\begin{aligned}
q_N &= \frac{\bar{p} - c_A}{b + k^R} - \left[\frac{k_A + k^R}{b + k^R} \right] \frac{1}{D} \{ [3b^2 + 2b (k + k_N + k^R) + k (k_N + k^R)] [\bar{p} - c_A] \\
&\quad + b [b + k^R] [a - c] - [2b + k] [b + k^R] [a - c_N] \} \\
&= \frac{1}{D [b + k^R]} \{ [\bar{p} - c_A] D \\
&\quad - [k_A + k^R] [3b^2 + 2b (k + k_N + k^R) + k (k_N + k^R)] [\bar{p} - c_A]
\end{aligned}$$

$$\begin{aligned}
& -b [b + k^R] [k_A + k^R] [a - c] \\
& + [2b + k] [k_A + k^R] [b + k^R] [a - c_N] \}.
\end{aligned} \tag{38}$$

(2) implies:

$$\begin{aligned}
D - [k_A + k^R] [3b^2 + 2b(k + k_N + k^R) + k(k_N + k^R)] \\
& = [2b + k] [k_N(k_A + k^R) + k_A k^R] + b k_A [3b + 2k] - b^2 [b + k] \\
& \quad - [k_A + k^R] [3b^2 + 2b(k + k_N + k^R) + k(k_N + k^R)] \\
& = [k_A + k^R] [(2b + k)k_N - 3b^2 - 2b(k + k_N + k^R) - k(k_N + k^R)] \\
& \quad + [2b + k] k_A k^R + 3b^2 k_A + 2b k k_A - b^3 - b^2 k \\
& = [k_A + k^R] [-3b^2 - 2b k - 2b k^R - k k^R] \\
& \quad + 2b k_A k^R + k k_A k^R + 3b^2 k_A + 2b k k_A - b^3 - b^2 k \\
& = -3b^2 k_A - 3b^2 k^R - 2b k k_A - 2b k k^R - 2b k_A k^R - 2b (k^R)^2 - k k_A k^R \\
& \quad - k (k^R)^2 + 2b k_A k^R + k k_A k^R + 3b^2 k_A + 2b k k_A - b^3 - b^2 k \\
& = -3b^2 k^R - 2b k k^R - 2b k_A k^R - 2b (k^R)^2 - k (k^R)^2 \\
& \quad + 2b k_A k^R - b^3 - b^2 k \\
& = -b^2 k^R - 2b^2 k^R - b k k^R - b k k^R - 2b (k^R)^2 - k (k^R)^2 - b^2 b - b k b \\
& = -b^2 [b + k^R] - 2b^2 k^R - b k [b + k^R] - b k k^R - 2b (k^R)^2 - k (k^R)^2 \\
& = -b^2 [b + k^R] - 2b k^R [b + k^R] - b k [b + k^R] - k k^R [b + k^R] \\
& = -[b + k^R] [b^2 + 2b k^R + b k + k k^R] \\
& = -[b + k^R] [b(b + 2k^R) + k(b + k^R)].
\end{aligned} \tag{39}$$

(38) and (39) imply:

$$\begin{aligned}
q_N = \frac{1}{D} \{ [2b + k] [k_A + k^R] [a - c_N] - b [k_A + k^R] [a - c] \\
- [b(b + 2k^R) + k(b + k^R)] [\bar{p} - c_A] \}.
\end{aligned} \tag{40}$$

Observe that:

$$3b^2 + 2b[k + k_N + k^R] + k[k_N + k^R] - [b(b + 2k^R) + k(b + k^R)]$$

$$\begin{aligned}
&= 3b^2 + 2bk + 2bk_N + 2bk^R + kk_N + kk^R - b^2 - 2bk^R - bk - kk^R \\
&= 2b^2 + bk + 2bk_N + kk_N = b[2b+k] + k_N[2b+k] = [2b+k][b+k_N]. \quad (41)
\end{aligned}$$

Further observe that:

$$\begin{aligned}
b[b+k^R] - b[k_A+k^R] &= b[b-k_A] \quad \text{and} \\
[2b+k][k_A+k^R] - [2b+k][b+k^R] &= [2b+k][k_A-b]. \quad (42)
\end{aligned}$$

(37) and (40) – (42) imply:

$$\begin{aligned}
q_A + q_N &= \frac{1}{D} \{ [2b+k][b+k_N][\bar{p}-c_A] - b[k_A-b][a-c] \\
&\quad + [2b+k][k_A-b][a-c_N] \}. \quad (43)
\end{aligned}$$

(32) and (43) imply:

$$\begin{aligned}
q &= \frac{a-c}{2b+k} - \left[\frac{b}{2b+k} \right] \frac{1}{D} \{ [2b+k][b+k_N][\bar{p}-c_A] - b[k_A-b][a-c] \\
&\quad + [2b+k][k_A-b][a-c_N] \} \\
&= \frac{D+b^2[k_A-b]}{D[2b+k]} [a-c] \\
&\quad - \frac{b}{D} \{ [b+k_N][\bar{p}-c_A] + [2b+k][k_A-b][a-c_N] \}. \quad (44)
\end{aligned}$$

(2) implies:

$$\begin{aligned}
D+b^2[k_A-b] &= [2b+k][k_N(k_A+k^R) + k_A k^R] + b k_A [3b+2k] \\
&\quad - b^2[b+k] + b^2[k_A-b] \\
&= [2b+k][k_N(k_A+k^R) + k_A k^R] + 4b^2 k_A + 2bk k_A - b^2[2b+k] \\
&= [2b+k][k_N(k_A+k^R) + k_A k^R] + 2b k_A [2b+k] - b^2[2b+k] \\
&= [2b+k][k_N(k_A+k^R) + k_A k^R + 2b k_A - b^2]. \quad (45)
\end{aligned}$$

(44) and (45) imply:

$$\begin{aligned}
q &= \frac{1}{D} \{ [k_N(k_A+k^R) + k_A k^R + 2b k_A - b^2][a-c] \\
&\quad - b[k_A-b][a-c_N] - b[b+k_N][\bar{p}-c_A] \}. \quad (46)
\end{aligned}$$

Observe that:

$$\begin{aligned}
[2b+k][b+k_N] - b[b+k_N] &= [b+k][b+k_N]; \\
[2b+k][k_A-b] - b[k_A-b] &= [b+k][k_A-b]; \quad \text{and}
\end{aligned}$$

$$\begin{aligned}
& k_N [k_A + k^R] + k_A k^R + 2b k_A - b^2 - b[k_A - b] \\
&= k_N [k_A + k^R] + k_A k^R + b k_A = k^R [k_A + k_N] + k_A [b + k_N].
\end{aligned} \tag{47}$$

(43), (46), and (47) imply:

$$\begin{aligned}
Q = q + q_A + q_N &= \frac{1}{D} \{ [b + k][b + k_N][\bar{p} - c_A] + [b + k][k_A - b][a - c_N] \\
&\quad + [k^R(k_A + k_N) + k_A(b + k_N)][a - c] \}.
\end{aligned} \tag{48}$$

It remains to show that $q_A > 0$ and $\bar{p} \leq P(Q)$ when $\bar{p} \in (\bar{p}_0, \bar{p}_d]$. (37) implies that $q_A > 0$ if:

$$\begin{aligned}
& b[b + k_N][a - c] + [\bar{p} - c_A][2bk + 2bk_N + 2bk^R + k k_N + k k^R + 3b^2] \\
&\quad - [a - c_N][bk + 2bk^R + k k^R + 2b^2] > 0 \\
\Leftrightarrow c_A + \frac{[a - c_N][bk + 2bk^R + k k^R + 2b^2] - b[b + k_N][a - c]}{2bk + 2bk_N + 2bk^R + k k_N + k k^R + 3b^2} &< \bar{p} \\
\Leftrightarrow \bar{p} > c_A + \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2} [b + k^R] &= \bar{p}_0.
\end{aligned}$$

The equality here reflects (6). (48) implies:

$$Q = \frac{1}{D} [C_1 a + C_2 c + C_3 c_N + C_4 c_A - C_4 \bar{p}] \tag{49}$$

$$\begin{aligned}
\text{where } C_1 &\equiv [b + k][k_A - b] + k^R[k_A + k_N] + k_A[b + k_N] \\
&= bk_A + bk_A + k k_A - bk + k_A k^R + k_N k^R + bk_A + k_A k_N \\
&= 2bk_A + k k_A + k_A k_N + k_A k^R + k_N k^R - b^2 - bk;
\end{aligned}$$

$$C_2 \equiv -k^R[k_A + k_N] - k_A[b + k_N]; \quad C_3 \equiv -[b + k][k_A - b]; \quad \text{and}$$

$$C_4 \equiv -[b + k][b + k_N]. \tag{50}$$

(49) implies:

$$P(Q) = a - bQ = \frac{[D - bC_1]a - bC_2 - bC_3 c_N - bC_4 c_A + bC_4 \bar{p}}{D}. \tag{51}$$

(2) and (50) imply:

$$\begin{aligned}
D - bC_1 &= [2b + k][k_N(k_A + k^R) + k_A k^R] + bk_A[3b + 2k] - b^2[b + k] \\
&\quad - b[2bk_A + k k_A + k_A k_N + k_A k^R + k_N k^R - b^2 - bk] \\
&= 2bk_A k_N + 2bk_N k^R + 2bk_A k^R + k k_A k_N + k k_N k^R + k k_A k^R + 3b^2 k_A
\end{aligned}$$

$$\begin{aligned}
& + 2 b k k_A - b k - 2 b^2 k_A - b k k_A - b k_A k_N - b k_A k^R - b k_N k^R + b^2 k \\
& = b^2 k_A + b k k_A + b k_A k_N + b k_A k^R + b k_N k^R + k k_A k_N + k k_A k^R + k k_N k^R \\
& = [b + k] [b k_A + k_A k_N + k_A k^R + k_N k^R] \\
& = [b + k] [(b + k_A) (k_N + k_A) + k_N k_A - b k_N]. \tag{52}
\end{aligned}$$

(2) and (50) imply:

$$\begin{aligned}
D - b C_4 & = [2 b + k] [k_N (k_A + k^R) + k_A k^R] + b k_A [3 b + 2 k] - b^2 [b + k] \\
& \quad - b [b + k] [b + k_N] \\
& = 3 b^2 k_A - b^2 k - b^3 + 2 b k k_A + 2 b k_A k_N + 2 b k_A k^R + 2 b k_N k^R + k k_A k_N \\
& \quad + k k_A k^R + k k_N k^R + k_N b^2 + k_N k b + b^3 + b^2 k \\
& = 3 b^2 k_A + 2 b k k_A + 2 b k_A k_N + 2 b k_A k^R + 2 b k_N k^R + k k_A k_N \\
& \quad + k k_A k^R + k k_N k^R + k_N b^2 + k_N k b \\
& = b [b + k] [k_N + k_A] + [k_A k_N - k_N b] [2 b + k] \\
& \quad + [k_N + k_A] [2 b + k] [b + k^R]. \tag{53}
\end{aligned}$$

(51) implies:

$$\begin{aligned}
\bar{p} \leq P(Q) & = \frac{[D - b C_1] a - b c C_2 - b C_3 c_N - b C_4 c_A + b C_4 \bar{p}}{D} \\
\Leftrightarrow \bar{p} - \frac{b C_4}{D} \bar{p} & \leq \frac{[D - b C_1] a - b c C_2 - b C_3 c_N - b C_4 c_A}{D} \\
\Leftrightarrow \bar{p} \left[1 - \frac{b C_4}{D} \right] & \leq \frac{[D - b C_1] a - b c C_2 - b C_3 c_N - b C_4 c_A}{D} \tag{54}
\end{aligned}$$

$$\begin{aligned}
\Leftrightarrow \bar{p} [D - b C_4] & \leq [D - b C_1] a - b c C_2 - b C_3 c_N - b C_4 c_A \\
\Leftrightarrow \bar{p} & \leq \frac{[D - b C_1] a - b c C_2 - b C_3 c_N - b C_4 c_A}{D - b C_4}. \tag{55}
\end{aligned}$$

(54) reflects the fact that $D - b C_4 > 0$ because $C_4 < 0$ (from (50)), and because $D > 0$, by assumption.

(50), (52), (53), and (55) imply:

$$\begin{aligned}
\bar{p} \leq \frac{1}{D_2} \{ a [b + k] [(b + k^R) (k_N + k_A) + k_N k_A - b k_N] \\
+ b c [(k_N + k_A) (b + k^R) + k_A k_N - b k_N] \}
\end{aligned}$$

$$\begin{aligned}
& + b[b+k][k_A-b]c_N + b[k_N+b][b+k]c_A \} \\
\Leftrightarrow \quad \bar{p} & \leq \frac{1}{D_2} \{ [(b+k)a + bc] [(b+k^R)(k_N+k_A) + k_N k_A - b k_N] \\
& + b[b+k][k_A-b]c_N + b[k_N+b][b+k]c_A \} = \bar{p}_d. \tag{56}
\end{aligned}$$

The equality in (56) reflects (7). (55) and (56) imply that $\bar{p} \leq P(Q)$ (and $q_A > 0$) when $\bar{p} \in (\bar{p}_0, \bar{p}_d]$. $\square$

Lemma A3. Suppose $\bar{p} \in (\bar{p}_d, \bar{p}_b]$, where $\bar{p}_d < \bar{p}_b$. Then in equilibrium, $P(Q) = \bar{p}$. Furthermore:

$$\begin{aligned}
q_A &= \frac{b[b+k][c_N - c_A] + k_N[a - \bar{p}][b+k] - b k_N[\bar{p} - c]}{b[b+k][k_N + k_A]}; \\
q_N &= \frac{k_A[b+k][a - \bar{p}] - b k_A[\bar{p} - c] - b[b+k][c_N - c_A]}{b[b+k][k_N + k_A]}; \\
Q^R &\equiv q_A + q_N = \frac{[b+k][a - \bar{p}] - b[\bar{p} - c]}{b[b+k]}; \\
q &= \frac{\bar{p} - c}{b+k}; \quad \text{and} \quad Q \equiv \frac{a - \bar{p}}{b}. \tag{57}
\end{aligned}$$

Proof. (4) implies that R 's problem, [P-R], can be written as:

$$\begin{aligned}
\text{Maximize}_{q_A, Q^R} \Pi_R &\equiv [P_A(q + Q^R) - c_A] q_A + [P(Q^R + q) - c_N] [Q^R - q_A] \\
&\quad - \frac{k_A}{2} [q_A]^2 - \frac{k_N}{2} [Q^R - q_A]^2 - \frac{k^R}{2} [Q^R]^2 \\
\text{where } P_A(q + Q^R) &= \begin{cases} \bar{p} & \text{if } P(q + Q^R) \geq \bar{p} \\ P(q + Q^R) & \text{if } \bar{p} > P(q + Q^R). \end{cases} \tag{58}
\end{aligned}$$

(58) implies that when $q_A > 0$ and there exists a range of $\bar{p}$ for which $P(Q) = \bar{p}$, the necessary conditions for a solution to R 's problem are:

$$\frac{\partial \Pi_R}{\partial q_A} = P_A(q + Q^R) - c_A - k_A q_A - [P(q + Q^R) - c_N] + k_N [Q^R - q_A] = 0; \tag{59}$$

$$\frac{\partial^+ \Pi_R}{\partial Q^R} \leq 0 \quad \text{and} \quad \frac{\partial^- \Pi_R}{\partial Q^R} \geq 0 \quad \text{for all } \bar{p} \in [\bar{p}_d, \bar{p}_b], \tag{60}$$

where: (i) $\frac{\partial^- \Pi_R}{\partial Q^R}$ denotes the left-sided derivative of Π_R with respect to Q^R , which is relevant when $P_A(\cdot) = \bar{p}$; and (ii) $\frac{\partial^+ \Pi_R}{\partial Q^R}$ denotes the right-sided derivative of Π_R with respect to Q^R , which is relevant when $P_A(\cdot) = P(Q)$. The first inequality in (60) indicates that R 's

profit declines if R increases Q^R so as to reduce $P(Q)$ below $\bar{p}$ (thereby rendering the cap nonbinding). The second inequality in (60) indicates that R 's profit declines if R reduces Q^R so as to increase $P(Q)$ above $\bar{p}$ (thereby causing the cap to bind). Together, the inequalities in (60) ensure that when $\bar{p} \in [\bar{p}_d, \bar{p}_b]$, R cannot increase its profit by changing Q^R so as to cause $P(Q)$ to differ from $\bar{p}$.

(12) implies:

$$\begin{aligned} a - bQ - bq - c - kq &= 0 \\ \Leftrightarrow \bar{p} - bq - c - kq &= 0 \quad \Leftrightarrow \quad q = \frac{\bar{p} - c}{b + k}. \end{aligned} \quad (61)$$

Because $\bar{p} = a - b[q + Q^R]$, (61) implies:

$$\begin{aligned} \bar{p} &= a - b \left[\frac{\bar{p} - c}{b + k} + Q^R \right] \quad \Leftrightarrow \quad bQ^R = a - \bar{p} - b \left[\frac{\bar{p} - c}{b + k} \right] \\ \Leftrightarrow \quad Q^R &= \frac{a - \bar{p}}{b} - \frac{\bar{p} - c}{b + k} = \frac{[a - \bar{p}][b + k] - b[\bar{p} - c]}{b[b + k]}. \end{aligned} \quad (62)$$

Because $\bar{p} = P_A(q + Q^R)$ in equilibrium, by assumption, (59) holds if:

$$\begin{aligned} \bar{p} - c_A - k_A q_A - [\bar{p} - c_N] + k_N [Q^R - q_A] &= 0 \\ \Leftrightarrow \quad c_N - c_A - k_A q_A + k_N Q^R - k_N q_A &= 0. \end{aligned} \quad (63)$$

(62) implies that (63) holds if:

$$\begin{aligned} c_N - c_A - k_A q_A + k_N \frac{[a - \bar{p}][b + k] - b[\bar{p} - c]}{b[b + k]} - k_N q_A &= 0 \\ \Leftrightarrow \quad q_A [k_N + k_A] &= c_N - c_A + k_N \frac{[a - \bar{p}][b + k] - b[\bar{p} - c]}{b[b + k]} \\ \Leftrightarrow \quad q_A &= \frac{c_N - c_A}{k_N + k_A} + k_N \frac{[a - \bar{p}][b + k] - b[\bar{p} - c]}{b[b + k][k_N + k_A]} \\ &= \frac{b[b + k][c_N - c_A] + k_N[a - \bar{p}][b + k] - b k_N[\bar{p} - c]}{b[b + k][k_N + k_A]}. \end{aligned} \quad (64)$$

(62) and (64) imply:

$$\begin{aligned} q_N = Q^R - q_A &= \frac{[a - \bar{p}][b + k] - b[\bar{p} - c]}{b[b + k]} \\ &\quad - \frac{b[b + k][c_N - c_A] + k_N[a - \bar{p}][b + k] - b k_N[\bar{p} - c]}{b[b + k][k_N + k_A]} \\ &= \frac{[a - \bar{p}][b + k][k_N + k_A] - b[\bar{p} - c][k_N + k_A]}{b[b + k][k_N + k_A]} \end{aligned}$$

$$\begin{aligned}
& - \frac{b[b+k][c_N - c_A] + k_N[a - \bar{p}][b+k] - b k_N[\bar{p} - c]}{b[b+k][k_N + k_A]} \\
& = \frac{k_A[b+k][a - \bar{p}] - b k_A[\bar{p} - c] - b[b+k][c_N - c_A]}{b[b+k][k_N + k_A]}. \tag{65}
\end{aligned}$$

(64) and (65) imply:

$$\begin{aligned}
Q^R \equiv q_A + q_N &= \frac{1}{b[b+k][k_N + k_A]} \{ [k_N + k_A][b+k][a - \bar{p}] \\
&\quad - b[k_N + k_A][\bar{p} - c] \} \\
&= \frac{[b+k][a - \bar{p}] - b[\bar{p} - c]}{b[b+k]}. \tag{66}
\end{aligned}$$

(61) and (66) imply:

$$Q \equiv Q^R + q = \frac{[b+k][a - \bar{p}] - b[\bar{p} - c]}{b[b+k]} + \frac{b[\bar{p} - c]}{b[b+k]} = \frac{a - \bar{p}}{b}.$$

(58) implies:

$$\begin{aligned}
\frac{\partial^+ \Pi_R}{\partial Q^R} &= -b q_A + a - 2b Q^R - b q - c_N + b q_A - k_N [Q^R - q_A] - k^R Q^R \\
&= a - 2b Q^R - b q - c_N - k_N [Q^R - q_A] - k^R Q^R \\
&= \bar{p} - b Q^R - c_N - k_N q_N - k^R Q^R = \bar{p} - [b + k^R] Q^R - c_N - k_N q_N; \tag{67}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial^- \Pi_R}{\partial Q^R} &= a - 2b Q^R - b q - c_N + b q_A - k_N [Q^R - q_A] - k^R Q^R \\
&= a - 2b Q^R - b q - c_N + b q_A - k_N q_N - k^R Q^R \\
&= \bar{p} - b Q^R - c_N + b q_A - k_N q_N - k^R Q^R \\
&= \bar{p} - [b + k^R] Q^R - c_N + b q_A - k_N q_N. \tag{68}
\end{aligned}$$

(67) and (68) imply that (60) can be written as:

$$\begin{aligned}
\bar{p} - [b + k^R] Q^R - c_N - k_N q_N &\leq 0 < \bar{p} - [b + k^R] Q^R - c_N + b q_A - k_N q_N \\
\Leftrightarrow [b + k^R] Q^R + c_N + k_N q_N - b q_A &< \bar{p} \leq [b + k^R] Q^R + c_N + k_N q_N. \tag{69}
\end{aligned}$$

(62) and (65) imply:

$$\begin{aligned}
\bar{p} &\leq [b + k^R] Q^R + c_N + k_N q_N \\
\Leftrightarrow [b + k^R] \frac{[a - \bar{p}][b+k] - b[\bar{p} - c]}{b[b+k]} + c_N &
\end{aligned}$$

$$\begin{aligned}
& + k_N \frac{k_A [b+k] [a-\bar{p}] - b k_A [\bar{p}-c] - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]} \geq \bar{p} \\
\Leftrightarrow & [b+k^R] \frac{a [b+k] - \bar{p} [2b+k] + b c}{b [b+k]} + c_N \\
& + k_N \frac{k_A [b+k] a - \bar{p} k_A [2b+k] + b k_A c - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]} \geq \bar{p} \\
\Leftrightarrow & [b+k^R] \frac{a [b+k] + b c}{b [b+k]} + c_N + k_N \frac{k_A [b+k] a + b k_A c - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]} \\
& \geq \bar{p} + \bar{p} \frac{k_N k_A [2b+k]}{b [b+k] [k_N + k_A]} + \bar{p} \frac{[2b+k] [b+k^R]}{b [b+k]} \\
\Leftrightarrow & [b+k^R] \frac{a [b+k] + b c}{b [b+k]} + c_N + k_N \frac{k_A [b+k] a + b k_A c - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]} \\
& \geq \bar{p} \left[1 + \frac{k_N k_A [2b+k]}{b [b+k] [k_N + k_A]} + \frac{[2b+k] [b+k^R]}{b [b+k]} \right] \\
\Leftrightarrow & [b+k^R] [a (b+k) + b c] [k_N + k_A] + c_N b [b+k] [k_N + k_A] \\
& + k_N [k_A (b+k) a + b k_A c - b (b+k) (c_N - c_A)] \\
& \geq \bar{p} [b (b+k) (k_N + k_A) + k_N k_A (2b+k) \\
& + (k_N + k_A) (2b+k) (b+k^R)] = \bar{p} D_3. \tag{70}
\end{aligned}$$

The last equality in (70) reflects (8). (70) implies:

$$\begin{aligned}
\bar{p} & \leq [b+k^R] Q^R + c_N + k_N q_N \\
\Leftrightarrow \bar{p} & \leq \frac{1}{D_3} \{ [b+k^R] [a (b+k) + b c] [k_N + k_A] + c_N b [b+k] [k_N + k_A] \\
& + k_N [k_A (b+k) a + b k_A c - b (b+k) (c_N - c_A)] \} \\
\Leftrightarrow \bar{p} & \leq \frac{1}{D_3} \{ [b+k^R] [a (b+k) + b c] [k_N + k_A] + c_N b [b+k] [k_N + k_A] \\
& + k_N k_A [b+k] a + b k_N k_A c - k_N b [b+k] [c_N - c_A] \} \\
\Leftrightarrow \bar{p} & \leq \frac{1}{D_3} \{ [b+k^R] [a (b+k) + b c] [k_N + k_A] + c_N b [b+k] k_A \\
& + k_N k_A [b+k] a + b k_A k_N c + k_N b [b+k] c_A
\end{aligned}$$

$$\begin{aligned}
& + k_N k_A [b+k] a + b k_A k_N c + k_N b [b+k] c_A \} \\
\Leftrightarrow \quad \bar{p} & \leq \frac{1}{D_3} \{ a [b+k] [(b+k^R) (k_N + k_A) + k_N k_A] \\
& + c [b (k_N + k_A) (b+k^R) + b k_A k_N] + c_N b [b+k] k_A + k_N b [b+k] c_A \} \\
\Leftrightarrow \quad \bar{p} & \leq \frac{1}{D_3} \{ a [b+k] [(b+k^R) (k_N + k_A) + k_N k_A] \\
& + b c [(k_N + k_A) (b+k^R) + k_A k_N] + c_N b [b+k] k_A + k_N b [b+k] c_A \} \\
\Leftrightarrow \quad \bar{p} & \leq \bar{p}_b. \tag{71}
\end{aligned}$$

(62), (64), and (65) imply:

$$\begin{aligned}
& [b+k^R] Q^R + c_N + k_N q_N - b q_A < \bar{p} \\
\Leftrightarrow \quad [b+k^R] & \frac{[a-\bar{p}][b+k] - b[\bar{p}-c]}{b[b+k]} + c_N \\
& + k_N \frac{k_A [b+k] [a-\bar{p}] - b k_A [\bar{p}-c] - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]} \\
& - b \frac{b [b+k] [c_N - c_A] + k_N [a-\bar{p}][b+k] - b k_N [\bar{p}-c]}{b [b+k] [k_N + k_A]} < \bar{p} \\
\Leftrightarrow \quad [b+k^R] & \frac{a [b+k] - \bar{p} [2b+k] + b c}{b [b+k]} + c_N \\
& + k_N \frac{k_A [b+k] a - \bar{p} k_A [2b+k] + b k_A c - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]} \\
& - b \frac{b [b+k] [c_N - c_A] + k_N a [b+k] - \bar{p} k_N [2b+k] + b k_N c}{b [b+k] [k_N + k_A]} < \bar{p} \\
\Leftrightarrow \quad [b+k^R] & \frac{a [b+k] + b c}{b [b+k]} + c_N + k_N \frac{k_A [b+k] a + b k_A c - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]} \\
& - b \frac{b [b+k] [c_N - c_A] + k_N a [b+k] + b k_N c}{b [b+k] [k_N + k_A]} \\
& < \bar{p} + \bar{p} \frac{[k_N k_A - k_N b] [2b+k]}{b [b+k] [k_N + k_A]} + \bar{p} \frac{[2b+k] [b+k^R]}{b [b+k]} \\
\Leftrightarrow \quad [b+k^R] & \frac{a [b+k] + b c}{b [b+k]} + c_N + k_N \frac{k_A [b+k] a + b k_A c - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]}
\end{aligned}$$

$$\begin{aligned}
& -b \frac{b[b+k][c_N - c_A] + k_N a[b+k] b k_N c}{b[b+k][k_N + k_A]} \\
& < \bar{p} \left[1 + \frac{[k_N k_A - k_N b][2b+k]}{b[b+k][k_N + k_A]} + \frac{[2b+k][b+k^R]}{b[b+k]} \right] \\
& \Leftrightarrow [b+k^R][a(b+k) + bc][k_N + k_A] + c_N b[b+k][k_N + k_A] \\
& \quad + k_N [k_A(b+k)a + b k_A c - b(b+k)(c_N - c_A)] \\
& \quad - b[b(b+k)(c_N - c_A) + k_N a(b+k) + b k_N c] \\
& < \bar{p} [b(b+k)(k_N + k_A) + k_N(k_A - b)(2b+k) \\
& \quad + (k_N + k_A)(2b+k)(b+k^R)] = \bar{p} D_2. \tag{72}
\end{aligned}$$

The last equality in (72) reflects (7). (7) and (72) imply:

$$\begin{aligned}
& [b+k^R] Q^R + c_N + k_N q_N - b q_A < \bar{p} \\
& \Leftrightarrow \bar{p} > \frac{1}{D_2} \{ [b+k^R][a(b+k) + bc][k_N + k_A] + c_N b[b+k][k_N + k_A] \\
& \quad + k_N [k_A(b+k)a + b k_A c - b(b+k)(c_N - c_A)] \\
& \quad - b[b(b+k)(c_N - c_A) + k_N a(b+k) + b k_N c] \} \\
& \Leftrightarrow \bar{p} > \frac{1}{D_2} \{ a[(b+k^R)(b+k)(k_N + k_A) + k_N k_A(b+k) - b(b+k)k_N] \\
& \quad + c[b(k_N + k_A)(b+k^R) + b k_A k_N - b^2 k_N] \\
& \quad + c_N b[b+k][k_A - b] + b[k_N + b][b+k]c_A \} \\
& \Leftrightarrow \bar{p} > \frac{1}{D_2} \{ a[b+k][(b+k^R)(k_N + k_A) + k_N k_A - b k_N] \\
& \quad + c b[(k_N + k_A)(b+k^R) + k_A k_N - b k_N] \\
& \quad + c_N b[b+k][k_A - b] + b[k_N + b][b+k]c_A \} = \bar{p}_d.
\end{aligned}$$

(7), (8), (67), (68), and (71) imply:

$$\begin{aligned}
\bar{p}_d &= [b+k^R] Q^R + c_N + k_N q_N - b q_A \text{ and} \\
\bar{p}_b &= [b+k^R] Q^R + c_N + k_N q_N. \tag{73}
\end{aligned}$$

(73) implies that $\bar{p}_d < \bar{p}_b$ because $q_A > 0$ when $\bar{p} > \bar{p}_0$. $\square$

Lemma A4. Suppose $\bar{p} > \bar{p}_b$. Then in equilibrium:

$$q_A = \frac{1}{D_3} \{ [a - c_A] [2bk + 2bk_N + 2bk^R + kk_N + kk^R + 3b^2] - [a - c_N] [2bk + 2bk^R + kk^R + 3b^2] - bk_N [a - c] \}; \quad (74)$$

$$q_N = \frac{1}{D_3} \{ [a - c_N] [2bk + 2bk_A + 2bk^R + kk_A + kk^R + 3b^2] - [a - c_A] [2bk + 2bk^R + kk^R + 3b^2] - bk_A [a - c] \}; \quad (75)$$

$$q = \frac{1}{D_3} \{ [a - c] [2bk_A + 2bk_N + k_A k_N + k_A k^R + k_N k^R] - bk_A [a - c_N] - bk_N [a - c_A] \}; \text{ and} \quad (76)$$

$$Q^R \equiv q_A + q_N = \frac{1}{D_3} \{ [a - c_A] k_N [2b + k] + [a - c_N] k_A [2b + k] - b [k_A + k_N] [a - c] \} \quad (77)$$

where D_3 is as specified in (8).

Proof. (4) implies that when the price cap does not bind, [P-R] is:

$$\begin{aligned} \text{Maximize}_{q_A, q_N} \quad & [a - b(q_A + q_N + q)] [q_A + q_N] - c_A q_A - \frac{k_A}{2} [q_A]^2 \\ & - c_N q_N - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_A + q_N]^2. \end{aligned} \quad (78)$$

Differentiating (78) with respect to q_A provides:

$$\begin{aligned} a - b[q_A + q_N + q] - b[q_A + q_N] - c_A - k_A q_A - k^R [q_A + q_N] &= 0 \\ \Leftrightarrow a - b[q_N + q] - b q_N - c_A - k^R q_N &= q_A [2b + k_A + k^R] \\ \Leftrightarrow q_A &= \frac{a - c_A - [2b + k^R] q_N - b q}{2b + k_A + k^R}. \end{aligned} \quad (79)$$

Corresponding differentiation of (78) with respect to q_N provides:

$$q_N = \frac{a - c_N - [2b + k^R] q_A - b q}{2b + k_N + k^R}. \quad (80)$$

(32) implies:

$$q = \frac{a - c}{2b + k} - \frac{b}{2b + k} [q_A + q_N]. \quad (81)$$

$$\text{Definitions. } K_A \equiv 2b + k_A + k^R \quad \text{and} \quad K_N \equiv 2b + k_N + k^R. \quad (82)$$

(79), (81), and (82) imply:

$$\begin{aligned}
q_A &= \frac{a - c_A}{K_A} - \frac{[2b + k^R] q_N}{K_A} - \frac{b}{K_A} \left[\frac{a - c - b(q_A + q_N)}{2b + k} \right] \\
\Rightarrow q_A &\left[1 - \frac{b^2}{[2b + k] K_A} \right] \\
&= \frac{[2b + k][a - c_A] - [2b + k^R][2b + k] q_N - b[a - c] + b^2 q_N}{[2b + k] K_A} \\
\Rightarrow q_A &\left[\frac{[2b + k] K_A - b^2}{[2b + k] K_A} \right] \\
&= \frac{[2b + k][a - c_A] - b[a - c] - ([2b + k^R][2b + k] - b^2) q_N}{[2b + k] K_A} \\
\Rightarrow q_A &= \frac{[2b + k][a - c_A] - b[a - c]}{D_A} - \frac{B}{D_A} q_N \\
&\text{where } D_A \equiv [2b + k] K_A - b^2 \quad \text{and} \quad B \equiv [2b + k^R][2b + k] - b^2. \quad (83)
\end{aligned}$$

(80) – (82) imply:

$$\begin{aligned}
q_N &= \frac{a - c_N}{K_N} - \frac{[2b + k^R] q_A}{K_N} - \frac{b}{K_N} \left[\frac{a - c - b(q_A + q_N)}{2b + k} \right] \\
\Rightarrow q_N &\left[1 - \frac{b^2}{[2b + k] K_N} \right] \\
&= \frac{[2b + k][a - c_N] - [2b + k^R][2b + k] q_A - b[a - c] + b^2 q_A}{[2b + k] K_N} \\
\Rightarrow q_N &\left[\frac{[2b + k] K_N - b^2}{[2b + k] K_N} \right] \\
&= \frac{[2b + k][a - c_N] - b[a - c] - ([2b + k^R][2b + k] - b^2) q_A}{[2b + k] K_N} \\
\Rightarrow q_N &= \frac{[2b + k][a - c_N] - b[a - c]}{D_N} - \frac{B}{D_N} q_A \\
&\text{where } K_N \equiv 2b + k_N + k^R \quad \text{and} \quad D_N \equiv [2b + k] K_N - b^2. \quad (84)
\end{aligned}$$

(83) and (84) imply:

$$q_A = \frac{[2b + k][a - c_A] - b[a - c]}{D_A} - \frac{B}{D_A D_N} \{ [2b + k][a - c_N] - b[a - c] - B q_A \}$$

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$$\begin{aligned}
\Rightarrow q_A \left[1 - \frac{B^2}{D_A D_N} \right] &= \frac{1}{D_A D_N} \left\{ [2b+k] D_N [a-c_A] - b D_N [a-c] \right. \\
&\quad \left. - B [2b+k] D_N [a-c_N] + b B [a-c] \right\} \\
\Rightarrow q_A [D_A D_N - B^2] &= [2b+k] D_N [a-c_A] + b [B - D_N] [a-c] \\
&\quad + [2b+k] B [a-c_N]. \tag{85}
\end{aligned}$$

(83) and (84) imply:

$$\begin{aligned}
D_A D_N - B^2 &= [(2b+k) K_A - b^2] [(2b+k) K_N - b^2] - [(2b+k^R)(2b+k) - b^2]^2 \\
&= [2b+k]^2 K_A K_N - b^2 [2b+k] K_A - b^2 [2b+k] K_N + b^4 \\
&\quad - [2b+k]^2 [2b+k^R]^2 + 2b^2 [2b+k] [2b+k^R] - b^4 \\
&= [2b+k] \{ [2b+k] K_A K_N - b^2 [K_A + K_N] + 2b^2 [2b+k^R] \\
&\quad - [2b+k] [2b+k^R]^2 \}. \tag{86}
\end{aligned}$$

(82) implies that the term in $\{\cdot\}$ in (86) is:

$$\begin{aligned}
&[2b+k] [2b+k^R+k_A] [2b+k^R+k_N] - b^2 [4b+k_A+k_N+2k^R] \\
&\quad + 2b^2 [2b+k^R] - [2b+k] [2b+k^R]^2 \\
&= [2b+k] \left\{ [2b+k^R]^2 + [k_A+k_N] [2b+k^R] + k_A k_N \right\} \\
&\quad + 2b^2 [2b+k^R] - [2b+k] [2b+k^R]^2 - b^2 [2(2b+k^R) + k_A + k_N] \\
&= [2b+k^R] \{ [2b+k] [k_A+k_N] + 2b^2 - 2b^2 \} + [2b+k] k_A k_N \\
&\quad + [2b+k] k_A k_N - b^2 [k_A + k_N] \\
&= [k_A + k_N] \{ [2b+k] [2b+k^R] - b^2 \} + [2b+k] k_A k_N \\
&= [k_A + k_N] \{ [2b+k] [b+k^R] + b [2b+k] - b^2 \} + [2b+k] k_A k_N \\
&= [k_A + k_N] \{ [2b+k] [b+k^R] + b [b+k] \} + [2b+k] k_A k_N = D_3. \tag{87}
\end{aligned}$$

The last equality in (87) reflects (8).

(82) and (84) imply:

$$\begin{aligned}
D_N &= [2b+k] [2b+k_N+k^R] - b^2 = 2b[2b+k] - b^2 + [2b+k] [k_N+k^R] \\
&= 3b^2 + 2bk + [2b+k] [k_N+k^R]. \tag{88}
\end{aligned}$$

(82) and (84) imply:

$$\begin{aligned} B - D_N &= [2b + k] [2b + k^R] - b^2 - \{ [2b + k] [2b + k_N + k^R] - b^2 \} \\ &= [2b + k] [2b + k^R - 2b - k_N - k^R] = - [2b + k] k_N. \end{aligned} \quad (89)$$

(83) and (85) – (89) imply that (74) holds. Furthermore, (74) and the symmetry of q_A and q_N in the analysis imply that (75) holds.

Observe that:

$$\begin{aligned} 3b^2 + 2bk + [2b + k] [k_N + k^R] &- [(2b + k) (2b + k^R) - b^2] \\ &= 4b^2 + 2bk + [2b + k] [k_N + k^R - (2b + k^R)] \\ &= 2b[2b + k] + [2b + k] [k_N - 2b] = [2b + k] k_N; \text{ and} \\ 3b^2 + 2bk + [2b + k] [k_A + k^R] &- [(2b + k) (2b + k^R) - b^2] \\ &= 4b^2 + 2bk + [2b + k] [k_A + k^R - (2b + k^R)] \\ &= 2b[2b + k] + [2b + k] [k_A - 2b] = [2b + k] k_A. \end{aligned} \quad (90)$$

(74), (75), and (90) imply that $Q^R = q_A + q_N$ is as specified in (77).

(77) and (81) imply:

$$\begin{aligned} q &= \frac{[a - c] D_3}{[2b + k] D_3} \\ &\quad - \frac{b}{[2b + k] D_3} \{ [a - c_A] k_N [2b + k] + [a - c_N] k_A [2b + k] \\ &\quad \quad - b [k_A + k_N] [a - c] \} \\ &= \frac{1}{[2b + k] D_3} \{ [a - c] [D_3 + b^2 (k_A + k_N)] - [2b + k] b k_A [a - c_N] \\ &\quad \quad - [2b + k] b k_N [a - c_A] \}. \end{aligned} \quad (91)$$

(8) implies:

$$\begin{aligned} D_3 + b^2 [k_A + k_N] &= [2b + k] k_A k_N + [k_A + k_N] [b^2 + b(b + k) + (2b + k) (b + k^R)] \\ &= [2b + k] k_A k_N + [k_A + k_N] [2b^2 + bk + 2b^2 + 2bk^R + bk + k k^R] \\ &= [2b + k] k_A k_N + [k_A + k_N] [4b^2 + 2bk + 2bk^R + k k^R] \\ &= [2b + k] k_A k_N + [k_A + k_N] [2b(2b + k) + k^R(2b + k)] \\ &= [2b + k] \{ k_A k_N + [k_A + k_N] [2b + k^R] \}. \end{aligned} \quad (92)$$

(91) and (92) imply that q is as specified in (76).

(74) – (76) imply:

$$\begin{aligned} P(Q) &= a - b [q_A + q_N + q] \\ &= a - \frac{b}{D_3} [B_1 (a - c_A) + B_2 (a - c_N) + B_3 (a - c)] \end{aligned} \quad (93)$$

where $B_1 = k_N [b + k]$; $B_2 = k_A [b + k]$; and

$$B_3 = [b + k^R] [k_A + k_N] + k_A k_N. \quad (94)$$

(93) implies:

$$P(Q) = \frac{[D_3 - b(B_1 + B_2 + B_3)] a + b B_1 c_A + b B_2 c_N + b B_3 c}{D_3}. \quad (95)$$

(94) implies:

$$\begin{aligned} B_1 + B_2 + B_3 &= [k_A + k_N] [b + k] + [b + k^R] [k_A + k_N] + k_A k_N \\ &= [2b + k + k^R] [k_A + k_N] + k_A k_N. \end{aligned}$$

(8) and (94) imply:

$$\begin{aligned} D_3 - b [B_1 + B_2 + B_3] &= b [b + k] [k_N + k_A] + k_N k_A [2b + k] + [k_N + k_A] [2b + k] [b + k^R] \\ &\quad - b [2b + k + k^R] [k_A + k_N] - b k_N k_A \\ &= b [b + k] [k_N + k_A] + k_N k_A [2b + k] + [k_N + k_A] [2b^2 + kb + 2k^R b + k k^R] \\ &\quad - [2b^2 + kb + b k^R] [k_A + k_N] - b k_N k_A \\ &= b [b + k] [k_N + k_A] + k_N k_A [b + k] + [k_N + k_A] [b k^R + k k^R] \\ &= b [b + k] [k_N + k_A] + k_N k_A [b + k] + k^R [k_N + k_A] [b + k] \\ &= [b + k] [b (k_N + k_A) + k_N k_A + k^R (k_N + k_A)] \\ &= [b + k] [(b + k^R) (k_N + k_A) + k_N k_A]. \end{aligned} \quad (96)$$

(94), (95), and (96) imply that the price cap does not bind if:

$$\begin{aligned} \bar{p} &> \frac{a [D_3 - b(B_1 + B_2 + B_3)] + b B_1 c_A + b B_2 c_N + b B_3 c}{D_3} \\ &= \frac{1}{D_3} \{ a [b + k] [(b + k^R) (k_N + k_A) + k_N k_A] + b [b + k] k_N c_A \\ &\quad + b [b + k] k_A c_N + b c [(b + k^R) (k_A + k_N) + k_A k_N] \} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{D_3} \{ [(b+k)a + bc] [(b+k^R)(k_N + k_A) + k_N k_A] \\
&\quad + b[b+k]k_N c_A + b[b+k]k_A c_N \} = \bar{p}_b.
\end{aligned} \tag{97}$$

The last equality in (97) reflects (8). $\square$

Definitions

$q_{A1}(\bar{p}_0)$, $q_{N1}(\bar{p}_0)$, and $q_1(\bar{p}_0)$, respectively, denote the values of q_A , q_N , and q specified in Lemma A1, where $\bar{p} \leq \bar{p}_0$.

$q_{A2}(\bar{p}_0)$, $q_{N2}(\bar{p}_0)$, and $q_2(\bar{p}_0)$, respectively, denote the values of q_A , q_N , and q specified in Lemma A2, where $\bar{p} \in (\bar{p}_0, \bar{p}_d]$.

Lemma A5. $\lim_{\bar{p} \rightarrow \bar{p}_0} q_{A2}(\bar{p}) = q_{A1}(\bar{p}_0)$, $\lim_{\bar{p} \rightarrow \bar{p}_0} q_{N2}(\bar{p}) = q_{N1}(\bar{p}_0)$, and $\lim_{\bar{p} \rightarrow \bar{p}_0} q_2(\bar{p}) = q_1(\bar{p}_0)$.

Proof. (11), (12), and (19) imply that when $\bar{p} \leq \bar{p}_0$, q_N , q , and q_A are determined by:

$$\begin{aligned}
\frac{\partial \pi^R}{\partial q_N} &= a - 2bq_N - bq - c_N - k_N q_N - k^R q_N = 0; \\
\frac{\partial \pi}{\partial q} &= a - bq_N - 2bq - c - kq = 0; \\
q_A &= 0; \text{ and } \frac{\partial \pi^R}{\partial q_A} = \bar{p} - c_A - bq_N - k^R q_N \leq 0.
\end{aligned} \tag{98}$$

(19) implies that the weak inequality in (98) holds as an equality when $\bar{p} = \bar{p}_0$.

(25), (26), and (32) imply that when $\bar{p} \in (\bar{p}_0, \bar{p}_d]$, q_N , q , and q_A are determined by:

$$\begin{aligned}
\frac{\partial \pi^R}{\partial q_N} &= a - 2bq_N - bq - bq_A - c_N - k_N q_N - k^R [q_N + q_A] = 0; \\
\frac{\partial \pi}{\partial q} &= a - bq_N - bq_A - 2bq - c - kq = 0; \\
\frac{\partial \pi^R}{\partial q_A} &= \bar{p} - c_A - k_A q_A - bq_N - k^R [q_N + q_A] = 0.
\end{aligned} \tag{99}$$

(6) and (25) imply:

$$\begin{aligned}
\lim_{\bar{p} \rightarrow \bar{p}_0} q_{A2}(\bar{p}) &= \frac{1}{D} \{ [3b^2 + 2b(k + k_N + k^R) + k(k_N + k^R)] [\bar{p}_0 - c_A] \\
&\quad + b[b + k^R][a - c] - [2b + k][b + k^R][a - c_N] \} \\
&= \frac{1}{D} \{ [3b^2 + 2b(k + k_N + k^R) + k(k_N + k^R)]
\end{aligned}$$

$$\begin{aligned}
& \cdot [b + k^R] \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2} \\
& + b[b + k^R][a - c] - [2b + k][b + k^R][a - c_N] \} \\
& = \frac{1}{D} \{ [b + k^R][a - c_N][2b + k] - b[b + k^R][a - c] \\
& + b[b + k^R][a - c] - [2b + k][b + k^R][a - c_N] \} = 0. \quad (100)
\end{aligned}$$

(100) reflects the fact that:

$$\begin{aligned}
[2b + k_N + k^R][2b + k] - b^2 &= 3b^2 + 2bk + [k_N^R + k^R][2b + k] \\
&= 3b^2 + 2b[k + k_N + k^R] + k[k_N + k^R].
\end{aligned}$$

(100) implies that $\lim_{\bar{p} \rightarrow \bar{p}_0} q_{A2}(\bar{p}) = q_{A1}(\bar{p}_0)$. The equations in (99) coincide with the equations in (98) when $\bar{p} = \bar{p}_0$. Therefore, because (20), (21), and (23) imply that q_A , q_N , and q are continuous functions of $\bar{p}$, $\lim_{\bar{p} \rightarrow \bar{p}_0} q_{A2}(\bar{p}) = q_{A1}(\bar{p}_0)$, $\lim_{\bar{p} \rightarrow \bar{p}_0} q_{N2}(\bar{p}) = q_{N1}(\bar{p}_0)$, and $\lim_{\bar{p} \rightarrow \bar{p}_0} q_2(\bar{p}) = q_1(\bar{p}_0)$. $\square$

Lemma A6. $0 < \bar{p}_0 < \bar{p}_d < \bar{p}_b$.

Proof. The proof of Lemma A3 establishes that $\bar{p}_d < \bar{p}_b$. From (6):

$$\bar{p}_0 = \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2} [b + k^R] + c_A > 0. \quad (101)$$

The inequality in (101) holds because $[a - c_N][2b + k] - b[a - c] > 0$, from (3).

To prove that $\bar{p}_0 < \bar{p}_d$, let $Q_1(\bar{p})$ denote the value of $Q(\bar{p})$ specified in Lemma A1, and let $Q_2(\bar{p})$ denote the value of $Q(\bar{p})$ specified in Lemma A2. Lemma A5 implies:

$$Q_1(\bar{p}_0) = Q_2(\bar{p}_0). \quad (102)$$

Lemma A2 implies:

$$\bar{p} < P(Q_2(\bar{p})) \Leftrightarrow \bar{p} < \bar{p}_d. \quad (103)$$

(102) and (103) imply that if $\bar{p}_0 < P(Q_1(\bar{p}_0))$, then:

$$\bar{p}_0 < P(Q_2(\bar{p}_0)) \Leftrightarrow \bar{p}_0 < \bar{p}_d. \quad (104)$$

The first inequality in (104) holds because (102) implies that $P(Q_1(\bar{p}_0)) = P(Q_2(\bar{p}_0))$. The equivalence in (104) reflects (103). (104) implies that to establish that $\bar{p}_0 < \bar{p}_d$, it suffices to show that $\bar{p}_0 < P(Q_1(\bar{p}_0))$.

(6) implies:

$$\bar{p}_0 = c_A + \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2} [b + k^R]$$

$$\begin{aligned}
&= \frac{1}{[2b + k_N + k^R][2b + k] - b^2} \\
&\quad \cdot \{ c_A [(2b + k_N + k^R)(2b + k) - b^2] + [a - c_N][2b + k][b + k^R] \\
&\quad \quad - b[b + k^R][a - c] \}. \tag{105}
\end{aligned}$$

Recall from (15) that when $q_A = 0$ and the price cap binds, the equilibrium price is:

$$\begin{aligned}
P(Q) &= a - b[q + q_N] = a - b \frac{[a - c][b + k_N + k^R] + [a - c_N][b + k]}{[2b + k_N + k^R][2b + k] - b^2} \\
&= \frac{1}{[2b + k_N + k^R][2b + k] - b^2} \\
&\quad \cdot \{ a [(2b + k_N + k^R)(2b + k) - b^2] - b[a - c][b + k_N + k^R] \\
&\quad \quad - b[a - c_N][b + k] \}. \tag{106}
\end{aligned}$$

(105) and (106) imply that $\bar{p}_0 < P(Q)$ if:

$$\begin{aligned}
&a [(2b + k_N + k^R)(2b + k) - b^2] - b[a - c][b + k_N + k^R] - b[a - c_N][b + k] \\
&> c_A [(2b + k_N + k^R)(2b + k) - b^2] + [a - c_N][2b + k][b + k^R] \\
&\quad - b[b + k^R][a - c] \\
\Leftrightarrow &[a - c_A] [(2b + k_N + k^R)(2b + k) - b^2] - b[a - c]k_N \\
&\quad - [a - c_N][(b + k)b + (2b + k)(b + k^R)] > 0 \\
\Leftrightarrow &[a - c_A][2bk + 2bk_N + 2bk^R + kk_N + kk^R + 3b^2] \\
&\quad - bk_N[a - c] - [a - c_N][2bk + 2bk^R + kk^R + 3b^2] > 0 \\
\Leftrightarrow &[c_N - c_A][2bk + 2bk^R + kk^R + 3b^2] + k_N[(a - c_A)(2b + k) - b(a - c)] > 0.
\end{aligned}$$

The inequality here holds because $c_N \geq c_A$, by assumption and because (3) implies:

$$\begin{aligned}
&k_A[(a - c_N)(2b + k) - b(a - c)] > [c_N - c_A][2bk + 2bk^R + kk^R + 3b^2] \\
\Rightarrow &[a - c_N][2b + k] - b[a - c] > 0 \Rightarrow [a - c_A][2b + k] - b[a - c] > 0. \tag{107}
\end{aligned}$$

The last two inequalities in (107) hold because $c_N \geq c_A$, by assumption. $\square$ ■

Proposition 2. *In equilibrium: (i) $\frac{dq_A}{d\bar{p}} < 0$, $\frac{dq_N}{d\bar{p}} < 0$, $\frac{dq}{d\bar{p}} > 0$, $\frac{dQ}{d\bar{p}} < 0$, and $\frac{dP(Q)}{d\bar{p}} = 1$ for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$; and (ii) $\frac{dq_A}{d\bar{p}} > 0$, $\frac{dq_N}{d\bar{p}} < 0$, $\frac{dq}{d\bar{p}} < 0$, $\frac{dQ^R}{d\bar{p}} > 0$, $\frac{dQ}{d\bar{p}} > 0$, and $\frac{dP(Q)}{d\bar{p}} < 0$ for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$.*

Proof. Lemma A3 implies that for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$:

$$\begin{aligned}\frac{dq_A}{d\bar{p}} &= -\frac{k_N [b+k] + b k_N}{b [b+k] [k_N + k_A]} < 0; \\ \frac{dq_N}{d\bar{p}} &= -\frac{k_A [b+k] + b k_A}{b [b+k] [k_N + k_A]} < 0; \quad \frac{dq}{d\bar{p}} = \frac{1}{b+k} > 0; \\ \frac{dQ}{d\bar{p}} &= -\frac{1}{b} < 0 \Rightarrow \frac{dP(Q)}{d\bar{p}} = -b \left[-\frac{1}{b} \right] = 1.\end{aligned}$$

Lemma A2 implies that for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\begin{aligned}\frac{dq_A}{d\bar{p}} &= \frac{3b^2 + 2b [k + k_N + k^R] + k [k_N + k^R]}{D} > 0; \\ \frac{dq_N}{d\bar{p}} &= -\frac{b [b + 2k^R] + k [b + k^R]}{D} < 0; \quad \frac{dQ^R}{d\bar{p}} = \frac{[2b+k][b+k_N]}{D} > 0; \\ \frac{dq}{d\bar{p}} &= -\frac{b [b + k_N]}{D} < 0; \quad \text{and} \quad \frac{dQ}{d\bar{p}} = \frac{[b+k][b+k_N]}{D} > 0. \quad \blacksquare\end{aligned}\tag{108}$$

Proposition 3. For $\bar{p} \in (\bar{p}_d, \bar{p}_b)$: (i) $V(\bar{p})$ is a strictly concave function of $\bar{p}$; (ii) $\frac{\partial V(\bar{p})}{\partial \bar{p}} \leq 0 \Leftrightarrow \bar{p} \geq \bar{p}_{VM}$ where $\bar{p}_{VM} \in [\bar{p}_d, \bar{p}_b]$; and (iii) $\bar{p}_{VM} = \bar{p}_d$ if $\Phi_1 \geq 0$, whereas $\bar{p}_{VM} > \bar{p}_d$ if $\Phi_1 < 0$, where

$$\begin{aligned}\Phi_1 &\equiv \left[k^R + \frac{b^2}{2b+k} \right] [k_A + k_N] A + 2b [b+k] c_A [k_N + b] \\ &\quad + [2b(b+k) c_N + A k_N] [k_A - b] \quad \text{where} \quad A \equiv a [b+k] + b c.\end{aligned}\tag{109}$$

Proof. (62) implies that for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$, R 's revenue is:

$$V(\bar{p}) = \bar{p} \left[\frac{a(b+k) + bc - \bar{p}(2b+k)}{b[b+k]} \right] = \frac{[a(b+k) + bc] \bar{p} - [2b+k] \bar{p}^2}{b[b+k]}.\tag{110}$$

The value of $\bar{p}$ at which $V(\bar{p})$ in (110) is maximized is determined by:

$$a[b+k] + bc - 2[2b+k] \bar{p} = 0 \Rightarrow \bar{p} = \frac{a[b+k] + bc}{2[2b+k]} \equiv \bar{p}_{VM}.\tag{111}$$

From (8):

$$\bar{p}_b = \frac{[a(b+k) + bc] [(b+k^R)(k_N + k_A) + k_N k_A] + b c_N [b+k] k_A + b k_N [b+k] c_A}{b[b+k][k_N + k_A] + k_N k_A [2b+k] + [k_N + k_A] [2b+k] [b+k^R]}$$

$$= \frac{[(b+k^R)(k_N+k_A)+k_N k_A] \frac{a[b+k]+bc}{b[b+k]} + c_N k_A + k_N c_A}{k_N+k_A+[(b+k^R)(k_N+k_A)+k_N k_A] \frac{2b+k}{b[b+k]}}. \quad (112)$$

(111) and (112) imply that $\bar{p}_{VM} < \bar{p}_b$ if:

$$\frac{a[b+k]+bc}{2[2b+k]} < \frac{[(b+k^R)(k_N+k_A)+k_N k_A] \frac{a[b+k]+bc}{b[b+k]} + c_N k_A + k_N c_A}{k_N+k_A+[(b+k^R)(k_N+k_A)+k_N k_A] \frac{2b+k}{b[b+k]}}. \quad (113)$$

The inequality in (113) holds if:

$$\frac{[(b+k^R)(k_N+k_A)+k_N k_A] \frac{a[b+k]+bc}{b[b+k]}}{k_N+k_A+[(b+k^R)(k_N+k_A)+k_N k_A] \frac{2b+k}{b[b+k]}} > \frac{a[b+k]+bc}{2[2b+k]}. \quad (114)$$

Define $z \equiv [(b+k^R)(k_N+k_A)+k_N k_A] \frac{1}{b[b+k]}$. Then the inequality in (114) holds if:

$$\begin{aligned} & \frac{z[a(b+k)+bc]}{k_N+k_A+z[2b+k]} > \frac{a[b+k]+bc}{2[2b+k]} \\ \Leftrightarrow & \frac{z}{k_N+k_A+z[2b+k]} > \frac{1}{2[2b+k]} \\ \Leftrightarrow & 2z[2b+k] > k_N+k_A+z[2b+k] \Leftrightarrow [2b+k]z > k_N+k_A \\ \Leftrightarrow & [(b+k^R)(k_N+k_A)+k_N k_A] \frac{2b+k}{b[b+k]} > k_N+k_A \\ \Leftrightarrow & \frac{[2b+k][b+k^R]}{b[b+k]} [k_N+k_A] + k_N k_A \left[\frac{2b+k}{b(b+k)} \right] > k_N+k_A \\ \Leftrightarrow & \frac{[2b+k][b+k^R]-b[b+k]}{b[b+k]} [k_N+k_A] + k_N k_A \left[\frac{2b+k}{b(b+k)} \right] > 0. \quad (115) \end{aligned}$$

The inequality in (115) always holds because:

$$\begin{aligned} [2b+k][b+k^R]-b[b+k] &= 2b^2+2bk^R+bk+kk^R-b^2-bk \\ &= b^2+2bk^R+kk^R > 0. \end{aligned}$$

(115) implies that $\bar{p}_{VM} < \bar{p}_b$.

(110) and (111) imply that for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$, $V(\bar{p})$ is a strictly concave function that attains its maximum at $\bar{p}_{VM}$. Therefore, $\frac{\partial V(\bar{p})}{\partial \bar{p}} < 0$ for $\bar{p} \in (\bar{p}_{VM}, \bar{p}_b)$.

(7) and (111) imply that $\bar{p}_d \geq \bar{p}_{VM}$ if and only if:

$$\begin{aligned} & \frac{1}{b[b+k][k_N+k_A]+[k_A k_N-k_N b][2b+k]+[k_N+k_A][2b+k][b+k^R]} \\ & \cdot \{[(b+k)a+bc][(b+k^R)(k_N+k_A)+k_N k_A-bk_N]\} \end{aligned}$$

$$\begin{aligned}
& + b[b+k][k_A-b]c_N + b[k_N+b][b+k]c_A \} \\
\geq & \frac{a[b+k] + bc}{2[b+k]} \\
\Leftrightarrow & \frac{1}{\frac{b[b+k]}{2b+k}[k_N+k_A] + [k_A k_N - k_N b] + [k_N+k_A][b+k^R]} \\
& \cdot \{ 2[(b+k)a+bc][(b+k^R)(k_N+k_A) + k_N k_A - b k_N] \\
& + 2b[b+k][k_A-b]c_N + 2b[k_N+b][b+k]c_A \} \\
\geq & a[b+k] + bc \\
\Leftrightarrow & 2[(b+k)a+bc][(b+k^R)(k_N+k_A) + k_N k_A - b k_N] + 2b[b+k][k_A-b]c_N \\
& + 2b[k_N+b][b+k]c_A \\
\geq & [(b+k)a+bc] \frac{b[b+k]}{2b+k}[k_N+k_A] + [a(b+k)+bc][k_A k_N - b k_N] \\
& + [a(b+k)+bc][k_N+k_A][b+k^R] \\
\Leftrightarrow & [(b+k)a+bc][(b+k^R)(k_N+k_A) + k_N k_A - b k_N] + 2b[b+k][k_A-b]c_N \\
& + 2b[k_N+b][b+k]c_A \geq [a(b+k)+bc] \frac{b[b+k]}{2b+k}[k_N+k_A] \\
\Leftrightarrow & [(b+k)a+bc] \left[\left(b - \frac{b(b+k)}{2b+k} + k^R \right) (k_N+k_A) + k_N k_A - b k_N \right] \\
& + 2b[b+k][k_A-b]c_N + 2b[k_N+b][b+k]c_A \geq 0 \\
\Leftrightarrow & [(b+k)a+bc] \left[\left(\frac{2b^2+kb}{2b+k} - \frac{b(b+k)}{2b+k} + k^R \right) (k_N+k_A) + k_N k_A - b k_N \right] \\
& + 2b[b+k][k_A-b]c_N + 2b[k_N+b][b+k]c_A \geq 0 \\
\Leftrightarrow & [(b+k)a+bc] \left[\left(\frac{b^2}{2b+k} + k^R \right) (k_N+k_A) + k_N k_A - b k_N \right] \\
& + 2b[b+k][k_A-b]c_N + 2b[k_N+b][b+k]c_A \geq 0 \\
\Leftrightarrow & [(b+k)a+bc] \left[\left(\frac{b^2}{2b+k} + k^R \right) (k_N+k_A) + k_N k_A - b k_N \right] \\
& + 2b[b+k][k_A c_N + k_N c_A - b(c_N - c_A)] \geq 0
\end{aligned}$$

$$\Leftrightarrow \left[\frac{(b+k)a+bc}{2b+k} \right] \left[(b^2+k^R[2b+k])(k_N+k_A) + k_N k_A (2b+k) - b k_N (2b+k) \right] \\ + 2b[b+k][k_A c_N + k_N c_A - b(c_N - c_A)] \geq 0. \quad (116)$$

Observe that:

$$\begin{aligned} & [b^2+k^R(2b+k)][k_N+k_A] + k_N k_A [2b+k] - b k_N [2b+k] \\ &= b^2[k_N+k_A] + k^R[2b+k][k_N+k_A] + k_N[2b+k][k_A-b] \\ &= b^2[k_N+k_A] - b[2b+k][k_N+k_A] + b[2b+k][k_N+k_A] \\ &\quad + k^R[2b+k][k_N+k_A] + k_N[2b+k][k_A-b] \\ &= -b[b+k][k_N+k_A] + [b+k^R][2b+k][k_N+k_A] + k_N[2b+k][k_A-b] \\ &= -2b[b+k][k_N+k_A] + D_2. \end{aligned} \quad (117)$$

The last equality in (117) reflects (7). (116) and (117) imply:

$$\bar{p}_d \geq \bar{p}_{VM} \Leftrightarrow \tilde{\Phi}_1 \geq 0, \\ \text{where } \tilde{\Phi}_1 \equiv \left[\frac{(b+k)a+bc}{2b+k} \right] \{ D_2 - 2b[b+k][k_N+k_A] \} \\ + 2b[b+k][k_A c_N + k_N c_A - b(c_N - c_A)]. \quad (118)$$

(7) implies:

$$\begin{aligned} D_2 - 2b[b+k][k_N+k_A] &= -b[b+k][k_N+k_A] + k_N[k_A-b][2b+k] \\ &\quad + [2b+k][b+k^R][k_N+k_A] \\ &= [k_A+k_N][2b^2+2bk^R+bk+k^R-b^2-bk] + k_N[k_A-b][2b+k] \\ &= [k_A+k_N][b^2+2bk^R+k^R] + k_N[k_A-b][2b+k]. \end{aligned} \quad (119)$$

(119) implies:

$$\begin{aligned} & \left[\frac{(b+k)a+bc}{2b+k} \right] [D_2 - 2b(b+k)(k_N+k_A)] \\ &= \left[\frac{(b+k)a+bc}{2b+k} \right] [k_A+k_N][b^2+k^R(2b+k)] + [(b+k)a+bc]k_N[k_A-b]. \end{aligned} \quad (120)$$

(109) and (120) imply:

$$\tilde{\Phi}_1 = \frac{b^2}{2b+k} [k_A+k_N][(b+k)a+bc] + k^R[k_A+k_N][(b+k)a+bc]$$

$$\begin{aligned}
& + [(b+k)a + bc] k_N [k_A - b] + 2b[b+k] c_N [k_A - b] \\
& + 2b[b+k] c_A [k_N + b] \\
= & \left[k^R + \frac{b^2}{2b+k} \right] [k_A + k_N] [(b+k)a + bc] + 2b[b+k] c_A [k_N + b] \\
& + \{ 2b[b+k] c_N + [(b+k)a + bc] k_N \} [k_A - b] \equiv \Phi_1. \blacksquare
\end{aligned}$$

Proposition 4. $\bar{p}_b - \bar{p}_d$ increases as: (i) c_A , k_A , or k^R declines; (ii) c or c_N increases; or (iii) k_N increases if $k_A - b$ is sufficiently small.

Proof. (7) and (8) imply:

$$\bar{p}_d = \frac{N_2}{D_2} \quad \text{and} \quad \bar{p}_b = \frac{N_3}{D_2 + b[2b+k]k_N}$$

$$\text{where } N_3 \equiv [a(b+k) + bc] [(b+k^R)(k_N + k_A) + k_N k_A]$$

$$+ b c_N [b+k] k_A + b k_N [b+k] c_A \quad \text{and}$$

$$N_2 \equiv [a(b+k) + bc] [(b+k^R)(k_N + k_A) + k_N k_A]$$

$$+ b c_N [b+k] k_A + b k_N [b+k] c_A$$

$$- b k_N [a(b+k) + bc] - b^2 [b+k] c_N + b^2 [b+k] c_A$$

$$= N_3 - b k_N [a(b+k) + bc] - b^2 [b+k] [c_N - c_A]. \quad (121)$$

To prove that $\frac{\partial(\bar{p}_b - \bar{p}_d)}{\partial k^R} < 0$, let $q_A(\bar{p})$ denote R 's equilibrium output using A 's input when the price cap is $\bar{p} \in [\bar{p}_d, \bar{p}_b]$. Let $q_N(\bar{p})$ denote R 's corresponding output when R does not employ A 's input. Also let $Q^R(\bar{p}) = q_A(\bar{p}) + q_N(\bar{p})$. (73) implies:

$$\bar{p}_b = [b + k^R] Q^R(\bar{p}_b) + c_N + k_N q_N(\bar{p}_b)$$

where, from (57):

$$q_N(\bar{p}_b) = \frac{k_A [b+k] [a - \bar{p}_b] - b k_A [\bar{p}_b - c] - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]} \quad \text{and}$$

$$Q^R(\bar{p}_b) = \frac{[b+k] [a - \bar{p}_b] - b [\bar{p}_b - c]}{b [b+k]}. \quad (122)$$

(122) implies that $q_N(\bar{p}_b)$ and $Q^R(\bar{p}_b)$ vary with k^R only through $\bar{p}_b$. Therefore, (122) implies:

$$\frac{\partial \bar{p}_b}{\partial k^R} = Q^R(\bar{p}_b) + [b + k^R] \frac{\partial Q^R(\bar{p}_b)}{\partial \bar{p}_b} \frac{\partial \bar{p}_b}{\partial k^R} + k_N \frac{\partial q_N(\bar{p}_b)}{\partial \bar{p}_b} \frac{\partial \bar{p}_b}{\partial k^R}. \quad (123)$$

(122) also implies:

$$\begin{aligned}\frac{\partial q_N(\bar{p}_b)}{\partial \bar{p}_b} &= -\frac{k_A [b+k] + b k_A}{b [b+k] [k_N + k_A]} \equiv D_N < 0; \\ \frac{\partial Q^R(\bar{p}_b)}{\partial \bar{p}_b} &= -\frac{2b+k}{b [b+k]} \equiv D_R < 0.\end{aligned}\tag{124}$$

(123) and (124) imply:

$$\begin{aligned}\frac{\partial \bar{p}_b}{\partial k^R} &= Q^R(\bar{p}_b) + [b+k^R] D_R \frac{\partial \bar{p}_b}{\partial k^R} + k_N D_N \frac{\partial \bar{p}_b}{\partial k^R} \\ \Rightarrow \frac{\partial \bar{p}_b}{\partial k^R} [1 - (b+k^R) D_R - k_N D_N] &= Q^R(\bar{p}_b) \\ \Rightarrow \frac{\partial \bar{p}_b}{\partial k^R} &= \frac{Q^R(\bar{p}_b)}{1 - [b+k^R] D_R - k_N D_N} > 0.\end{aligned}\tag{125}$$

The inequality in (125) holds because $D_R < 0$ and $D_N < 0$, from (124).

(73) implies:

$$\bar{p}_d = [b+k^R] Q^R(\bar{p}_d) + c_N + k_N q_N(\bar{p}_d) - b q_A(\bar{p}_d)$$

where, from (57):

$$\begin{aligned}q_A(\bar{p}_d) &= \frac{b [b+k] [c_N - c_A] + k_N [a - \bar{p}] [b+k] - b k_N [\bar{p} - c]}{b [b+k] [k_N + k_A]}, \\ q_N(\bar{p}_d) &= \frac{k_A [b+k] [a - \bar{p}_d] - b k_A [\bar{p}_d - c] - b [b+k] [c_N - c_A]}{b [b+k] [k_N + k_A]}; \text{ and} \\ Q^R(\bar{p}_d) &= \frac{[b+k] [a - \bar{p}_d] - b [\bar{p}_d - c]}{b [b+k]}.\end{aligned}\tag{126}$$

(126) implies that $q_A(\bar{p}_d)$, $q_N(\bar{p}_d)$, and $Q^R(\bar{p}_d)$ vary with k^R only through $\bar{p}_d$. Therefore, (126) implies:

$$\begin{aligned}\frac{\partial \bar{p}_d}{\partial k^R} &= Q^R(\bar{p}_d) + [b+k^R] \frac{\partial Q^R(\bar{p}_d)}{\partial \bar{p}_d} \frac{\partial \bar{p}_d}{\partial k^R} \\ &\quad + k_N \frac{\partial q_N(\bar{p}_d)}{\partial \bar{p}_d} \frac{\partial \bar{p}_d}{\partial k^R} - b \frac{\partial q_A(\bar{p}_d)}{\partial \bar{p}_d} \frac{\partial \bar{p}_d}{\partial k^R}.\end{aligned}\tag{127}$$

(126) also implies:

$$\frac{\partial q_A(\bar{p}_d)}{\partial \bar{p}_d} = -\frac{k_N [b+k] + b k_N}{b [b+k] [k_N + k_A]} \equiv D_A < 0;$$

$$\begin{aligned}
\frac{\partial q_N(\bar{p}_d)}{\partial \bar{p}_d} &= -\frac{k_A [b+k] + b k_A}{b [b+k] [k_N + k_A]} \equiv D_N < 0; \\
\frac{\partial Q^R(\bar{p}_d)}{\partial \bar{p}_d} &= -\frac{2b+k}{b [b+k]} \equiv D_R < 0.
\end{aligned} \tag{128}$$

(127) and (128) imply:

$$\begin{aligned}
\frac{\partial \bar{p}_d}{\partial k^R} &= Q^R(\bar{p}_d) + [b+k^R] D_R \frac{\partial \bar{p}_d}{\partial k^R} + k_N D_N \frac{\partial \bar{p}_d}{\partial k^R} - b D_A \frac{\partial \bar{p}_d}{\partial k^R} \\
\Rightarrow \frac{\partial \bar{p}_d}{\partial k^R} [1 - (b+k^R) D_R - k_N D_N + b D_A] &= Q^R(\bar{p}_d) \\
\Rightarrow \frac{\partial \bar{p}_d}{\partial k^R} &= \frac{Q^R(\bar{p}_d)}{1 - [b+k^R] D_R - k_N D_N + b D_A}.
\end{aligned} \tag{129}$$

(128) implies:

$$\begin{aligned}
-b D_R + b D_A &= b [-D_R + D_A] = b \left[\frac{2b+k}{b(b+k)} - \frac{k_N(b+k) + b k_N}{b(b+k)(k_N + k_A)} \right] \\
&= b \left[\frac{2b+k}{b(b+k)} - \left(\frac{k_N}{k_N + k_A} \right) \frac{2b+k}{b(b+k)} \right] \\
&= b \left[\frac{2b+k}{b(b+k)} \right] \left[1 - \frac{k_N}{k_N + k_A} \right] > 0
\end{aligned} \tag{130}$$

Because $D_N < 0$ and $D_R < 0$ from (128), (130) implies:

$$\begin{aligned}
1 - [b+k^R] D_R - k_N D_N + b D_A &= 1 - k^R D_R - k_N D_N - b D_R + b D_A \\
&> 1 - k^R D_R - k_N D_N > 0.
\end{aligned} \tag{131}$$

Because $D_A < 0$ from (128), (131) implies:

$$1 - [b+k^R] D_R - k_N D_N > 0. \tag{132}$$

(129) and (131) imply:

$$\frac{\partial \bar{p}_d}{\partial k^R} = \frac{Q^R(\bar{p}_d)}{1 - [b+k^R] D_R - k_N D_N + b D_A} > 0. \tag{133}$$

(125) and (131) – (133) imply:

$$\frac{\partial \bar{p}_b}{\partial k^R} - \frac{\partial \bar{p}_d}{\partial k^R} = \frac{Q^R(\bar{p}_b)}{1 - [b+k^R] D_R - k_N D_N} - \frac{Q^R(\bar{p}_d)}{1 - [b+k^R] D_R - k_N D_N + b D_A} < 0$$

$$\begin{aligned}
&\Leftrightarrow \frac{Q^R(\bar{p}_b)}{1 - [b + k^R] D_R - k_N D_N} < \frac{Q^R(\bar{p}_d)}{1 - [b + k^R] D_R - k_N D_N + b D_A} \\
&\Leftrightarrow \frac{Q^R(\bar{p}_b)}{Q^R(\bar{p}_d)} < \frac{1 - [b + k^R] D_R - k_N D_N}{1 - [b + k^R] D_R - k_N D_N + b D_A}.
\end{aligned} \tag{134}$$

(62) implies that $Q^R(\bar{p}_b) < Q^R(\bar{p}_d)$. Therefore:

$$\frac{Q^R(\bar{p}_b)}{Q^R(\bar{p}_d)} < 1. \tag{135}$$

Furthermore, because $1 - [b + k^R] D_R - k_N D_N + b D_A > 0$ from (131):

$$\begin{aligned}
&\frac{1 - [b + k^R] D_R - k_N D_N}{1 - [b + k^R] D_R - k_N D_N + b D_A} > 1 \\
&\Leftrightarrow 1 - [b + k^R] D_R - k_N D_N > 1 - [b + k^R] D_R - k_N D_N + b D_A \\
&\Leftrightarrow D_A < 0.
\end{aligned} \tag{136}$$

(128) implies that the last inequality in (136) holds. (135) and (136) imply that (134) holds. Therefore, because $\bar{p}_b > \bar{p}_d > 0$ from Proposition 1, (125) and (134) imply that $\frac{\partial(\bar{p}_b - \bar{p}_d)}{\partial k^R} < 0$.

To prove that $\frac{\partial(\bar{p}_b - \bar{p}_d)}{\partial c_N} > 0$, observe that (7) and (8) imply:

$$\frac{\partial \bar{p}_d}{\partial c_N} = \frac{b[b + k][k_A - b]}{D_2} \quad \text{and} \quad \frac{\partial \bar{p}_b}{\partial c_N} = \frac{b[b + k]k_A}{D_2 + b k_N[2b + k]}. \tag{137}$$

(137) implies:

$$\begin{aligned}
\frac{\partial \bar{p}_b}{\partial c_N} - \frac{\partial \bar{p}_d}{\partial c_N} &= \frac{b[b + k]k_A}{D_2 + b k_N[2b + k]} - \frac{b[b + k][k_A - b]}{D_2} > 0 \\
&\Leftrightarrow \frac{k_A}{D_2 + b k_N[2b + k]} > \frac{k_A - b}{D_2} \\
&\Leftrightarrow D_2 k_A > [D_2 + b k_N(2b + k)][k_A - b] \\
&\Leftrightarrow D_2 k_A > D_2 k_A - b D_2 + b k_N[2b + k][k_A - b] \\
&\Leftrightarrow D_2 - k_N[2b + k][k_A - b] > 0.
\end{aligned} \tag{138}$$

The inequality in (138) holds because, from (7):

$$\begin{aligned}
&D_2 - k_N[2b + k][k_A - b] \\
&= b[b + k][k_N + k_A] + [2b + k]k_N[k_A - b]
\end{aligned}$$

$$\begin{aligned}
& + [2b + k][k_N + k_A][b + k^R] - k_N[2b + k][k_A - b] \\
& = b[b + k][k_N + k_A] + [2b + k][k_N + k_A][b + k^R] > 0.
\end{aligned}$$

To prove that $\frac{\partial(\bar{p}_b - \bar{p}_d)}{\partial c_A} < 0$, observe that (7) and (8) imply:

$$\frac{\partial \bar{p}_d}{\partial c_A} = \frac{b[b + k][k_N + b]}{D_2} \quad \text{and} \quad \frac{\partial \bar{p}_b}{\partial c_A} = \frac{b[b + k]k_N}{D_2 + b k_N[2b + k]}. \quad (139)$$

(139) implies:

$$\begin{aligned}
\frac{\partial \bar{p}_b}{\partial c_A} - \frac{\partial \bar{p}_d}{\partial c_A} &= \frac{b[b + k]k_N}{D_2 + b k_N[2b + k]} - \frac{b[b + k][k_N + b]}{D_2} < 0 \\
&\Leftrightarrow \frac{b[b + k]k_N}{D_2 + b k_N[2b + k]} < \frac{b[b + k][k_N + b]}{D_2} \\
&\Leftrightarrow b D_2 [b + k] k_N < [D_2 + b k_N (2b + k)] b [b + k] [k_N + b] \\
&\Leftrightarrow D_2 k_N < [D_2 + b k_N (2b + k)] [k_N + b] \\
&\Leftrightarrow D_2 k_N < D_2 k_N + b D_2 + b k_N [2b + k] [k_N + b] \\
&\Leftrightarrow D_2 + k_N [2b + k] [k_N + b] > 0.
\end{aligned} \quad (140)$$

The inequality in (140) holds because, from (7):

$$\begin{aligned}
D_2 &= b[b + k][k_N + k_A] + [2b + k] \{ k_N [k_A - b] + [k_N + k_A][b + k^R] \} \\
&= b[b + k][k_N + k_A] + [2b + k] \{ k_N [k_A + k^R] + k_A [b + k^R] \} > 0.
\end{aligned}$$

To prove that $\frac{\partial(\bar{p}_b - \bar{p}_d)}{\partial k_A} < 0$, we introduce the following:

Definition. $Y_1 \equiv b[b + k] \{ k_N [a(b + k) + bc - (2b + k)c_A] \\ + [c_N - c_A][b(b + k) + (2b + k)(b + k^R)] \}.$ (141)

Observe that:

$$Y_1 > 0. \quad (142)$$

(142) holds because $c_N \geq c_A$ by assumption, and (3) implies:

$$[a - c_A][2b + k] - b[a - c] > 0 \Rightarrow a[b + k] + bc - [2b + k]c_A > 0.$$

(8) implies:

$$(D_3)^2 \frac{\partial \bar{p}_b}{\partial k_A} = D_3 \{ [a(b + k) + bc][b + k^R + k_N] + b[b + k]c_N \}$$

$$\begin{aligned}
& - \left\{ b[b+k] + [2b+k] [k_N + b + k^R] \right\} \\
& \quad \cdot \left\{ [a(b+k) + bc] [(b+k^R)(k_A + k_N) + k_A k_N] \right. \\
& \quad \quad \left. + b[b+k] [c_N k_A + c_A k_N] \right\} \\
= & \left\{ [a(b+k) + bc] [b+k^R + k_N] + b[b+k] c_N \right\} \\
& \quad \cdot \left\{ b[b+k] [k_A + k_N] + [2b+k] [k_A k_N + (k_A + k_N)(b+k^R)] \right\} \\
& - \left\{ b[b+k] + [2b+k] [k_N + b + k^R] \right\} \\
& \quad \cdot \left\{ [a(b+k) + bc] [(b+k^R)(k_A + k_N) + k_A k_N] \right. \\
& \quad \quad \left. + b[b+k] [c_N k_A + c_A k_N] \right\} \\
= & [a(b+k) + bc] [b+k^R + k_N] b[b+k] [k_A + k_N] \\
& + [a(b+k) + bc] [b+k^R + k_N] [2b+k] [(b+k^R)(k_A + k_N) + k_A k_N] \\
& + b[b+k] c_N b[b+k] [k_A + k_N] \\
& + b[b+k] c_N [2b+k] [(b+k^R)(k_A + k_N) + k_A k_N] \\
& - b[b+k] [a(b+k) + bc] [(b+k^R)(k_A + k_N) + k_A k_N] \\
& - b[b+k] b[b+k] [c_N k_A + c_A k_N] \\
& - [2b+k] [k_N + b + k^R] [a(b+k) + bc] [(b+k^R)(k_A + k_N) + k_A k_N] \\
& - [2b+k] [k_N + b + k^R] b[b+k] [c_N k_A + c_A k_N] \equiv \Phi. \tag{143}
\end{aligned}$$

(143) implies:

$$\Phi = [a(b+k) + bc] \Phi_A + b[b+k] \Phi_B \tag{144}$$

where

$$\begin{aligned}
\Phi_A & \equiv b[b+k] [b+k^R + k_N] [k_A + k_N] \\
& + [2b+k] [b+k^R + k_N] [(b+k^R)(k_A + k_N) + k_A k_N] \\
& - b[b+k] [(b+k^R)(k_A + k_N) + k_A k_N] \\
& - [2b+k] [b+k^R + k_N] [(b+k^R)(k_A + k_N) + k_A k_N] \\
= & b[b+k] \left\{ [b+k^R + k_N] [k_A + k_N] - [(b+k^R)(k_A + k_N) + k_A k_N] \right\} \\
= & b[b+k] \{ k_N [k_A + k_N] - k_A k_N \} = b[b+k] (k_N)^2 \quad \text{and} \tag{145}
\end{aligned}$$

$$\begin{aligned}
\Phi_B &\equiv c_N b [b+k] [k_A + k_N] + c_N [2b+k] [(b+k^R)(k_A + k_N) + k_A k_N] \\
&\quad - b [b+k] [c_N k_A + c_A k_N] - [2b+k] [k_N + b + k^R] [c_N k_A + c_A k_N] \\
&= b [b+k] k_N [c_N - c_A] + [2b+k] \Phi_C
\end{aligned} \tag{146}$$

where

$$\begin{aligned}
\Phi_C &\equiv c_N [(b+k^R)(k_A + k_N) + k_A k_N] - [k_N + b + k^R] [c_N k_A + c_A k_N] \\
&= c_N [(b+k^R)(k_A + k_N) + k_A k_N - k_A (k_N + b + k^R)] - c_A k_N [k_N + b + k^R] \\
&= c_N k_N [b+k^R] - c_A k_N [k_N + b + k^R] = [b+k^R] k_N [c_N - c_A] - c_A (k_N)^2.
\end{aligned} \tag{147}$$

(146) and (147) imply:

$$\begin{aligned}
\Phi_B &= b [b+k] k_N [c_N - c_A] + [2b+k] \{ [b+k^R] k_N [c_N - c_A] - c_A (k_N)^2 \} \\
&= k_N [c_N - c_A] \{ b [b+k] + [2b+k] [b+k^R] \} - [2b+k] c_A (k_N)^2.
\end{aligned} \tag{148}$$

(141), (144), (145), and (148) imply:

$$\begin{aligned}
\Phi &= [a(b+k) + b c] b [b+k] (k_N)^2 \\
&\quad + b [b+k] k_N [c_N - c_A] \{ b [b+k] + [2b+k] [b+k^R] \} \\
&\quad - b [b+k] [2b+k] c_A (k_N)^2 \\
&= b [b+k] k_N \{ [a(b+k) + b c] k_N - [2b+k] c_A k_N \\
&\quad \quad + [c_N - c_A] [b(b+k) + (2b+k)(b+k^R)] \} \\
&= b [b+k] k_N \{ k_N [a(b+k) + b c - (2b+k) c_A] \\
&\quad \quad + [c_N - c_A] [b(b+k) + (2b+k)(b+k^R)] \} = k_N Y_1.
\end{aligned} \tag{149}$$

(142), (143), and (149) imply that $\frac{\partial \bar{p}_b}{\partial k_A} = \frac{k_N Y_1}{(D_3)^2} > 0$.

(7) implies:

$$\begin{aligned}
(D_2)^2 \frac{\partial \bar{p}_d}{\partial k_A} &= D_2 \{ [a(b+k) + b c] [b+k^R + k_N] + b [b+k] c_N \} \\
&\quad - \{ b [b+k] + [2b+k] [k_N + b + k^R] \} \\
&\quad \cdot \{ [a(b+k) + b c] [(b+k^R)(k_A + k_N) + k_A k_N - b k_N] \\
&\quad \quad + b [b+k] [c_N (k_A - b) + c_A (k_N + b)] \}
\end{aligned}$$

$$\begin{aligned}
&= \{ [a(b+k) + bc] [b + k^R + k_N] + b[b+k] c_N \} \\
&\quad \cdot \{ b[b+k] [k_A + k_N] + [2b+k] [k_N (k_A + k^R) + k_A (b + k^R)] \} \\
&\quad - \{ b[b+k] + [2b+k] [k_N + b + k^R] \} \\
&\quad \cdot \{ [a(b+k) + bc] [k_A (b + k_N) + k^R (k_A + k_N)] \\
&\quad \quad + b[b+k] [c_N (k_A - b) + c_A (k_N + b)] \} \\
&= [a(b+k) + bc] [b + k^R + k_N] b[b+k] [k_N + k_A] \\
&\quad + [a(b+k) + bc] [b + k^R + k_N] [2b+k] [k_N (k_A + k^R) + k_A (b + k^R)] \\
&\quad + b[b+k] c_N b[b+k] [k_N + k_A] \\
&\quad + b[b+k] c_N [2b+k] [k_N (k_A + k^R) + k_A (b + k^R)] \\
&\quad - b[b+k] [a(b+k) + bc] [k_A (b + k_N) + k^R (k_A + k_N)] \\
&\quad - b[b+k] b[b+k] [c_N (k_A - b) + c_A (k_N + b)] \\
&\quad - [2b+k] [k_N + b + k^R] [a(b+k) + bc] [k_A (b + k_N) + k^R (k_A + k_N)] \\
&\quad - [2b+k] [k_N + b + k^R] b[b+k] [c_N (k_A - b) + c_A (k_N + b)] \equiv F. \tag{150}
\end{aligned}$$

(150) implies:

$$F = [a(b+k) + bc] F_1 + b[b+k] F_2 \tag{151}$$

where

$$\begin{aligned}
F_1 &\equiv [b + k^R + k_N] b[b+k] [k_N + k_A] \\
&\quad + [2b+k] [b + k^R + k_N] [k_N (k_A + k^R) + k_A (b + k^R)] \\
&\quad - b[b+k] [k_A (b + k_N) + k^R (k_A + k_N)] \\
&\quad - [2b+k] [k_N + b + k^R] [k_A (b + k_N) + k^R (k_A + k_N)] \\
&= b[b+k] \{ [b + k^R + k_N] [k_N + k_A] - [k_A (b + k_N) + k^R (k_A + k_N)] \} \\
&= b[b+k] \{ [b + k_N] [k_N + k_A] - k_A [b + k_N] \} = b[b+k] [b + k_N] k_N \tag{152}
\end{aligned}$$

and

$$\begin{aligned}
F_2 &\equiv b[b+k] [k_N + k_A] c_N + c_N [2b+k] [k_N (k_A + k^R) + k_A (b + k^R)] \\
&\quad - b[b+k] [c_N (k_A - b) + c_A (k_N + b)]
\end{aligned}$$

$$\begin{aligned}
& - [2b + k] [k_N + b + k^R] [c_N (k_A - b) + c_A (k_N + b)] \\
= & b [b + k] [c_N (k_A + k_N) - c_N (k_A - b) - c_A (k_N + b)] \\
& + [2b + k] \{ c_N [k_N (k_A + k^R) + k_A (b + k^R)] \\
& \quad - [k_N + b + k^R] [c_N (k_A - b) + c_A (k_N + b)] \} \\
= & b [b + k] [c_N k_N + b c_N - c_A (k_N + b)] \\
& + [2b + k] \{ c_N [k_N (k_A + k^R) - k_N (k_A - b) + k_A (b + k^R) \\
& \quad - (k_A - b) (b + k^R)] - c_A [k_N + b] [k_N + b + k^R] \} \\
= & b [b + k] [k_N (c_N - c_A) + b (c_N - c_A)] \\
& + [2b + k] \{ c_N [k_N (b + k^R) + b (b + k^R)] - c_A [k_N + b] k_N \\
& \quad - c_A [k_N + b] [b + k^R] \} \\
= & b [b + k] [c_N - c_A] [k_N + b] \\
& + [2b + k] \{ [c_N - c_A] [k_N + b] [b + k^R] - c_A [k_N + b] k_N \} \\
= & b [b + k] [c_N - c_A] [k_N + b] \\
& + [2b + k] [k_N + b] [(b + k^R) (c_N - c_A) - c_A k_N] .
\end{aligned} \tag{153}$$

(141), (151), (152), and (153) imply:

$$\begin{aligned}
F &= [a(b + k) + bc] b [b + k] [b + k_N] k_N \\
&+ b [b + k] \{ b [b + k] [c_N - c_A] [k_N + b] \\
&\quad + [2b + k] [k_N + b] [(b + k^R) (c_N - c_A) - c_A k_N] \} \\
= & b [b + k] [b + k_N] \{ k_N [a(b + k) + bc - (2b + k) c_A] \\
&\quad + [c_N - c_A] [b(b + k) + (2b + k) (b + k^R)] \} \\
= & [b + k_N] Y_1 .
\end{aligned} \tag{154}$$

(142), (150), and (154) imply that $\frac{\partial \bar{p}_d}{\partial k_A} = \frac{[k_N + b] Y_1}{(D_2)^2} > 0$.

(141), (143), (149), (150), and (154) imply:

$$\frac{\partial (\bar{p}_b - \bar{p}_d)}{\partial k_A} = Y_1 \left[\frac{k_N}{(D_3)^2} - \frac{k_N + b}{(D_2)^2} \right] < 0 .$$

The inequality here holds because: (i) $Y_1 > 0$, from (142); (ii) $k_N < k_N + b$; and (iii) $D_3 > D_2 > 0$, from (7) and (8).

To prove that $\frac{\partial(\bar{p}_b - \bar{p}_d)}{\partial k_N} > 0$ if $k_A - b$ is sufficiently small, we introduce the following:

Definition. $Y_2 \equiv b[b+k] \{ k_A [a(b+k) + bc - (2b+k)c_N] - [c_N - c_A] [b(b+k) + (2b+k)(b+k^R)] \}.$ (155)

Observe that:

$$Y_2 > 0. \quad (156)$$

(156) follows from (3) and (155) because:

$$\begin{aligned} a[b+k] + bc - [2b+k]c_N &= a[2b+k] - ab + bc - [2b+k]c_N \\ &= [a - c_N][2b+k] - b[a - c] \quad \text{and} \\ b[b+k] + [2b+k][b+k^R] &= b^2 + bk + 2b^2 + 2bk^R + bk + k^R \\ &= 3b^2 + 2bk + 2bk^R + k^R = 3b^2 + 2b[k + k^R] + k^R. \end{aligned}$$

(8) implies:

$$\begin{aligned} (D_3)^2 \frac{\partial \bar{p}_b}{\partial k_N} &= D_3 \{ [a(b+k) + bc] [b + k^R + k_A] + b[b+k]c_A \} \\ &\quad - \{ b[b+k] + [2b+k][k_A + b + k^R] \} \\ &\quad \cdot \{ [a(b+k) + bc] [(b+k^R)(k_A + k_N) + k_A k_N] \\ &\quad + b[b+k][c_N k_A + c_A k_N] \} \\ &= \{ [a(b+k) + bc] [b + k^R + k_A] + b[b+k]c_A \} \\ &\quad \cdot \{ b[b+k][k_A + k_N] + [2b+k][k_A k_N + (k_A + k_N)(b + k^R)] \} \\ &\quad - \{ b[b+k] + [2b+k][k_A + b + k^R] \} \\ &\quad \cdot \{ [a(b+k) + bc] [(b+k^R)(k_A + k_N) + k_A k_N] \\ &\quad + b[b+k][c_N k_A + c_A k_N] \} \\ &= [a(b+k) + bc] [b + k^R + k_A] b[b+k][k_A + k_N] \\ &\quad + [a(b+k) + bc] [b + k^R + k_A] [2b+k] [(b+k^R)(k_A + k_N) + k_A k_N] \\ &\quad + b[b+k]c_A b[b+k][k_A + k_N] \end{aligned}$$

$$\begin{aligned}
& + b [b + k] c_A [2b + k] [(b + k^R)(k_A + k_N) + k_A k_N] \\
& - b [b + k] [a(b + k) + b c] [(b + k^R)(k_A + k_N) + k_A k_N] \\
& - b [b + k] b [b + k] [c_N k_A + c_A k_N] \\
& - [2b + k] [k_A + b + k^R] [a(b + k) + b c] [(b + k^R)(k_A + k_N) + k_A k_N] \\
& - [2b + k] [k_A + b + k^R] b [b + k] [c_N k_A + c_A k_N] \equiv \Lambda.
\end{aligned} \tag{157}$$

(157) implies:

$$\Lambda = [a(b + k) + b c] \Lambda_1 + b [b + k] \Lambda_2 \tag{158}$$

where

$$\begin{aligned}
\Lambda_1 & \equiv b [b + k] [b + k^R + k_A] [k_A + k_N] \\
& + [2b + k] [b + k^R + k_A] [(b + k^R)(k_A + k_N) + k_A k_N] \\
& - b [b + k] [(b + k^R)(k_A + k_N) + k_A k_N] \\
& - [2b + k] [b + k^R + k_A] [(b + k^R)(k_A + k_N) + k_A k_N] \\
& = b [b + k] \{ [b + k^R + k_A] [k_A + k_N] - [(b + k^R)(k_A + k_N) + k_A k_N] \} \\
& = b [b + k] \{ k_A [k_A + k_N] - k_A k_N \} = b [b + k] (k_A)^2 \quad \text{and}
\end{aligned} \tag{159}$$

$$\begin{aligned}
\Lambda_2 & \equiv c_A b [b + k] [k_A + k_N] + c_A [2b + k] [(b + k^R)(k_A + k_N) + k_A k_N] \\
& - b [b + k] [c_N k_A + c_A k_N] - [2b + k] [k_A + b + k^R] [c_N k_A + c_A k_N] \\
& = -b [b + k] k_A [c_N - c_A] + [2b + k] \Lambda_3
\end{aligned} \tag{160}$$

where

$$\begin{aligned}
\Lambda_3 & \equiv c_A [(b + k^R)(k_A + k_N) + k_A k_N] - [k_A + b + k^R] [c_N k_A + c_A k_N] \\
& = c_A [(b + k^R)(k_A + k_N) + k_A k_N - k_N (k_A + b + k^R)] - c_N k_A [k_A + b + k^R] \\
& = c_A k_A [b + k^R] - c_N k_A [k_A + b + k^R] = - [b + k^R] k_A [c_N - c_A] - c_N (k_A)^2.
\end{aligned} \tag{161}$$

(160) and (161) imply:

$$\begin{aligned}
\Lambda_2 & = -b [b + k] k_A [c_N - c_A] - [2b + k] \{ [b + k^R] k_A [c_N - c_A] - c_N (k_A)^2 \} \\
& = -k_A [c_N - c_A] \{ b [b + k] + [2b + k] [b + k^R] \} - [2b + k] c_N (k_A)^2.
\end{aligned} \tag{162}$$

(155), (158), (159), and (162) imply:

$$\begin{aligned}
\Lambda &= [a(b+k) + bc] b[b+k] (k_A)^2 \\
&\quad - b[b+k] k_A [c_N - c_A] \{ b[b+k] + [2b+k] [b+k^R] \} \\
&\quad - b[b+k] [2b+k] c_N (k_A)^2 \\
&= b[b+k] k_A \{ [a(b+k) + bc] k_A - [2b+k] c_N k_A \\
&\quad - [c_N - c_A] [b(b+k) + (2b+k) (b+k^R)] \} \\
&= b[b+k] k_A \{ k_A [a(b+k) + bc - (2b+k) c_N] \\
&\quad - [c_N - c_A] [b(b+k) + (2b+k) (b+k^R)] \} = k_A Y_2. \quad (163)
\end{aligned}$$

(156), (157), and (163) imply that $\frac{\partial \bar{p}_b}{\partial k_N} = \frac{k_A Y_2}{(D_3)^2} > 0$.

(7) implies:

$$\begin{aligned}
(D_2)^2 \frac{\partial \bar{p}_d}{\partial k_N} &= D_2 \{ [a(b+k) + bc] [b+k^R + k_A - b] + b[b+k] c_A \} \\
&\quad - \{ b[b+k] + [2b+k] [k_A - b + b + k^R] \} \\
&\quad \cdot \{ [a(b+k) + bc] [(b+k^R) (k_A + k_N) + k_A k_N - b k_N] \\
&\quad + b[b+k] [c_N (k_A - b) + c_A (k_N + b)] \} \\
&= \{ [a(b+k) + bc] [k^R + k_A] + b[b+k] c_A \} \\
&\quad \cdot \{ b[b+k] [k_A + k_N] + [2b+k] [k_N (k_A + k^R) + k_A (b + k^R)] \} \\
&\quad - \{ b[b+k] + [2b+k] [k_A + k^R] \} \\
&\quad \cdot \{ [a(b+k) + bc] [k_A (b + k_N) + k^R (k_A + k_N)] \\
&\quad + b[b+k] [c_N (k_A - b) + c_A (k_N + b)] \} \\
&= [a(b+k) + bc] [k^R + k_A] b[b+k] [k_A + k_N] \\
&\quad + [a(b+k) + bc] [k^R + k_A] [2b+k] [k_N (k_A + k^R) + k_A (b + k^R)] \\
&\quad + b[b+k] c_A b[b+k] [k_A + k_N] \\
&\quad + b[b+k] c_A [2b+k] [k_N (k_A + k^R) + k_A (b + k^R)]
\end{aligned}$$

$$\begin{aligned}
& -b[b+k][a(b+k)+bc][k_A(b+k_N)+k^R(k_A+k_N)] \\
& -b[b+k]b[b+k][c_N(k_A-b)+c_A(k_N+b)] \\
& -[2b+k][k_A+k^R][a(b+k)+bc][k_A(b+k_N)+k^R(k_A+k_N)] \\
& -[2b+k][k_A+k^R]b[b+k][c_N(k_A-b)+c_A(k_N+b)] \equiv \Gamma.
\end{aligned} \tag{164}$$

(164) implies:

$$\Gamma = [a(b+k)+bc]F_1 + b[b+k]F_2 \tag{165}$$

where

$$\begin{aligned}
\Gamma_1 & \equiv b[b+k][k^R+k_A][k_A+k_N] \\
& + [2b+k][k^R+k_A][k_N(k_A+k^R)+k_A(b+k^R)] \\
& - b[b+k][k_A(b+k_N)+k^R(k_A+k_N)] \\
& - [2b+k][k_A+k^R][k_A(b+k_N)+k^R(k_A+k_N)] \\
& = b[b+k]\{[k^R+k_A][k_A+k_N]-[k_A(b+k_N)+k^R(k_A+k_N)]\} \\
& = b[b+k]\{k_A[k_A+k_N]-k_A[b+k_N]\} = b[b+k][k_A-b]k_A
\end{aligned} \tag{166}$$

and

$$\begin{aligned}
\Gamma_2 & \equiv b[b+k][k_A+k_N]c_A + c_A[2b+k][k_N(k_A+k^R)+k_A(b+k^R)] \\
& - b[b+k][c_N(k_A-b)+c_A(k_N+b)] \\
& - [2b+k][k_A+k^R][c_N(k_A-b)+c_A(k_N+b)] \\
& = b[b+k][c_A(k_A+k_N)-c_N(k_A-b)-c_A(k_N+b)] \\
& + [2b+k]\{c_A[k_N(k_A+k^R)+k_A(b+k^R)] \\
& \quad - [k_A+k^R][c_N(k_A-b)+c_A(k_N+b)]\} \\
& = b[b+k][c_A(k_A-b)-c_N(k_A-b)] \\
& + [2b+k]\{c_A[k_N(k_A+k^R)+k_A(b+k^R)-(k_A+k^R)(k_N+b)] \\
& \quad - c_N[k_A+k^R][k_A-b]\} \\
& = -b[b+k][k_A-b][c_N-c_A] \\
& + [2b+k]\{c_A[k_A(b+k^R)-b(k_A+k^R)]-c_N[k_A+k^R][k_A-b]\}
\end{aligned}$$

$$\begin{aligned}
&= -b[b+k][k_A-b][c_N-c_A] \\
&\quad + [2b+k] \{ c_A k^R [k_A-b] - c_N k^R [k_A-b] - c_N k_A [k_A-b] \} \\
&= [k_A-b] \{ -b[b+k][c_N-c_A] - [2b+k] k^R [c_N-c_A] - [2b+k] k_A c_N \} \\
&= -[k_A-b] \{ [c_N-c_A] [b(b+k) + (2b+k) k^R] + [2b+k] k_A c_N \}. \tag{167}
\end{aligned}$$

(155), (165), (166), and (167) imply:

$$\begin{aligned}
\Gamma &= [a(b+k) + bc] b[b+k][k_A-b] k_A \\
&\quad - b[b+k][k_A-b] \{ [c_N-c_A] [b(b+k) + (2b+k) k^R] \\
&\quad \quad + [2b+k] k_A c_N \} \\
&= b[b+k][k_A-b] \{ [a(b+k) + bc] k_A - [c_N-c_A] [b(b+k) + (2b+k) k^R] \\
&\quad \quad - [2b+k] k_A c_N \} \\
&= [k_A-b] b[b+k] \{ k_A [a(b+k) + bc - (2b+k) c_N] \\
&\quad \quad - [c_N-c_A] [b(b+k) + (2b+k) k^R] \} \\
&= [k_A-b] Y_2. \tag{168}
\end{aligned}$$

(164) and (168) imply that $\frac{\partial \bar{p}_d}{\partial k_N} = \frac{[k_A-b] Y_2}{(D_2)^2} \gtrless 0 \Leftrightarrow k_A \gtrless b$.

(155), (156), (157), (163), (164), and (168) imply:

$$\frac{\partial (\bar{p}_b - \bar{p}_d)}{\partial k_N} = Y_2 \left[\frac{k_A}{(D_3)^2} - \frac{k_A-b}{(D_2)^2} \right] > 0 \text{ if } k_A-b \text{ is sufficiently small.} \tag{169}$$

To prove that $\frac{\partial (\bar{p}_b - \bar{p}_d)}{\partial c} > 0$, observe that (7) implies:

$$\begin{aligned}
\frac{\partial \bar{p}_d}{\partial c} &= \frac{b}{D_2} [(b+k^R)(k_A+k_N) + k_N(k_A-b)] \\
&= \frac{b}{D_2} [k_A(b+k^R) + k_N(b+k^R+k_A-b)] \\
&= \frac{b}{D_2} [k_A(b+k^R) + k_N(k_A+k^R)] > 0. \tag{170}
\end{aligned}$$

Furthermore, (8) implies:

$$\frac{\partial \bar{p}_b}{\partial c} = \frac{1}{D_3} \{ b[(b+k^R)(k_A+k_N) + k_N k_A] \}$$

$$\begin{aligned}
&= \frac{b}{D_3} \left[(b + k^R) (k_A + k_N) + k_N k_A \right] \\
&= \frac{b}{D_3} \left[k_A (b + k^R) + k_N (k_A + k^R + b) \right] > 0.
\end{aligned} \tag{171}$$

(170) and (171) imply:

$$\begin{aligned}
\frac{\partial (\bar{p}_b - \bar{p}_d)}{\partial c} &\stackrel{s}{=} \frac{k_A [b + k^R] + k_N [k_A + k^R + b]}{D_3} - \frac{k_A [b + k^R] + k_N [k_A + k^R]}{D_2} > 0 \\
&\Leftrightarrow \frac{k_A [b + k^R] + k_N [k^R + b + k_A]}{D_3} > \frac{k_A [b + k^R] + k_N [k_A + k^R]}{D_2} \\
&\Leftrightarrow \frac{k_A [b + k^R] + k_N [k^R + b + k_A]}{D_2 + b k_N [2b + k]} > \frac{k_A [b + k^R] + k_N [k_A + k^R]}{D_2} \\
&\Leftrightarrow \frac{Z + b k_N}{D_2 + b k_N [2b + k]} > \frac{Z}{D_2} \text{ where } Z \equiv k_A [b + k^R] + k_N [k_A + k^R].
\end{aligned} \tag{172}$$

(172) implies:

$$\begin{aligned}
\frac{\partial (\bar{p}_b - \bar{p}_d)}{\partial c} > 0 &\Leftrightarrow Z D_2 + b k_N D_2 > Z D_2 + Z b k_N [2b + k] \\
&\Leftrightarrow b k_N D_2 > Z b k_N [2b + k] \Leftrightarrow D_2 > Z [2b + k] \\
&\Leftrightarrow D_2 > [k_A (b + k^R) + k_N (k_A + k^R)] [2b + k] \\
&\Leftrightarrow b [b + k] [k_N + k_A] + k_N [k_A - b] [2b + k] + [k_N + k_A] [2b + k] [b + k^R] \\
&\quad > [k_A (b + k^R) + k_N (k_A + k^R)] [2b + k] \\
&\Leftrightarrow b [b + k] [k_N + k_A] + k_N [k_A - b] [2b + k] + [k_N + k_A] [2b + k] [b + k^R] \\
&\quad > [(b + k^R) (k_A + k_N) + k_N (k_A - b)] [2b + k] \\
&\Leftrightarrow b [b + k] [k_N + k_A] > 0. \blacksquare
\end{aligned}$$

Recall that welfare is:

$$W(\bar{p}) \equiv S(\bar{p}) - r [\bar{p} q_A + (a - b [q_A + q_N + q]) q_N] = S(\bar{p}) - r V(\bar{p}) \tag{173}$$

where $r > 0$ is a parameter and $S(\cdot)$ denotes consumer surplus. The gross value that consumers derive from Q units of output is:

$$\frac{1}{2} [a - P(Q)] Q + P(Q) Q = \frac{1}{2} [a + P(Q)] Q = \frac{1}{2} [a + a - b Q] Q = a Q - \frac{b}{2} Q^2.$$

Therefore, consumer surplus when the price cap is $\bar{p}$ is:

$$S(\bar{p}) = aQ - \frac{b}{2} Q^2 - \bar{p} q_A - P(Q) [q_N + q]. \quad (174)$$

Lemma 1. *Equilibrium consumer surplus, $S(\bar{p})$, is a strictly decreasing, strictly convex function of $\bar{p}$ for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$.*

Proof. (57) implies that when $\bar{p} \in (\bar{p}_d, \bar{p}_b)$, so $P(Q) = \bar{p}$:

$$Q = \frac{a - \bar{p}}{b} \Rightarrow \frac{\partial Q}{\partial \bar{p}} = -\frac{1}{b}. \quad (175)$$

(174) and (175) imply that for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$, where $P(Q) = \bar{p}$:

$$\begin{aligned} \frac{\partial S(\bar{p})}{\partial \bar{p}} &= a \frac{\partial Q}{\partial \bar{p}} - bQ \frac{\partial Q}{\partial \bar{p}} - Q - \bar{p} \frac{\partial Q}{\partial \bar{p}} = -\frac{a}{b} + Q - Q + \frac{\bar{p}}{b} \\ &= -\frac{a - \bar{p}}{b} < 0 \Rightarrow \frac{\partial^2 S(\bar{p})}{\partial (\bar{p})^2} = \frac{1}{b} > 0. \blacksquare \end{aligned} \quad (176)$$

Lemma A7. $V(\bar{p}_0) < V(\bar{p}_b)$.

Proof. Lemmas A1 and A3 imply that because $q_A(\bar{p}_0) = 0$ and $P(Q(\bar{p}_b)) = \bar{p}_b$:

$$\begin{aligned} V(\bar{p}_0) &= \bar{p}_0 q_N(\bar{p}_0) = \bar{p}_0 \frac{[a - c_N][2b + k] - b[a - c]}{[2b + k_N + k^R][2b + k] - b^2}; \\ V(\bar{p}_b) &= \bar{p}_b Q^R(\bar{p}_b) = \bar{p}_b \frac{[b + k][a - \bar{p}_b] - b[\bar{p}_b - c]}{b[b + k]}. \end{aligned} \quad (177)$$

$$\text{Definition. } D_N \equiv [2b + k_N + k^R][2b + k] - b^2. \quad (178)$$

Because $\bar{p}_0 < \bar{p}_b$, (177) and (178) imply that $V(\bar{p}_0) < V(\bar{p}_b)$ if:

$$\begin{aligned} q_N(\bar{p}_0) &= \frac{[a - c_N][2b + k] - b[a - c]}{D_N} < \frac{[b + k][a - \bar{p}_b] - b[\bar{p}_b - c]}{b[b + k]} = Q^R(\bar{p}_b) \\ \Leftrightarrow \frac{a[b + k] + ab - c_N[2b + k] - ba + bc}{D_N} &< \frac{[b + k]a - [b + k]\bar{p}_b - b\bar{p}_b + bc}{b[b + k]} \\ \Leftrightarrow \frac{a[b + k] + bc - c_N[2b + k]}{D_N} &< \frac{[b + k]a + bc - [2b + k]\bar{p}_b}{b[b + k]} \\ \Leftrightarrow \frac{a[b + k] + bc - c_N[2b + k]}{D_N} b[b + k] - [b + k]a - bc &< -[2b + k]\bar{p}_b \end{aligned}$$

$$\begin{aligned}
&\Leftrightarrow \frac{a[b+k] + bc}{2b+k} - \frac{a[b+k] + bc - c_N[2b+k]}{[2b+k]D_N} b[b+k] > \bar{p}_b \\
&\Leftrightarrow \frac{a[b+k] + bc}{2b+k} - \frac{[a(b+k) + bc]b[b+k] - c_N[2b+k]b[b+k]}{[2b+k]D_N} > \bar{p}_b \\
&\Leftrightarrow \frac{1}{[2b+k]D_N} \{ [a(b+k) + bc] [(2b+k_N + k^R)(2b+k) - b^2 - b(b+k)] \\
&\quad + c_N[2b+k]b[b+k] \} > \bar{p}_b \\
&\Leftrightarrow \frac{1}{[2b+k]D_N} \{ [a(b+k) + bc] [(2b+k_N + k^R)(2b+k) - b(2b+k)] \\
&\quad + c_N[2b+k]b[b+k] \} > \bar{p}_b \\
&\Leftrightarrow \frac{[a(b+k) + bc][2b+k_N + k^R - b] + c_N b[b+k]}{D_N} > \bar{p}_b \\
&\Leftrightarrow \frac{[a(b+k) + bc][b+k_N + k^R] + c_N b[b+k]}{D_N} > \bar{p}_b. \tag{179}
\end{aligned}$$

(8) implies:

$$\bar{p}_b = \frac{[a(b+k) + bc][(b+k^R)(k_N + k_A) + k_N k_A] + b c_N [b+k] k_A + b k_N [b+k] c_A}{b[b+k][k_N + k_A] + k_N k_A [2b+k] + [k_N + k_A][2b+k][b+k^R]}. \tag{180}$$

As established in the proof of Proposition 4 (just below (149)), $\bar{p}_b$ is increasing in k_A . Therefore, (180) implies that because $k_A \leq k_N$ by assumption:

$$\bar{p}_b \leq \frac{[a(b+k) + bc][2k_N(b+k^R) + (k_N)^2] + b c_N [b+k] k_N + b k_N [b+k] c_A}{2b[b+k]k_N + (k_N)^2[2b+k] + 2k_N[2b+k][b+k^R]}. \tag{181}$$

(8) implies that $\bar{p}_b$ is increasing in c_A . Therefore, because $c_A \leq c_N$ by assumption, (181) implies:

$$\begin{aligned}
\bar{p}_b &\leq \frac{[a(b+k) + bc][2k_N(b+k^R) + (k_N)^2] + 2b c_N [b+k] k_N}{2b[b+k]k_N + (k_N)^2[2b+k] + 2k_N[2b+k][b+k^R]} \\
&= \frac{[a(b+k) + bc][2(b+k^R) + k_N] + 2b c_N [b+k]}{2b[b+k] + k_N[2b+k] + 2[2b+k][b+k^R]} \\
&= \frac{[a(b+k) + bc][b+k^R + \frac{k_N}{2}] + b c_N [b+k]}{b[b+k] + \frac{k_N}{2}[2b+k] + [2b+k][b+k^R]} \\
&= \frac{[a(b+k) + bc][b+k^R + \frac{k_N}{2}] + b c_N [b+k]}{[2b+k][b+k^R + \frac{k_N}{2}] + b[b+k]}
\end{aligned}$$

$$= \frac{[a(b+k) + bc][b + k^R + \frac{k_N}{2}] + bc_N[b+k]}{[2b+k][2b + k^R + \frac{k_N}{2}] - b^2}. \quad (182)$$

The last equality in (182) holds because:

$$\begin{aligned} & [2b+k] \left[b + k^R + \frac{k_N}{2} \right] + b[b+k] \\ &= [2b+k] \left[2b + k^R + \frac{k_N}{2} \right] - b[2b+k] + b[b+k] \\ &= [2b+k] \left[2b + k^R + \frac{k_N}{2} \right] - 2b^2 - bk + b^2 + bk \\ &= [2b+k] \left[2b + k^R + \frac{k_N}{2} \right] - b^2. \end{aligned}$$

(178), (179), and (182) imply that the Lemma holds if:

$$\begin{aligned} & \frac{[a(b+k) + bc][b + k^R + \frac{k_N}{2}] + bc_N[b+k]}{[2b+k][2b + k^R + \frac{k_N}{2}] - b^2} \\ & < \frac{[a(b+k) + bc][b + k^R + k_N] + bc_N[b+k]}{[2b+k][2b + k^R + k_N] - b^2}. \end{aligned} \quad (183)$$

Definition. $f(x) \equiv \frac{A[b + k^R + x] + bc_N[b+k]}{[2b+k][2b + k^R + x] - b^2}$ where $A \equiv a[b+k] + bc$. (184)

(184) implies that (183) holds if $\frac{\partial f}{\partial x} > 0$. (178) and (184) imply:

$$\begin{aligned} \frac{\partial f(\cdot)}{\partial x} &\stackrel{s}{=} \{ [2b+k][2b + k^R + x] - b^2 \} A \\ &\quad - [2b+k] \{ A[b + k^R + x] + bc_N[b+k] \} \\ &= A \{ [2b+k][2b + k^R + x - (b + k^R + x)] - b^2 \} - b[b+k][2b+k]c_N \\ &= A \{ b[2b+k] - b^2 \} - b[b+k][2b+k]c_N \\ &= Ab[b+k] - b[b+k][2b+k]c_N \stackrel{s}{=} A - [2b+k]c_N \\ &= a[b+k] + bc - [2b+k]c_N > 0. \end{aligned}$$

The inequality here holds because (3) implies:

$$[a - c_N][2b+k] - b[a - c] > 0$$

$$\begin{aligned}
&\Rightarrow [a - c_N][b + k] + b[a - c_N] - b[a - c] > 0 \\
&\Rightarrow [a - c_N][b + k] + b[c - c_N] > 0 \\
&\Rightarrow a[b + k] - c_N[b + k] + b[c - c_N] > 0 \\
&\Rightarrow a[b + k] + bc - [2b + k]c_N > 0. \blacksquare
\end{aligned}$$

Proposition A1. $\bar{p}^* \in [\bar{p}_0, \bar{p}_d]$.

Proof. Proposition 3 and Lemma 1 imply that $W(\cdot)$ is a strictly convex function of $\bar{p}$ for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$. Therefore, $\bar{p}^* \notin (\bar{p}_d, \bar{p}_b)$. Lemma A1 implies that $W(\bar{p}) = W(\bar{p}_0)$ for all $\bar{p} < \bar{p}_0$. Lemma A4 implies that $W(\bar{p}) = W(\bar{p}_b)$ for all $\bar{p} > \bar{p}_b$. Therefore, $\bar{p}^* \in [\bar{p}_0, \bar{p}_d] \cup \bar{p}_b$.

It remains to show that $\bar{p}^* \neq \bar{p}_b$. The proof of Lemma A7 establishes that:

$$Q^R(\bar{p}_0) < Q^R(\bar{p}_b) \quad (185)$$

where $Q^R(\bar{p})$ is R 's total output when the price cap is $\bar{p}$. Lemma A6 and Proposition 2 imply:

$$Q^R(\bar{p}_b) < Q^R(\bar{p}_d). \quad (186)$$

(185) and (186) imply that $Q^R(\bar{p}_0) < Q^R(\bar{p}_b) < Q^R(\bar{p}_d)$. $Q^R(\bar{p})$ is continuous and monotonically increasing in $\bar{p}$ for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$ (from Lemma A2). Therefore, the intermediate value theorem implies that there exists a $\bar{p}_E \in (\bar{p}_0, \bar{p}_d)$ such that:

$$Q^R(\bar{p}_E) = Q^R(\bar{p}_b). \quad (187)$$

(12) implies that the rival's output q is determined by:

$$a - b[Q^R(\bar{p}) + q(\bar{p})] - c - bq(\bar{p}) - kq(\bar{p}) = 0. \quad (188)$$

(187) and (188) imply:

$$q(\bar{p}_E) = q(\bar{p}_b). \quad (189)$$

(187) and (189) imply:

$$Q(\bar{p}_E) = Q(\bar{p}_b) \text{ and } P(Q(\bar{p}_E)) = P(Q(\bar{p}_b)). \quad (190)$$

R 's revenue is:

$$\begin{aligned}
V_2(\bar{p}_E) &= \bar{p}_E q_A(\bar{p}_E) + P(Q(\bar{p}_E)) q_N(\bar{p}_E) \\
&< P(Q(\bar{p}_E)) q_A(\bar{p}_E) + P(Q(\bar{p}_E)) q_N(\bar{p}_E) \\
&= P(Q(\bar{p}_E)) Q^R(\bar{p}_E) = P(Q(\bar{p}_b)) Q^R(\bar{p}_b) = V_3(\bar{p}_b).
\end{aligned} \quad (191)$$

The inequality in (191) holds because $\bar{p}_E < P(Q(\bar{p}_E))$, since $\bar{p}_E \in (\bar{p}_0, \bar{p}_d)$. The penultimate equality in (191) reflects (190). The last equality in (191) holds because $P(Q(\bar{p}_b)) = \bar{p}_b$.

(174) and (190) imply:

$$\begin{aligned}
S(\bar{p}_E) &= a Q(\bar{p}_E) - \frac{b}{2} Q(\bar{p}_E)^2 - P(Q(\bar{p}_E)) [q(\bar{p}_E) + q_N(\bar{p}_E)] - \bar{p}_E q_A(\bar{p}_E) \\
&> a Q(\bar{p}_E) - \frac{b}{2} Q(\bar{p}_E)^2 - P(Q(\bar{p}_E)) [q(\bar{p}_E) + q_N(\bar{p}_E) + q_A(\bar{p}_E)] \\
&= a Q(\bar{p}_E) - \frac{b}{2} Q(\bar{p}_b)^2 - P(Q(\bar{p}_E)) Q(\bar{p}_E) \\
&= a Q(\bar{p}_E) - \frac{b}{2} Q(\bar{p}_b)^2 - P(Q(\bar{p}_b)) Q(\bar{p}_b) = S(\bar{p}_b). \tag{192}
\end{aligned}$$

The inequality in (192) holds because $\bar{p}_E < P(Q(\bar{p}_E))$, since $\bar{p}_E \in (\bar{p}_0, \bar{p}_d)$. (191) and (192) imply that consumer surplus is higher and R 's revenue is lower when $\bar{p} = \bar{p}_E$ than when $\bar{p} = \bar{p}_b$. Therefore, $W(\bar{p}_E) > W(\bar{p}_b)$, so $\bar{p}^* \neq \bar{p}_b$. ■

Lemma 2. For $\bar{p} \in (\bar{p}_0, \bar{p}_d)$: (i) $V(\bar{p})$ is a strictly convex function of $\bar{p}$; (ii) $\frac{\partial V(\bar{p})}{\partial \bar{p}} \leq 0 \Leftrightarrow \bar{p} \leq \bar{p}_{Vm}$ where $\bar{p}_{Vm} \in [\bar{p}_0, \bar{p}_d]$; and (iii) $\bar{p}_{Vm} > \bar{p}_0$ if $\Phi_2 > 0$, where

$$\begin{aligned}
\Phi_2 \equiv & \{ k^R [2b + k] [k^R (2b + k) + 2b(3b + 2k)] \\
& + k_N [2b + k] [k^R (2b + k) + b^2] + b^2 [5b^2 + 6bk + 2k^2] \} c_N \\
& - \{ b [3b + 2k] + [2b + k] [k_N + k^R] \}^2 c_A \\
& - b [b^2 - k k_N + (2b + k) k^R] [a(b + k) + bc]. \tag{193}
\end{aligned}$$

Corollary to Lemma 2. $\left. \frac{\partial V(\bar{p})}{\partial \bar{p}} \right|_{\bar{p}=\bar{p}_0} < 0$ if $\Phi_2 > 0$.

Proof of Lemma 2 and its Corollary.

Define:

$$\tilde{V}_2(\bar{p}) \equiv q_{A2}(\bar{p}) \bar{p} + q_{N2}(\bar{p}) P(Q_2(\bar{p})) \tag{194}$$

where $q_{A2}(\bar{p})$ and $q_{N2}(\bar{p})$ are as defined in (20) and (21), respectively. Observe that $\tilde{V}_2(\bar{p}) = V(\bar{p})$ for $\bar{p} \in [\bar{p}_0, \bar{p}_d]$.

Because $P(Q_2) = a - bQ_2$, (194) implies:

$$\frac{\partial \tilde{V}_2(\bar{p})}{\partial \bar{p}} = q_{A2} + \bar{p} \frac{\partial q_{A2}}{\partial \bar{p}} + P(Q_2) \frac{\partial q_{N2}}{\partial \bar{p}} - b q_{N2} \frac{\partial Q_2}{\partial \bar{p}}. \tag{195}$$

(2) and Lemma A2 imply:

$$\frac{\partial^2 q_{A2}}{\partial (\bar{p})^2} = \frac{\partial^2 q_{N2}}{\partial (\bar{p})^2} = \frac{\partial^2 Q_2}{\partial (\bar{p})^2} = \frac{\partial^2 Q_2}{\partial (\bar{p})^2} = 0. \tag{196}$$

(195) and (196) imply:

$$\begin{aligned}\frac{\partial^2 \tilde{V}_2(\bar{p})}{\partial (\bar{p})^2} &= \frac{\partial q_{A2}}{\partial \bar{p}} + \frac{\partial q_{A2}}{\partial \bar{p}} - b \frac{\partial Q_2}{\partial \bar{p}} \frac{\partial q_{N2}}{\partial \bar{p}} - b \frac{\partial q_{N2}}{\partial \bar{p}} \frac{\partial Q_2}{\partial \bar{p}} \\ &= 2 \frac{\partial q_{A2}}{\partial \bar{p}} - 2b \frac{\partial Q_2}{\partial \bar{p}} \frac{\partial q_{N2}}{\partial \bar{p}} > 0.\end{aligned}\quad (197)$$

The inequality in (197) holds because $D > 0$ by assumption, so $\frac{\partial q_{A2}}{\partial \bar{p}} > 0$ from (20), $\frac{\partial Q_2}{\partial \bar{p}} > 0$ from (24), and $\frac{\partial q_{N2}}{\partial \bar{p}} < 0$ from (21).

$\bar{p}_{Vm} \equiv \arg \min_{\bar{p}} \{ \tilde{V}_2(\bar{p}) \}$ is unique and is determined by:

$$\left. \frac{\partial \tilde{V}_2(\bar{p}_{Vm})}{\partial \bar{p}} \right|_{\bar{p}=\bar{p}_{Vm}} = 0. \quad (198)$$

This is the case because (2), (20) – (24), and (195) imply that $\frac{\partial \tilde{V}_2(\bar{p})}{\partial \bar{p}}$ is a linear function of $\bar{p}$. Therefore, $\tilde{V}_2(\bar{p})$ is a quadratic function of $\bar{p}$. Consequently, (197) implies that $\tilde{V}_2(\bar{p})$ has a unique minimum that is determined by (198).

To prove the Corollary to Lemma 2 and thereby establish that $\bar{p}_{Vm} > \bar{p}_0$ when $\Phi_2 > 0$, observe that R 's revenue is:

$$V(\bar{p}) = \bar{p} q_A + P(Q) q_N = \bar{p} q_A + [a - bQ] q_N. \quad (199)$$

(199) implies that the Corollary to Lemma 2 holds if:

$$\frac{\partial^+ V(\bar{p}_0)}{\partial \bar{p}} = q_A + \bar{p}_0 \frac{\partial q_A}{\partial \bar{p}} - b \frac{\partial Q}{\partial \bar{p}} q_N + P(Q) \frac{\partial q_N}{\partial \bar{p}} < 0, \quad (200)$$

where: (i) $\frac{\partial^+ V(\bar{p}_0)}{\partial \bar{p}} = \left. \frac{\partial V(\bar{p})}{\partial \bar{p}} \right|_{\bar{p}=\bar{p}_0}$ denotes the right-sided derivative of $V(\cdot)$; (ii) $\frac{\partial q_A}{\partial \bar{p}}$, $\frac{\partial q_N}{\partial \bar{p}}$, and $\frac{\partial Q}{\partial \bar{p}}$ pertain to the quantities identified in Lemma A2 (which prevail when $\bar{p} \in (\bar{p}_0, \bar{p}_d)$); and (iii) q_A , q_N , and Q are as defined in Lemma A1.

Define:

$$\begin{aligned}E &= 2b[2b + k] + [k_N + k^R][2b + k] - b^2 \\ &= 3b^2 + 2bk + [k_N + k^R][2b + k] \\ &= b[3b + 2k] + [2b + k][k_N + k^R].\end{aligned}\quad (201)$$

(201) and Lemma A2 imply that when $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\begin{aligned}\frac{\partial q_N}{\partial \bar{p}} &= - \frac{bk + 2bk^R + k k^R + b^2}{D}; \\ \frac{\partial q_A}{\partial \bar{p}} &= \frac{2bk + 2bk_N + 2bk^R + k k_N + k k^R + 3b^2}{D}\end{aligned}$$

$$\begin{aligned}
&= \frac{[2b + k_N + k^R][2b + k] - b^2}{D} = \frac{E}{D}; \\
\frac{\partial Q}{\partial \bar{p}} &= \frac{1}{D} \{ 2bk + 2bk_N + 2bk^R + kk_N + kk^R + 3b^2 \\
&\quad - [bk + 2bk^R + kk^R + b^2] - [b^2 + k_N b] \} \\
&= \frac{bk + bk_N + kk_N + b^2}{D} = \frac{[b + k][b + k_N]}{D}. \tag{202}
\end{aligned}$$

Lemma A1 implies that when $\bar{p} \leq \bar{p}_0$:

$$\begin{aligned}
q_N &= \frac{[a - c_N][2b + k] - b[a - c]}{E}, \quad q = \frac{[a - c][2b + k_N + k^R] - b[a - c_N]}{E}, \text{ and} \\
P(Q) &= a - b[q_N + q] = a - b \frac{[a - c_N][b + k] + [b + k_N + k^R][a - c]}{E} \\
&= \frac{aE - b[a - c_N][b + k] - b[b + k_N + k^R][a - c]}{E}. \tag{203}
\end{aligned}$$

(200) – (203) imply that because $q_A = 0$ when $\bar{p} = \bar{p}_0$ (from Lemma A1):

$$\begin{aligned}
\frac{\partial^+ V(\bar{p}_0)}{\partial \bar{p}} &= \bar{p}_0 \frac{E}{D} - b \left[\frac{(b + k)(b + k_N)}{D} \right] \left[\frac{(a - c_N)(2b + k) - b(a - c)}{E} \right] \\
&\quad - \left[\frac{aE - b(a - c_N)(b + k) - b(b + k_N + k^R)(a - c)}{E} \right] \\
&\quad \cdot \left[\frac{bk + 2bk^R + kk^R + b^2}{D} \right] \\
&= \frac{1}{DE} \{ \bar{p}_0 E^2 - b[b + k][b + k_N][(a - c_N)(2b + k) - b(a - c)] \\
&\quad - [aE - b(a - c_N)(b + k) - b(b + k_N + k^R)(a - c)] \\
&\quad \cdot [bk + 2bk^R + kk^R + b^2] \}. \tag{204}
\end{aligned}$$

(6) and (201) imply:

$$E \bar{p}_0 = c_A E + [a - c_N][2b + k][b + k^R] - b[a - c][b + k^R]. \tag{205}$$

(201), (205), and (206) imply:

$$\begin{aligned}
\frac{\partial^+ V(\bar{p}_0)}{\partial \bar{p}} &= \frac{1}{DE} \{ c_A E^2 + E[(a - c_N)(2b + k)(b + k^R) - b(a - c)(b + k^R)] \\
&\quad - b[b + k][b + k_N][(a - c_N)(2b + k) - b(a - c)] \}
\end{aligned}$$

$$\begin{aligned}
& - \left[a E - b (a - c_N) (b + k) - b (b + k_N + k^R) (a - c) \right] \left[b k + 2 b k^R + k k^R + b^2 \right] \} \\
& = \frac{1}{D E} \{ c_A E^2 + \left[E (b + k^R) - b (b + k) (b + k_N) \right] \left[(a - c_N) (2 b + k) - b (a - c) \right] \\
& \quad - \left[a E - b (a - c_N) (b + k) - b (b + k_N + k^R) (a - c) \right] \left[b k + 2 b k^R + k k^R + b^2 \right] \} \\
& = \frac{1}{D E} \left[c_A E^2 - \tilde{E} \right]. \tag{207}
\end{aligned}$$

where

$$\begin{aligned}
\tilde{E} & = \left[a E - b (a - c_N) (b + k) - b (b + k_N + k^R) (a - c) \right] \left[b k + 2 b k^R + k k^R + b^2 \right] \\
& \quad - \left[E (b + k^R) - b (b + k) (b + k_N) \right] \left[(a - c_N) (2 b + k) - b (a - c) \right] \\
& = \left[a E - b (a - c_N) (b + k) - b (b + k_N + k^R) (a - c) \right] \left[b k + 2 b k^R + k k^R + b^2 \right] \\
& \quad + b [b + k] [b + k_N] [(a - c_N) (2 b + k) - b (a - c)] \\
& \quad - E [b + k^R] [a - c_N] [2 b + k] + E [b + k^R] b [a - c] \\
& = \left[a E - b (a - c_N) (b + k) - b (b + k_N + k^R) (a - c) \right] \left[b k + 2 b k^R + k k^R + b^2 \right] \\
& \quad + b [b + k] [b + k_N] [(a - c_N) (2 b + k) - b (a - c)] \\
& \quad - E [b + k^R] a [2 b + k] + E [b + k^R] b [a - c] + c_N E [b + k^R] [2 b + k]. \tag{208}
\end{aligned}$$

(201) implies:

$$\begin{aligned}
[b + k^R] [2 b + k] & = [2 b + k_N + k^R] [2 b + k] - [b + k_N] [2 b + k] \\
& = E + b^2 - [b + k_N] [2 b + k] = E - [(b + k_N) (2 b + k) - b^2]. \tag{209}
\end{aligned}$$

(201), (208), and (209) imply:

$$\begin{aligned}
\tilde{E} & = \hat{E} + c_N E^2, \text{ where} \\
\hat{E} & \equiv \left[a E - b (a - c_N) (b + k) - b (b + k_N + k^R) (a - c) \right] \left[b k + 2 b k^R + k k^R + b^2 \right] \\
& \quad + b [b + k] [b + k_N] [(a - c_N) (2 b + k) - b (a - c)] \\
& \quad - E [b + k^R] a [2 b + k] + E [b + k^R] b [a - c] \\
& \quad - E [(b + k_N) (2 b + k) - b^2] c_N. \tag{210}
\end{aligned}$$

(207) and (210) imply:

$$\frac{\partial^+ V(\bar{p}_0)}{\partial \bar{p}} = \frac{1}{D E} \left[c_A E^2 - \left(\hat{E} + c_N E^2 \right) \right] = - \frac{1}{D E} \left[(c_N - c_A) E^2 + \hat{E} \right]$$

$$< 0 \text{ if } c_N - c_A > -\frac{\hat{E}}{E^2} \Leftrightarrow \Phi_2 \equiv E^2 [c_N - c_A] + \hat{E} > 0. \quad (211)$$

(211) reflects the facts that $E > 0$ (from (201)) and $D > 0$ (by assumption).

It remains to demonstrate that Φ_2 is as specified in (193). (201) and (210) imply:

$$\hat{E} = \psi_1 E + \psi_2, \text{ where} \quad (212)$$

$$\begin{aligned} \psi_1 \equiv & a [bk + 2bk^R + kk^R + b^2] - a [b + k^R] [2b + k] + b [b + k^R] [a - c] \\ & - [(b + k_N)(2b + k) - b^2] c_N, \text{ and} \end{aligned} \quad (213)$$

$$\begin{aligned} \psi_2 \equiv & - [b(a - c_N)(b + k) + b(b + k_N + k^R)(a - c)] [bk + 2bk^R + kk^R + b^2] \\ & + b[b + k][b + k_N][(a - c_N)(2b + k) - b(a - c)]. \end{aligned} \quad (214)$$

(213) implies:

$$\begin{aligned} \psi_1 = & a [bk + 2bk^R + kk^R + b^2 - (b + k^R)(2b + k) + b(b + k^R)] \\ & - b[b + k^R]c - [(b + k_N)(2b + k) - b^2]c_N \\ = & a [bk + 2bk^R + kk^R + b^2 - 2b^2 - bk - 2bk^R - kk^R + b^2 + bk^R] \\ & - b[b + k^R]c - [(b + k_N)(2b + k) - b^2]c_N \\ = & abk^R - b[b + k^R]c - [(b + k_N)(2b + k) - b^2]c_N. \end{aligned} \quad (215)$$

(214) implies:

$$\begin{aligned} \psi_2 = & - \{ b[b + k][a - c_N] + b[b + k_N + k^R][a - c] \} [bk + 2bk^R + kk^R + b^2] \\ & + b[b + k][b + k_N][2b + k][a - c_N] - b^2[b + k][b + k_N][a - c] \\ = & [a - c_N] \{ b[b + k][b + k_N][2b + k] - b[b + k][bk + 2bk^R + kk^R + b^2] \} \\ & - [a - c] \{ b^2[b + k][b + k_N] + b[b + k_N + k^R][bk + 2bk^R + kk^R + b^2] \} \\ = & [a - c_N]b[b + k] \{ [b + k_N][2b + k] - [bk + 2bk^R + kk^R + b^2] \} \\ & - b[a - c] \{ b[b + k][b + k_N] + [b + k_N + k^R][bk + 2bk^R + kk^R + b^2] \}. \end{aligned} \quad (216)$$

The coefficient on $[a - c_N]b[b + k]$ in (216) is:

$$\begin{aligned} & 2b^2 + bk + 2bk_N + kk_N - bk - 2bk^R - kk^R - b^2 \\ = & b^2 + 2bk_N + kk_N - 2bk^R - kk^R = b^2 + [2b + k][k_N - k^R]. \end{aligned} \quad (217)$$

The coefficient on $-b[a - c]$ in (216) is:

$$\begin{aligned}
& b[b + k][b + k_N] + [b + k_N][bk + 2bk^R + kk^R + b^2] \\
& \quad + k^R[bk + 2bk^R + kk^R + b^2] \\
& = [b + k_N][b^2 + bk + bk + 2bk^R + kk^R + b^2] + k^R[bk + 2bk^R + kk^R + b^2] \\
& = [b + k_N][2b^2 + 2bk + 2bk^R + kk^R] + k^R[bk + 2bk^R + kk^R + b^2] \\
& = [b + k_N][2b^2 + 2bk] + k^R[(b + k_N)(2b + k) + bk + 2bk^R + kk^R + b^2] \\
& = 2b[b + k][b + k_N] + k^R[2b^2 + bk + 2bk_N + kk_N + bk + 2bk^R + kk^R + b^2] \\
& = 2b[b + k][b + k_N] + k^R[3b^2 + 2bk + 2bk_N + kk_N + 2bk^R + kk^R] \\
& = 2b[b + k][b + k_N] + k^R[3b^2 + 2bk + (k_N + k^R)(2b + k)] \\
& = 2b[b + k][b + k_N] + k^R E. \tag{218}
\end{aligned}$$

The last equality in (218) reflects (201).

(212) and (215) – (218) imply:

$$\begin{aligned}
\hat{E} & = E \{ [a - c]bk^R - b^2c - [(b + k_N)(2b + k) - b^2]c_N \} \\
& \quad + \{ b^2 + [2b + k][k_N - k^R] \} b[b + k][a - c_N] \\
& \quad - \{ 2b[b + k][b + k_N] + k^R E \} b[a - c] \\
& = -E \{ b^2c + [(b + k_N)(2b + k) - b^2]c_N \} \\
& \quad + b[b + k] \{ b^2 + [2b + k][k_N - k^R] \} [a - c_N] \\
& \quad - 2b^2[b + k][b + k_N][a - c]. \tag{219}
\end{aligned}$$

(201) and (219) imply:

$$\begin{aligned}
\Phi_2 & \equiv E^2[c_N - c_A] + \hat{E} \\
& = \{ b[3b + 2k] + [2b + k][k_N + k^R] \}^2 [c_N - c_A] \\
& \quad - \{ b[3b + 2k] + [2b + k][k_N + k^R] \} \\
& \quad \cdot \{ b^2c + [(b + k_N)(2b + k) - b^2]c_N \} \\
& \quad + b[b + k] \{ b^2 + [2b + k][k_N - k^R] \} [a - c_N] - 2b^2[b + k][b + k_N][a - c]. \tag{220}
\end{aligned}$$

Observe that:

$$[b + k_N][2b + k] - b^2 = b^2 + bk + k_N[2b + k] = b[b + k] + [2b + k]k_N. \quad (221)$$

(220) and (221) imply:

$$\begin{aligned}
\Phi_2 &= \{b[3b + 2k] + [2b + k][k_N + k^R]\}^2 [c_N - c_A] \\
&\quad - \{b[3b + 2k] + [2b + k][k_N + k^R]\} \\
&\quad \cdot \{b^2c + [b(b + k) + (2b + k)k_N]c_N\} \\
&\quad + b[b + k]\{b^2 + [2b + k][k_N - k^R]\}[a - c_N] \\
&\quad - 2b^2[b + k][b + k_N][a - c] \\
&= \{b[3b + 2k] + [2b + k][k_N + k^R]\} \\
&\quad \cdot \left[\{b[3b + 2k] + [2b + k][k_N + k^R]\}[c_N - c_A] \right. \\
&\quad \quad \left. - b^2c - b[b + k]c_N - [2b + k]k_Nc_N \right] \\
&\quad + b[b + k]\{b^2[a - c_N] + [2b + k][k_N - k^R][a - c_N] \\
&\quad \quad - 2b[b + k_N][a - c]\} \\
&= \{b[3b + 2k] + [2b + k][k_N + k^R]\} \\
&\quad \cdot \left[\{b[3b + 2k] + [2b + k][k_N + k^R] - b[b + k] - [2b + k]k_N\}c_N \right. \\
&\quad \quad \left. - b^2c - \{b[3b + 2k] + [2b + k][k_N + k^R]\}c_A \right] \\
&\quad + b[b + k]\left[\{b^2 + [2b + k][k_N - k^R] - 2b[b + k_N]\}a \right. \\
&\quad \quad \left. - \{b^2 + [2b + k][k_N - k^R]\}c_N + 2b[b + k_N]c \right]. \quad (222)
\end{aligned}$$

Observe that:

$$\begin{aligned}
&b[3b + 2k] + [2b + k][k_N + k^R] - b[b + k] - [2b + k]k_N \\
&= b[3b + 2k - b - k] + [2b + k]k^R = [2b + k][b + k^R]. \quad (223)
\end{aligned}$$

Further observe that:

$$\begin{aligned}
&b^2 + [2b + k][k_N - k^R] - 2b[b + k_N] \\
&= b^2 + [2b + k - 2b]k_N - [2b + k]k^R - 2b^2
\end{aligned}$$

$$= -b^2 + k k_N - [2b + k] k^R = -[b^2 - k k_N + (2b + k) k^R]. \quad (224)$$

(222) – (224) imply:

$$\begin{aligned} E^2 [c_N - c_A] + \hat{E} \\ &= \{ b [3b + 2k] + [2b + k] [k_N + k^R] \} \\ &\quad \cdot \left\{ [2b + k] [b + k^R] c_N - b^2 c \right. \\ &\quad \left. - \{ b [3b + 2k] + [2b + k] [k_N + k^R] \} c_A \right\} \\ &\quad - b [b + k] \left\{ [b^2 - k k_N + (2b + k) k^R] a \right. \\ &\quad \left. + [b^2 + (2b + k) (k_N - k^R)] c_N - 2b [b + k_N] c \right\}. \quad (225) \end{aligned}$$

The coefficient on c_N in (225) is:

$$\begin{aligned} &b [2b + k] [3b + 2k] [b + k^R] + [2b + k]^2 [b + k^R] [k_N + k^R] \\ &- b^3 [b + k] - b [b + k] [2b + k] [k_N - k^R] \\ &= k^R \{ b [2b + k] [3b + 2k] + [2b + k]^2 [b + k^R] + b [b + k] [2b + k] \} \\ &\quad + k_N \{ [2b + k]^2 [b + k^R] - b [b + k] [2b + k] \} \\ &\quad + b^2 [2b + k] [3b + 2k] - b^3 [b + k] \\ &= k^R [2b + k] \{ b [3b + 2k] + [2b + k] [b + k^R] + b [b + k] \} \\ &\quad + k_N [2b + k] \{ [2b + k] [b + k^R] - b [b + k] \} \\ &\quad + b^2 \{ [2b + k] [3b + 2k] - b [b + k] \} \\ &= k^R [2b + k] \{ k^R [2b + k] + b [3b + 2k + 2b + k + b + k] \} \\ &\quad + k_N [2b + k] \{ k^R [2b + k] + b [2b + k - b - k] \} \\ &\quad + b^2 [6b^2 + 7bk + 2k^2 - b^2 - bk] \\ &= k^R [2b + k] [k^R (2b + k) + 2b (3b + 2k)] \\ &\quad + k_N [2b + k] [k^R (2b + k) + b^2] + b^2 [5b^2 + 6bk + 2k^2]. \quad (226) \end{aligned}$$

The coefficient on c in (225) is:

$$\begin{aligned}
& 2b^2 [b+k] [b+k_N] - b^2 \{ b [3b+2k] + [2b+k] [k_N+k^R] \} \\
&= b^2 \{ 2 [b+k] [b+k_N] - b [3b+2k] - [2b+k] [k_N+k^R] \} \\
&= b^2 \{ 2 [b^2 + bk + bk_N + k k_N] - 3b^2 - 2bk - 2bk_N - k k_N - [2b+k] k^R \} \\
&= b^2 \{ -b^2 + k k_N - [2b+k] k^R \} = -b^2 [b^2 - k k_N + (2b+k) k^R]. \tag{227}
\end{aligned}$$

(201) implies that the coefficient on c_A in (225) is $-E^2$. Therefore, (201) and (225) – (227) imply:

$$\begin{aligned}
\Phi_2 &= \{ k^R [2b+k] [k^R (2b+k) + 2b(3b+2k)] \\
&\quad + k_N [2b+k] [k^R (2b+k) + b^2] + b^2 [5b^2 + 6bk + 2k^2] \} c_N \\
&\quad - b^2 [b^2 - k k_N + (2b+k) k^R] c - E^2 c_A \\
&\quad - b [b+k] [b^2 - k k_N + (2b+k) k^R] a \\
&= \{ k^R [2b+k] [k^R (2b+k) + 2b(3b+2k)] \\
&\quad + k_N [2b+k] [k^R (2b+k) + b^2] + b^2 [5b^2 + 6bk + 2k^2] \} c_N \\
&\quad - b [b^2 - k k_N + (2b+k) k^R] [a(b+k) + bc] \\
&\quad - \{ b [3b+2k] + [2b+k] [k_N+k^R] \}^2 c_A. \tag{228}
\end{aligned}$$

It remains to prove that $\bar{p}_{Vm} < \bar{p}_d$, which is established by demonstrating that $\left. \frac{\partial^- V(\bar{p})}{\partial \bar{p}} \right|_{\bar{p}=\bar{p}_d} > 0$. Define $V_2(\bar{p}) \equiv \bar{p} q_A(\cdot) + P(Q(\cdot)) q_N(\cdot)$ for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$. Because $P(Q) = a - bQ$:

$$\frac{\partial^- V_2(\bar{p}_d)}{\partial \bar{p}} = q_A + \bar{p}_d \frac{\partial q_A}{\partial \bar{p}} + P(Q) \frac{\partial q_N}{\partial \bar{p}} - b q_N \frac{\partial Q}{\partial \bar{p}} \tag{229}$$

where q_A , q_N , and Q are as specified in Lemma A2, evaluated at $\bar{p} = \bar{p}_d$. Because $\bar{p}_d = P(Q)$, (229) implies:

$$\frac{\partial^- V_2(\bar{p}_d)}{\partial \bar{p}} = q_A + \bar{p}_d \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right] - b q_N \frac{\partial Q}{\partial \bar{p}}. \tag{230}$$

(68) implies:

$$\begin{aligned}
\bar{p}_d &= [b+k^R] Q^R + c_N + k_N q_N - b q_A \\
&= [b+k^R] q_A + [b+k^R] q_N + c_N + k_N q_N - b q_A \\
&= k^R q_A + [b+k_N+k^R] q_N + c_N. \tag{231}
\end{aligned}$$

(230) and (231) imply:

$$\begin{aligned}
\frac{\partial^- V_2(\bar{p}_d)}{\partial \bar{p}} &= q_A - b q_N \frac{\partial Q}{\partial \bar{p}} + [k^R q_A + (b + k_N + k^R) q_N + c_N] \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right] \\
&= q_A + [k^R q_A + (k_N + k^R) q_N + c_N] \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right] \\
&\quad + b q_N \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right] - b q_N \frac{\partial Q}{\partial \bar{p}} \\
&= q_A + [k^R q_A + (k_N + k^R) q_N + c_N] \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right] \\
&\quad + b q_N \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} - \frac{\partial Q}{\partial \bar{p}} \right] \\
&= q_A + [k^R q_A + (k_N + k^R) q_N + c_N] \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right] - b q_N \frac{\partial q}{\partial \bar{p}} > 0. \quad (232)
\end{aligned}$$

The inequality holds here because $\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} = \frac{\partial Q^R}{\partial \bar{p}} > 0$ (from (22)) and $\frac{\partial q}{\partial \bar{p}} < 0$ (from (23)). ■

Lemma 3. For $\bar{p} \in (\bar{p}_0, \bar{p}_d)$: (i) $S(\bar{p})$ is a strictly concave function of $\bar{p}$; (ii) $\frac{\partial S(\bar{p})}{\partial \bar{p}} \geq 0 \Leftrightarrow \bar{p} \leq \bar{p}_{SM}$ where $\bar{p}_{SM} \in (\bar{p}_0, \bar{p}_d]$; and (iii) $\bar{p}_{S_2M} > \bar{p}_{Vm}$.

Proof. As in (174), define:

$$\tilde{S}_2(\bar{p}) \equiv a Q_2(\bar{p}) - \frac{b}{2} Q_2(\bar{p})^2 - q_{A2}(\bar{p}) \bar{p} - [q_2(\bar{p}) + q_{N2}(\bar{p})] P(Q_2(\bar{p})) \quad (233)$$

where $q_{A2}(\bar{p})$, $q_{N2}(\bar{p})$, $q_2(\bar{p})$, and $Q_2(\bar{p})$ are as defined in (20), (21), (23), and (24), respectively. Observe that $\tilde{S}_2(\bar{p}) = S(\bar{p})$ for $\bar{p} \in [\bar{p}_0, \bar{p}_d]$.

(233) implies that because $P(Q_2) = a - b Q_2$ and $Q_2 = q_{A2} + q_{N2} + q_2$:

$$\begin{aligned}
\frac{\partial \tilde{S}_2(\bar{p})}{\partial \bar{p}} &= a \frac{\partial Q_2}{\partial \bar{p}} - b Q_2 \frac{\partial Q_2}{\partial \bar{p}} - q_{A2} - \bar{p} \frac{\partial q_{A2}}{\partial \bar{p}} - P(Q_2) \left[\frac{\partial q_{N2}}{\partial \bar{p}} + \frac{\partial q_2}{\partial \bar{p}} \right] + b \frac{\partial Q_2}{\partial \bar{p}} [q_{N2} + q_2] \\
&= P(Q_2) \left[\frac{\partial Q_2}{\partial \bar{p}} - \frac{\partial q_{N2}}{\partial \bar{p}} - \frac{\partial q_2}{\partial \bar{p}} \right] - \bar{p} \frac{\partial q_{A2}}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} [q_{N2} + q_2] - q_{A2} \\
&= [P(Q_2) - \bar{p}] \frac{\partial q_{A2}}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} [q_{N2} + q_2] - q_{A2}. \quad (234)
\end{aligned}$$

(196) and (234) imply that because $P(Q_2) = a - b Q_2$:

$$\frac{\partial^2 \tilde{S}_2(\bar{p})}{\partial (\bar{p})^2} = \left[-b \frac{\partial Q_2}{\partial \bar{p}} - 1 \right] \frac{\partial q_{A2}}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} \left[\frac{\partial q_{N2}}{\partial \bar{p}} + \frac{\partial q_2}{\partial \bar{p}} \right] - \frac{\partial q_{A2}}{\partial \bar{p}} < 0. \quad (235)$$

The inequality in (235) holds because $D > 0$ by assumption, so $\frac{\partial q_{A2}}{\partial \bar{p}} > 0$ from (20), $\frac{\partial Q_2}{\partial \bar{p}} > 0$ from (24), $\frac{\partial q_{N2}}{\partial \bar{p}} < 0$ from (21), and $\frac{\partial q_2}{\partial \bar{p}} < 0$ from (23).

$\bar{p}_{SM} \equiv \arg \max_{\bar{p}} \{ \tilde{S}_2(\bar{p}) \}$ is unique and is determined by:

$$\frac{\partial \tilde{S}_2(\bar{p}_{SM})}{\partial \bar{p}} \equiv \frac{\partial \tilde{S}_2(\bar{p})}{\partial \bar{p}} \bigg|_{\bar{p}=\bar{p}_{SM}} = 0. \quad (236)$$

This is the case because (2), (20) – (24), and (234) imply that $\frac{\partial \tilde{S}_2(\bar{p})}{\partial \bar{p}}$ is a linear function of $\bar{p}$. Therefore, $\tilde{S}_2(\bar{p})$ is a quadratic function of $\bar{p}$. Consequently, (235) implies that $\tilde{S}_2(\bar{p})$ has a unique maximum that is determined by (236).

To prove that $\bar{p}_{SM} > \bar{p}_{Vm}$, let:

$$H(\bar{p}) \equiv a Q_2 - \frac{b}{2} Q_2^2 - [a - b Q_2] q_2 \quad (237)$$

$$\Rightarrow \frac{\partial H(\bar{p})}{\partial \bar{p}} \equiv [a - b Q_2] \frac{\partial Q_2}{\partial \bar{p}} - [a - b Q_2] \frac{\partial q_2}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} q_2 \quad (238)$$

where q_2 and Q_2 are defined in (23) and (24). Differentiating (238) provides:

$$\frac{\partial^2 H(\bar{p})}{(\partial \bar{p})^2} \equiv -b \left(\frac{\partial Q_2}{\partial \bar{p}} \right)^2 + 2b \frac{\partial Q_2}{\partial \bar{p}} \frac{\partial q_2}{\partial \bar{p}} < 0. \quad (239)$$

The inequality in (239) holds because $\frac{\partial Q_2}{\partial \bar{p}} > 0$ and $\frac{\partial q_2}{\partial \bar{p}} < 0$, from (23) and (24). (238) implies:

$$\frac{\partial H(\bar{p}_d)}{\partial \bar{p}} \equiv \frac{\partial H(\bar{p})}{\partial \bar{p}} \bigg|_{\bar{p}=\bar{p}_d} = \bar{p}_d \frac{\partial Q_2}{\partial \bar{p}} - \bar{p}_d \frac{\partial q_2}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} q_2(\bar{p}_d) > 0. \quad (240)$$

The inequality in (240) holds because $\frac{\partial Q_2}{\partial \bar{p}} > 0$ and $\frac{\partial q_2}{\partial \bar{p}} < 0$, from (23) and (24). The concavity of $H(\bar{p})$ established in (239), along with (240), imply:

$$\frac{\partial H(\bar{p})}{\partial \bar{p}} > 0 \text{ for all } \bar{p} < \bar{p}_d \Rightarrow \frac{\partial H(\bar{p}_{Vm})}{\partial \bar{p}} > 0. \quad (241)$$

The implication in (241) holds because $\bar{p}_{Vm} < \bar{p}_d$, from Lemma 2.

(195) and (236) imply:

$$\frac{\partial \tilde{V}_2(\bar{p}_{Vm})}{\partial \bar{p}} = [a - b Q_2(\cdot)] \frac{\partial q_{N2}(\cdot)}{\partial \bar{p}} - b \frac{\partial Q_2(\cdot)}{\partial \bar{p}} q_{N2}(\cdot)$$

$$+ q_{A2}(\cdot) + \bar{p}_{V_m} \frac{\partial q_{A2}(\cdot)}{\partial \bar{p}} = 0 \quad (242)$$

where $q_{A2}(\cdot)$, $q_{N2}(\cdot)$, and $Q_2(\cdot)$ are defined in (20), (21), and (24), and evaluated at $\bar{p}_{V_m}$.

(234) implies:

$$\begin{aligned} \frac{\partial \tilde{S}_2(\bar{p})}{\partial \bar{p}} &= [a - b Q_2] \frac{\partial Q_2}{\partial \bar{p}} - [a - b Q_2] \frac{\partial q_2}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} q_2 \\ &\quad - [a - b Q_2] \frac{\partial q_{N2}}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} q_{N2} - q_{A2} - \bar{p} \frac{\partial q_{A2}}{\partial \bar{p}} \end{aligned} \quad (243)$$

where q_{A2} , q_{N2} , q_2 , and Q_2 are defined in (20), (21), (23), and (24). (243) implies:

$$\begin{aligned} \frac{\partial \tilde{S}_2(\bar{p}_{V_m})}{\partial \bar{p}} &= [a - b Q_2(\bar{p}_{V_m})] \frac{\partial Q_2}{\partial \bar{p}} - [a - b Q_2(\bar{p}_{V_m})] \frac{\partial q_2}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} q_2(\bar{p}_{V_m}) \\ &\quad - [a - b Q_2(\bar{p}_{V_m})] \frac{\partial q_{N2}}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} q_{N2}(\bar{p}_{V_m}) - q_{A2}(\bar{p}_{V_m}) - \bar{p}_{V_m} \frac{\partial q_{A2}}{\partial \bar{p}} \\ &= [a - b Q_2(\bar{p}_{V_m})] \frac{\partial Q_2}{\partial \bar{p}} - [a - b Q_2(\bar{p}_{V_m})] \frac{\partial q_2}{\partial \bar{p}} + b \frac{\partial Q_2}{\partial \bar{p}} q_2(\bar{p}_{V_m}) \\ &= \frac{\partial H(\bar{p}_{V_m})}{\partial \bar{p}} > 0. \end{aligned} \quad (244)$$

The last equality in (244) reflects (242). The inequality in (244) reflects (241).

(235) implies that $\tilde{S}_2(\bar{p})$ is a strictly concave function of $\bar{p}$. Therefore, $\bar{p}_{V_m} < \bar{p}_{SM}$ because: (i) $\frac{\partial \tilde{S}_2(\bar{p}_{SM})}{\partial \bar{p}} = 0$ from (236); and (ii) $\frac{\partial \tilde{S}_2(\bar{p}_{V_m})}{\partial \bar{p}} > 0$, from (244).

To prove that $\bar{p}_{SM} > \bar{p}_0$, it suffices to establish that $\frac{\partial^+ S_2(\bar{p}_0)}{\partial \bar{p}} \equiv \frac{\partial^+ S_2(\bar{p}_0)}{\partial \bar{p}} \Big|_{\bar{p}=\bar{p}_0} > 0$. Lemma A1 implies that $q_A = 0$ when $\bar{p} = \bar{p}_0$. Therefore, (174) implies:

$$\begin{aligned} \frac{\partial^+ \tilde{S}_2(\bar{p}_0)}{\partial \bar{p}} &= [a - b Q] \frac{\partial Q}{\partial \bar{p}} - \bar{p}_0 \frac{\partial q_A}{\partial \bar{p}} - P(Q) \left[\frac{\partial q_N}{\partial \bar{p}} + \frac{\partial q}{\partial \bar{p}} \right] + b [q_N + q] \frac{\partial Q}{\partial \bar{p}} \\ &= P(Q) \frac{\partial Q}{\partial \bar{p}} - \bar{p}_0 \frac{\partial q_A}{\partial \bar{p}} - P(Q) \left[\frac{\partial q_N}{\partial \bar{p}} + \frac{\partial q}{\partial \bar{p}} \right] + b [q_N + q] \frac{\partial Q}{\partial \bar{p}} \\ &= [P(Q) - \bar{p}_0] \frac{\partial q_A}{\partial \bar{p}} + b [q_N + q] \frac{\partial Q}{\partial \bar{p}} > 0. \end{aligned} \quad (245)$$

The inequality in (245) holds because $D > 0$ by assumption, so $\frac{\partial q_A}{\partial \bar{p}} > 0$ from (20), $\frac{\partial Q}{\partial \bar{p}} > 0$ from (24), and $P(Q) > \bar{p}_0$ when $\bar{p} \in (\bar{p}_0, \bar{p}_d)$. ■

Proposition 5. $\bar{p}^* \in (\bar{p}_0, \bar{p}_d]$ if $\Phi_2 \geq 0$. $\bar{p}^* = \bar{p}_0$ if $\Phi_2 < 0$ and r is sufficiently large.

Proof. The first conclusion in the Proposition follows from Proposition A1 because (173) implies that when if $\Phi_2 \geq 0$:

$$\frac{\partial^+ W_2(\bar{p}_0)}{\partial \bar{p}} \equiv \left. \frac{\partial^+ W_2(\bar{p})}{\partial \bar{p}} \right|_{\bar{p}=\bar{p}_0} = \frac{\partial^+ S_2(\bar{p}_0)}{\partial \bar{p}} - r \frac{\partial^+ V_2(\bar{p}_0)}{\partial \bar{p}} > 0. \quad (246)$$

The inequality in (246) holds because when $\Phi_2 \geq 0$: (i) $\frac{\partial^+ V_2(\bar{p}_0)}{\partial \bar{p}} \leq 0$ from (211); and (ii) $\frac{\partial^+ S_2(\bar{p}_0)}{\partial \bar{p}} > 0$ from (245).

The second conclusion in the Proposition holds if $V(\bar{p}_0) < V(\bar{p})$ for all $\bar{p} > \bar{p}_0$ when r is sufficiently large and $\Phi_2 < 0$. (211) and (220) imply:

$$\left. \frac{\partial^+ V(\bar{p})}{\partial \bar{p}} \right|_{\bar{p}=\bar{p}_0} > 0 \text{ when } \Phi_2 < 0. \quad (247)$$

$V(\bar{p})$ is a strictly convex function of $\bar{p}$ for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$, from Lemma 2. Therefore, (247) implies that $V(\bar{p})$ is a strictly increasing function of $\bar{p}$ for $\bar{p} \in [\bar{p}_0, \bar{p}_d]$ under the maintained conditions. Consequently:

$$V(\bar{p}_0) < V(\bar{p}) \text{ for all } \bar{p} \in (\bar{p}_0, \bar{p}_d]. \quad (248)$$

Lemma A7 implies that under the maintained conditions:

$$V(\bar{p}_0) < V(\bar{p}_b). \quad (249)$$

(110) implies that $V(\bar{p})$ is a strictly concave function of $\bar{p}$ for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$. Therefore, (248) and (249) imply:

$$V(\bar{p}) > V(\bar{p}_0) \text{ for all } \bar{p} \in (\bar{p}_d, \bar{p}_b]. \quad (250)$$

The conclusion follows from (248), (250), and Proposition A1. ■

Proposition A2. $\bar{p}^* \in [\bar{p}_{Vm}, \bar{p}_{SM}]$. Furthermore: (i) $\bar{p}^* < \bar{p}_{SM}$ when $\bar{p}_{SM} < \bar{p}_d$ and $d > 0$; (ii) $\bar{p}^* > \bar{p}_{Vm}$ when $\bar{p}_{Vm} > \bar{p}_0$; (iii) $\bar{p}^* \rightarrow \bar{p}_{SM}$ as $r \rightarrow 0$; and (iv) $\bar{p}^* \rightarrow \bar{p}_{Vm}$ as $r \rightarrow \infty$.

Proof. To prove that $\bar{p}^* \leq \bar{p}_{SM}$, suppose that $\bar{p}^* > \bar{p}_{SM}$. $\tilde{S}_2(\bar{p})$ is a strictly concave function of $\bar{p}$, from Lemma 3. Therefore, because $\bar{p}^* > \bar{p}_{SM}$, (236) implies:

$$\frac{\partial \tilde{S}_2(\bar{p}^*)}{\partial \bar{p}} < \frac{\partial \tilde{S}_2(\bar{p}_{SM})}{\partial \bar{p}} = 0. \quad (251)$$

$\tilde{V}_2(\bar{p})$ is a strictly convex function of $\bar{p}$, from Lemma 2. Therefore, because $\bar{p}_{Vm} < \bar{p}_{SM}$ from Lemma 3 and because $\bar{p}^* > \bar{p}_{SM}$ by assumption, (198) implies:

$$\frac{\partial \tilde{V}_2(\bar{p}^*)}{\partial \bar{p}} > \frac{\partial \tilde{V}_2(\bar{p}_{SM})}{\partial \bar{p}} > \frac{\partial \tilde{V}_2(\bar{p}_{Vm})}{\partial \bar{p}} = 0. \quad (252)$$

(251) and (252) imply that R 's revenue declines and consumer surplus increases as $\bar{p}$ declines below $\bar{p}^*$. Therefore, $\bar{p}^*$ is not the welfare-maximizing value of $\bar{p}$. Hence, by contradiction, $\bar{p}^* \leq \bar{p}_{SM}$.

To prove that $\bar{p}^* \geq \bar{p}_{Vm}$, suppose that $\bar{p}^* < \bar{p}_{Vm}$. $\tilde{V}_2(\bar{p})$ is a strictly convex function of $\bar{p}$, from Lemma 2. Therefore, because $\bar{p}_{Vm} < \bar{p}_{SM}$ from Lemma 3, (198) implies:

$$\frac{\partial \tilde{V}_2(\bar{p}^*)}{\partial \bar{p}} < \frac{\partial \tilde{V}_2(\bar{p}_{Vm})}{\partial \bar{p}} = 0. \quad (253)$$

$\tilde{S}_2(\bar{p})$ is a strictly concave function of $\bar{p}$, from Lemma 3. Therefore, because $\bar{p}_{Vm} < \bar{p}_{SM}$ from Lemma 3 and because $\bar{p}^* < \bar{p}_{Vm}$ by assumption, (236) implies:

$$\frac{\partial \tilde{S}_2(\bar{p}^*)}{\partial \bar{p}} > \frac{\partial \tilde{S}_2(\bar{p}_{Vm})}{\partial \bar{p}} > \frac{\partial \tilde{S}_2(\bar{p}_{SM})}{\partial \bar{p}} = 0. \quad (254)$$

(253) and (254) imply that R 's revenue declines and consumer surplus increases as $\bar{p}$ increases above $\bar{p}^*$. Therefore, $\bar{p}^*$ is not the welfare-maximizing value of $\bar{p}$. Hence, by contradiction, $\bar{p}^* \geq \bar{p}_{Vm}$.

To prove conclusion (i) in the Proposition, define $\tilde{W}_2(\cdot) \equiv \tilde{S}_2(\cdot) - r \tilde{V}_2(\cdot)$ and observe that when $\bar{p}_{SM} < \bar{p}_d$ and $r > 0$:

$$\begin{aligned} \left. \frac{\partial \tilde{W}_2(\bar{p})}{\partial \bar{p}} \right|_{\bar{p}=\bar{p}_{SM}} &= \frac{\partial \tilde{S}_2(\bar{p}_{SM})}{\partial \bar{p}} - r \frac{\partial \tilde{V}_2(\bar{p}_{SM})}{\partial \bar{p}} \\ &= -r \frac{\partial \tilde{V}_2(\bar{p}_{S2M})}{\partial \bar{p}} < -r \frac{\partial \tilde{V}_2(\bar{p}_{Vm})}{\partial \bar{p}} = 0. \end{aligned} \quad (255)$$

The inequality in (255) holds because: (i) $\bar{p}_{S2M} > \bar{p}_{Vm}$, from Lemma 3; and (ii) $\tilde{V}_2(\cdot)$ is a strictly convex function of $\bar{p}$, from Lemma 2. (255) implies that $\bar{p}_{SM} > \bar{p}^*$ because $\tilde{W}_2(\cdot)$ is a strictly concave function of $\bar{p}$ (because $\tilde{S}_2(\cdot)$ is a strictly concave function of $\bar{p}$ and $\tilde{V}_2(\cdot)$ is a strictly convex function of $\bar{p}$).

To prove conclusion (ii) in the Proposition, observe that when $\bar{p}_{V2m} > \bar{p}_0$:

$$\begin{aligned} \left. \frac{\partial \tilde{W}_2(\bar{p})}{\partial \bar{p}} \right|_{\bar{p}=\bar{p}_{Vm}} &= \frac{\partial \tilde{S}_2(\bar{p}_{Vm})}{\partial \bar{p}} - r \frac{\partial \tilde{V}_2(\bar{p}_{Vm})}{\partial \bar{p}} \\ &= \frac{\partial \tilde{S}_2(\bar{p}_{V2m})}{\partial \bar{p}} > \frac{\partial \tilde{S}_2(\bar{p}_{SM})}{\partial \bar{p}} = 0. \end{aligned} \quad (256)$$

The inequality in (256) holds because: (i) $\bar{p}_{S2M} > \bar{p}_{Vm}$, from Lemma 3; and (ii) $\tilde{S}_2(\cdot)$ is a strictly concave function of $\bar{p}$, from Lemma 3. (256) implies that $\bar{p}^* > \bar{p}_{Vm}$ because $\tilde{W}_2(\cdot)$ is a strictly concave function of $\bar{p}$.

Conclusions (iii) and (iv) in the Proposition follow immediately from (173) because $\bar{p}^* \in (\bar{p}_0, \bar{p}_d)$ is a non-increasing function of r . This is the case because (173) implies that when $\bar{p}^* \in (\bar{p}_0, \bar{p}_d)$:

$$\begin{aligned}
\frac{\partial S(\bar{p}^*)}{\partial \bar{p}} - r \frac{\partial \tilde{V}(\bar{p}^*)}{\partial \bar{p}} &= 0 \Rightarrow \frac{\partial^2 \tilde{S}(\bar{p}^*)}{\partial (\bar{p})^2} \frac{\partial \bar{p}^*}{\partial r} - \frac{\partial \tilde{V}(\bar{p}^*)}{\partial \bar{p}} - r \frac{\partial^2 \tilde{V}(\bar{p}^*)}{\partial (\bar{p})^2} \frac{\partial \bar{p}^*}{\partial r} = 0 \\
&\Rightarrow \frac{\partial \bar{p}^*}{\partial r} = \frac{\frac{\partial \tilde{V}(\bar{p}^*)}{\partial \bar{p}}}{\frac{\partial^2 \tilde{S}(\bar{p}^*)}{\partial (\bar{p})^2} - r \frac{\partial^2 \tilde{V}(\bar{p}^*)}{\partial (\bar{p})^2}} \stackrel{s}{=} - \frac{\partial \tilde{V}(\bar{p}^*)}{\partial \bar{p}}. \tag{257}
\end{aligned}$$

The last conclusion in (257) holds because Lemmas 2 and 3 imply that $\frac{\partial^2 \tilde{S}(\bar{p}^*)}{\partial (\bar{p})^2} < 0$ and $\frac{\partial^2 \tilde{V}(\bar{p}^*)}{\partial (\bar{p})^2} > 0$ when $\bar{p}^* \in (\bar{p}_0, \bar{p}_d)$.

It remains to prove that $\frac{\partial \tilde{V}_2(\bar{p}^*)}{\partial \bar{p}} \geq 0$. To do so, suppose that $\frac{\partial \tilde{V}_2(\bar{p}^*)}{\partial \bar{p}} < 0$. Then:

$$\bar{p}^* < \bar{p}_{Vm}. \tag{258}$$

(258) holds because: (i) $\tilde{V}_2(\bar{p})$ is a strictly convex function of $\bar{p}$, from Lemma 2; and (ii) $\frac{\partial \tilde{V}_2(\bar{p}_{Vm})}{\partial \bar{p}} = 0$, from (198). Furthermore, because $\tilde{S}_2(\bar{p})$ is a strictly concave function of $\bar{p}$, from Lemma 3:

$$\frac{\partial \tilde{S}_2(\bar{p})}{\partial \bar{p}} > 0 \text{ for all } \bar{p} < \bar{p}_{SM}. \tag{259}$$

Observe that:

$$\bar{p}^* < \bar{p}_{Vm} < \bar{p}_{SM}. \tag{260}$$

The first inequality in (260) reflects (258). The second inequality in (260) reflects Lemma 3. (236), (259), and (260) imply:

$$\frac{\partial \tilde{S}_2(\bar{p}^*)}{\partial \bar{p}} > 0. \tag{261}$$

$\frac{\partial \tilde{S}_2(\bar{p}^*)}{\partial \bar{p}} > 0$ (from (261)), $\frac{\partial \tilde{V}_2(\bar{p}^*)}{\partial \bar{p}} < 0$ (by assumption), and $\bar{p}^* \in (\bar{p}_0, \bar{p}_d)$ (by assumption) imply that consumer surplus increases and R 's revenue declines as $\bar{p}$ increases above $\bar{p}^*$. Therefore, $\bar{p}^*$ cannot be the welfare-maximizing value of $\bar{p}$. Hence, by contradiction, $\frac{\partial \tilde{V}_2(\bar{p}^*)}{\partial \bar{p}} \geq 0$. Consequently, (257) implies that $\frac{\partial \bar{p}^*}{\partial r} \leq 0$. ■

Proposition 6. When $\bar{p}^* \in (\bar{p}_0, \bar{p}_d)$: (i) $\frac{d\bar{p}^*}{dc_A} > 0$; (ii) $\frac{d\bar{p}^*}{dk_A} > 0$; (iii) $\frac{d\bar{p}^*}{dc} > 0$; and (iv) $\frac{d\bar{p}^*}{dc_N} < 0$.

Proof. (174) implies that consumer surplus is:

$$\begin{aligned}
S &= aQ - \frac{1}{2}bQ^2 - p[q + q_N] - \bar{p}q_A \\
&= aQ - \frac{1}{2}bQ^2 - p[q + q_N + q_A] + [p - \bar{p}]q_A \\
&= aQ - \frac{1}{2}bQ^2 - [a - bQ]Q + [p - \bar{p}]q_A \\
&= \frac{1}{2}bQ^2 + [p - \bar{p}]q_A = \frac{1}{2}bQ^2 + [a - bQ - \bar{p}]q_A
\end{aligned}$$

$$= \frac{b}{2} Q^2 + [a - \bar{p}] q_A - b Q q_A. \quad (262)$$

(262) implies that $\bar{p}^*$ is the solution to:

$$\text{Maximize}_{\bar{p}} W = \frac{b}{2} Q^2 + [a - \bar{p}] q_A - b Q q_A - r \bar{p} q_A - r a q_N + r b Q q_N. \quad (263)$$

(108) and (263) imply that for $\bar{p} \in [\bar{p}_0, \bar{p}_d]$:

$$\begin{aligned} \frac{dW}{d\bar{p}} &= b Q \left[\frac{[b+k][b+k_N]}{D} \right] + [a - \bar{p}] \left[\frac{3b^2 + 2b[k+k_N+k^R] + k[k_N+k^R]}{D} \right] \\ &\quad - q_A - b Q \left[\frac{3b^2 + 2b[k+k_N+k^R] + k[k_N+k^R]}{D} \right] - b q_A \left[\frac{[b+k][b+k_N]}{D} \right] \\ &\quad - r q_A - r \bar{p} \left[\frac{3b^2 + 2b[k+k_N+k^R] + k[k_N+k^R]}{D} \right] \\ &\quad - r a \left[-\frac{b[b+2k^R] + k[b+k^R]}{D} \right] \\ &\quad + r b Q \left[-\frac{b[b+2k^R] + k[b+k^R]}{D} \right] + r b q_N \left[\frac{[b+k][b+k_N]}{D} \right] = 0 \\ \Leftrightarrow &b[b+k][b+k_N] Q + [a - \bar{p}] \{ 3b^2 + 2b[k+k_N+k^R] + k[k_N+k^R] \} \\ &- D q_A - b \{ 3b^2 + 2b[k+k_N+k^R] + k[k_N+k^R] \} Q - b[b+k][b+k_N] q_A \\ &- r D q_A - r \bar{p} \{ 3b^2 + 2b[k+k_N+k^R] + k[k_N+k^R] \} \\ &+ r a \{ b[b+2k^R] + k[b+k^R] \} \\ &- r b \{ b[b+2k^R] + k[b+k^R] \} Q + r b[b+k][b+k_N] q_N = 0 \\ \Leftrightarrow &\{ b[b+k][b+k_N] - b[3b^2 + 2b(k+k_N+k^R) + k(k_N+k^R)] \\ &- r b[b(b+2k^R) + k(b+k^R)] \} Q \\ &- \{ D + b[b+k][b+k_N] + r D \} q_A + r b[b+k][b+k_N] q_N \\ &- \{ 3b^2 + 2b[k+k_N+k^R] + k[k_N+k^R] \\ &\quad + r[3b^2 + 2b(k+k_N+k^R) + k(k_N+k^R)] \} \bar{p} \end{aligned}$$

$$\begin{aligned}
& + \{ 3b^2 + 2b[k + k_N + k^R] + k[k_N + k^R] \\
& + r[b(b + 2k^R) + k(b + k^R)] \} a = 0.
\end{aligned} \tag{264}$$

The coefficient on Q in (264) is:

$$\begin{aligned}
& b[b + k][b + k_N] - b[3b^2 + 2b(k + k_N + k^R) + k(k_N + k^R)] \\
& - r b[b(b + 2k^R) + k(b + k^R)] \\
& = b[b^2 + b k_N + b k + k k_N - 3b^2 - 2b k - 2b k_N - 2b k^R - k k_N - k k^R \\
& - r b(b + 2k^R) - r k(b + k^R)] \\
& = b[-2b^2 - b k_N - b k - 2b k^R - k k^R - b^2 r - 2b r k^R - b r k - r k k^R] \\
& = -b[2b^2 + b k + b k_N + 2b k^R + k k^R + b^2 r \\
& + 2b r k^R + b r k + r k k^R] < 0.
\end{aligned} \tag{265}$$

(2) implies that the coefficient on $-q_A$ in (264) is:

$$\begin{aligned}
& [1 + r]D + b[b + k][b + k_N] \\
& = [1 + r] \{ [2b + k][k_N(k_A + k^R) + k_A k^R] + b k_A[3b + 2k] - b^2[b + k] \} \\
& + b[b + k][b + k_N] > 0 \text{ because } D > 0.
\end{aligned} \tag{266}$$

(264) – (266) imply that if $\bar{p}^* \in (\bar{p}_0, \bar{p}_d)$, $\bar{p}^*$ is determined by:

$$G - g \bar{p}^* = 0, \text{ where} \tag{267}$$

$$\begin{aligned}
G \equiv & r b[b + k][b + k_N] q_N \\
& + \{ 3b^2 + 2b[k + k_N + k^R] + k[k_N + k^R] \\
& + r[b(b + 2k^R) + k(b + k^R)] \} a \\
& - b[2b^2 + b k + b k_N + 2b k^R + k k^R + b^2 r \\
& + 2b r k^R + b r k + r k k^R] Q \\
& - \{ [1 + r]D + b[b + k][b + k_N] \} q_A, \text{ and}
\end{aligned}$$

$$\begin{aligned}
g \equiv & \{ 3b^2 + 2b[k + k_N + k^R] + k[k_N + k^R] \\
& + r[3b^2 + 2b(k + k_N + k^R) + k(k_N + k^R)] \} > 0.
\end{aligned} \tag{268}$$

To prove that $\frac{d\bar{p}^*}{dc_A} > 0$, observe from (268) that $\frac{dg}{d\bar{p}} = 0$. Therefore, (267) implies that for parameter x :

$$[G_x - \bar{p}^* g_x] dx + [G_{\bar{p}} - g] d\bar{p}^* = 0 \Rightarrow \frac{d\bar{p}^*}{dx} = \frac{G_x - \bar{p} g_x}{g - G_{\bar{p}}}. \quad (269)$$

(2) and (268) imply that because $D > 0$:

$$\begin{aligned} G_{c_A} &= r b [b + k] [b + k_N] \frac{dq_N}{dc_A} \\ &\quad - b [2b^2 + b k + b k_N + 2b k^R + k k^R + b^2 r \\ &\quad \quad + 2b r k^R + b r k + r k k^R] \frac{dQ}{dc_A} \\ &\quad - \{ [1 + r] D + b [b + k] [b + k_N] \} \frac{dq_A}{dc_A} \\ &> 0 \text{ if } \frac{dq_A}{dc_A} < 0, \frac{dq_N}{dc_A} > 0, \frac{dQ}{dc_A} < 0. \end{aligned} \quad (270)$$

(270) implies that because $D > 0$:

$$G_{c_A} > 0. \quad (271)$$

(2) and (268) imply that because $D > 0$:

$$\begin{aligned} G_{\bar{p}} &= r b [b + k] [b + k_N] \frac{dq_N}{d\bar{p}} \\ &\quad - b [2b^2 + b k + b k_N + 2b k^R + k k^R + b^2 r \\ &\quad \quad + 2b r k^R + b r k + r k k^R] \frac{dQ}{d\bar{p}} \\ &\quad - \{ [1 + r] D + b [b + k] [b + k_N] \} \frac{dq_A}{d\bar{p}} \\ &< 0 \text{ if } \frac{dq_A}{d\bar{p}} > 0, \frac{dq_N}{d\bar{p}} < 0, \text{ and } \frac{dQ}{d\bar{p}} > 0. \end{aligned} \quad (272)$$

(108) implies that because $D > 0$:

$$\frac{dq_A}{d\bar{p}} > 0, \frac{dq_N}{d\bar{p}} < 0, \text{ and } \frac{dQ}{d\bar{p}} > 0. \quad (273)$$

(272) and (273) imply that because $D > 0$:

$$G_{\bar{p}} < 0. \quad (274)$$

(268) implies:

$$g_{c_A} = 0. \quad (275)$$

(268), (269), (271), (274), and (275) imply that because $D > 0$:

$$\frac{d\bar{p}^*}{dc_A} = \frac{G_{c_A}}{g - G_{\bar{p}}} > 0.$$

To prove that $\frac{d\bar{p}^*}{dc} > 0$, observe that (2) and (268) imply that because $D > 0$:

$$\begin{aligned}
G_c &= r b [b + k] [b + k_N] \frac{dq_N}{dc} \\
&\quad - b [2b^2 + b k + b k_N + 2b k^R + k k^R + b^2 r \\
&\quad \quad + 2b r k^R + b r k + r k k^R] \frac{dQ}{dc} \\
&\quad - \{ [1 + r] D + b [b + k] [b + k_N] \} \frac{dq_A}{dc} \\
&> 0 \text{ if } \frac{dq_A}{dc} < 0, \frac{dq_N}{dc} > 0, \text{ and } \frac{dQ}{dc} < 0.
\end{aligned} \tag{276}$$

(276) implies that because $D > 0$:

$$G_c > 0. \tag{277}$$

(268) implies:

$$g_c = 0. \tag{278}$$

(268), (269), (274), (277), and (278) imply that because $D > 0$:

$$\frac{d\bar{p}^*}{dc} = \frac{G_c}{g - G_{\bar{p}}} > 0.$$

To prove that $\frac{\partial \bar{p}^*}{\partial c_N} < 0$, observe that (195) implies that for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\begin{aligned}
\frac{\partial V_2(\bar{p})}{\partial \bar{p}} &= q_A + \bar{p} \frac{\partial q_A}{\partial \bar{p}} + P(Q) \frac{\partial q_N}{\partial \bar{p}} - b q_N \frac{\partial Q}{\partial \bar{p}} \\
\Rightarrow \frac{\partial^2 V_2(\bar{p})}{\partial \bar{p} \partial c_N} &= \frac{\partial q_A}{\partial c_N} + \bar{p} \frac{\partial^2 q_A}{\partial \bar{p} \partial c_N} + P(Q) \frac{\partial^2 q_N}{\partial \bar{p} \partial c_N} \\
&\quad - b \frac{\partial Q}{\partial c_N} \frac{\partial q_N}{\partial \bar{p}} - b q_N \frac{\partial^2 Q}{\partial \bar{p} \partial c_N} - b \frac{\partial q_N}{\partial c_N} \frac{\partial Q}{\partial \bar{p}} \\
&= \frac{\partial q_A}{\partial c_N} - b \frac{\partial Q}{\partial c_N} \frac{\partial q_N}{\partial \bar{p}} - b \frac{\partial q_N}{\partial c_N} \frac{\partial Q}{\partial \bar{p}}.
\end{aligned} \tag{280}$$

The last equality in (280) holds because $\frac{\partial^2 q_A}{\partial \bar{p} \partial c_N} = \frac{\partial^2 q_N}{\partial \bar{p} \partial c_N} = \frac{\partial^2 Q}{\partial \bar{p} \partial c_N} = 0$ when $\bar{p} \in (\bar{p}_0, \bar{p}_d)$, from Lemma A2.

(2) and Lemma A2 imply that when $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\begin{aligned}
&\frac{\partial Q}{\partial c_N} \frac{\partial q_N}{\partial \bar{p}} + \frac{\partial q_N}{\partial c_N} \frac{\partial Q}{\partial \bar{p}} \\
&\stackrel{s}{=} [b + k] [k_A - b] [b (b + 2k^R) + k (b + k^R)] - [2b + k] [k_A + k^R] [b + k] [b + k_N] \\
&\stackrel{s}{=} [k_A - b] [b (b + 2k^R) + k (b + k^R)] - [2b + k] [k_A + k^R] [b + k_N]
\end{aligned}$$

$$\begin{aligned}
&= k_A [b(b + 2k^R) + k(b + k^R)] - b[b(b + 2k^R) + k(b + k^R)] \\
&\quad - k_A[2b + k][b + k_N] - k^R[b + k_N][2b + k] \\
&= k_A [b(b + 2k^R) + k(b + k^R) - (2b + k)(b + k_N)] \\
&\quad - b[b(b + 2k^R) + k(b + k^R)] - k^R[b + k_N][2b + k]. \tag{281}
\end{aligned}$$

The coefficient on k_A in (281) is:

$$\begin{aligned}
&b[b + 2k^R] + k[b + k^R] - [2b + k][b + k_N] \\
&= b^2 + 2bk^R + bk + k^R - 2b^2 - 2bk_N - kb - kk_N \\
&= 2bk^R + k^R - b^2 - 2bk_N - kk_N. \tag{282}
\end{aligned}$$

(281) and (282) imply that because $k_A \leq k_N$:

$$\begin{aligned}
&\frac{\partial Q}{\partial c_N} \frac{\partial q_N}{\partial \bar{p}} + \frac{\partial q_N}{\partial c_N} \frac{\partial Q}{\partial \bar{p}} \\
&\stackrel{s}{=} k_A [2bk^R + k^R - b^2 - 2bk_N - kk_N] - b[b(b + 2k^R) + k(b + k^R)] \\
&\quad - k^R[b + k_N][2b + k] \\
&\leq k_A [2bk^R + k^R - b^2 - 2bk_N - kk_N] - b[b(b + 2k^R) + k(b + k^R)] \\
&\quad - k^R[b + k_A][2b + k] \\
&= k_A [2bk^R + k^R - b^2 - 2bk_N - kk_N - k^R(2b + k)] \\
&\quad - b[b(b + 2k^R) + k(b + k^R)] - k^Rb[2b + k] \\
&= k_A [-b^2 - 2bk_N - kk_N] - b[b(b + 2k^R) + k(b + k^R)] - k^Rb[2b + k] < 0. \tag{283}
\end{aligned}$$

Because $\frac{\partial q_A}{\partial c_N} > 0$ when $\bar{p} \in (\bar{p}_0, \bar{p}_d)$, from Lemma A2, (280) and (283) imply:

$$\frac{\partial^2 V_2(\bar{p})}{\partial \bar{p} \partial c_N} = \frac{\partial q_A}{\partial c_N} - b \left[\frac{\partial Q}{\partial c_N} \frac{\partial q_N}{\partial \bar{p}} + \frac{\partial q_N}{\partial c_N} \frac{\partial Q}{\partial \bar{p}} \right] > 0. \tag{284}$$

(234) and (279) imply:

$$\begin{aligned}
\frac{\partial S_2(\bar{p})}{\partial \bar{p}} &= a \frac{\partial Q}{\partial \bar{p}} - bQ \frac{\partial Q}{\partial \bar{p}} - q_A - \bar{p} \frac{\partial q_A}{\partial \bar{p}} - P(Q) \left[\frac{\partial q_N}{\partial \bar{p}} + \frac{\partial q}{\partial \bar{p}} \right] + b \frac{\partial Q}{\partial \bar{p}} [q_N + q] \\
&= a \frac{\partial Q}{\partial \bar{p}} - bQ \frac{\partial Q}{\partial \bar{p}} - P(Q) \frac{\partial q}{\partial \bar{p}} + b \frac{\partial Q}{\partial \bar{p}} q
\end{aligned}$$

$$\begin{aligned}
& -q_A - \bar{p} \frac{\partial q_A}{\partial \bar{p}} - P(Q) \frac{\partial q_N}{\partial \bar{p}} + b \frac{\partial Q}{\partial \bar{p}} q_N \\
& = a \frac{\partial Q}{\partial \bar{p}} - b Q \frac{\partial Q}{\partial \bar{p}} - P(Q) \frac{\partial q}{\partial \bar{p}} + b \frac{\partial Q}{\partial \bar{p}} q - \frac{\partial V_2(\bar{p})}{\partial \bar{p}} \\
\Rightarrow \frac{\partial^2 S_2(\bar{p})}{\partial \bar{p} \partial c_N} & = a \frac{\partial^2 Q}{\partial \bar{p} \partial c_N} - b Q \frac{\partial^2 Q}{\partial \bar{p} \partial c_N} - b \frac{\partial Q}{\partial \bar{p}} \frac{\partial Q}{\partial c_N} - \frac{\partial^2 V_2(\bar{p})}{\partial \bar{p} \partial c_N} \\
& - P(Q) \frac{\partial^2 q}{\partial \bar{p} \partial c_N} + b \frac{\partial Q}{\partial c_N} \frac{\partial q}{\partial \bar{p}} + b \frac{\partial^2 Q}{\partial \bar{p} \partial c_N} q + b \frac{\partial Q}{\partial \bar{p}} \frac{\partial q}{\partial c_N}. \tag{285}
\end{aligned}$$

Lemma A2 implies that $\frac{\partial^2 Q}{\partial \bar{p} \partial c_N} = \frac{\partial^2 q}{\partial \bar{p} \partial c_N} = 0$ when $\bar{p} \in (\bar{p}_0, \bar{p}_d)$. Therefore, because $Q = Q^R + q$, (285) implies:

$$\begin{aligned}
\frac{\partial^2 S_2(\bar{p})}{\partial \bar{p} \partial c_N} & = -b \frac{\partial Q}{\partial \bar{p}} \frac{\partial Q}{\partial c_N} - \frac{\partial^2 V_2(\bar{p})}{\partial \bar{p} \partial c_N} + b \frac{\partial Q}{\partial c_N} \frac{\partial q}{\partial \bar{p}} + b \frac{\partial Q}{\partial \bar{p}} \frac{\partial q}{\partial c_N} \\
& = -b \frac{\partial Q}{\partial \bar{p}} \frac{\partial Q^R}{\partial c_N} - \frac{\partial^2 V_2(\bar{p})}{\partial \bar{p} \partial c_N} + b \frac{\partial Q}{\partial c_N} \frac{\partial q}{\partial \bar{p}}. \tag{286}
\end{aligned}$$

(280) and (286) imply that because $Q^R = q_A + q_N$:

$$\begin{aligned}
\frac{\partial^2 S_2(\bar{p})}{\partial \bar{p} \partial c_N} & = -b \frac{\partial Q}{\partial \bar{p}} \frac{\partial Q^R}{\partial c_N} - \left[\frac{\partial q_A}{\partial c_N} - b \frac{\partial Q}{\partial c_N} \frac{\partial q_N}{\partial \bar{p}} - b \frac{\partial q_N}{\partial c_N} \frac{\partial Q}{\partial \bar{p}} \right] + b \frac{\partial Q}{\partial c_N} \frac{\partial q}{\partial \bar{p}} \\
& = -b \frac{\partial Q}{\partial \bar{p}} \left[\frac{\partial Q^R}{\partial c_N} - \frac{\partial q_N}{\partial c_N} \right] - \frac{\partial q_A}{\partial c_N} + b \frac{\partial Q}{\partial c_N} \left[\frac{\partial q}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right] \\
& = -b \frac{\partial Q}{\partial \bar{p}} \frac{\partial q_A}{\partial c_N} - \frac{\partial q_A}{\partial c_N} + b \frac{\partial Q}{\partial c_N} \left[\frac{\partial q}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right]. \tag{287}
\end{aligned}$$

(2), (287), and Lemma A2 imply:

$$\begin{aligned}
\frac{\partial^2 S_2(\bar{p})}{\partial \bar{p} \partial c_N} & \stackrel{s}{=} -b[b+k][b+k_N][2b+k][b+k^R] - [2b+k][b+k^R]D \\
& + b[-(b+k)(k_A-b)][-b(b+k_N)-b(b+2k^R)-k(b+k^R)] \\
& = -b[b+k][b+k_N][2b+k][b+k^R] - [2b+k][b+k^R]D \\
& + b[b+k][k_A-b][b(b+k_N)+b(b+2k^R)+k(b+k^R)]. \tag{288}
\end{aligned}$$

(2) and (288) imply:

$$\frac{\partial^2 S_2(\bar{p})}{\partial \bar{p} \partial c_N} = -b[b+k][b+k_N][2b+k][b+k^R]$$

$$\begin{aligned}
& + b[b+k][k_A-b][b(b+k_N)+b(b+2k^R)+k(b+k^R)] \\
& - [2b+k][b+k^R] \\
& \cdot [(2b+k)(k_N[k_A+k^R]+k_Ak^R)+bk_A(3b+2k)-b^2(b+k)] \\
= & b[b+k][k_A-b][b(b+k_N)+b(b+2k^R)+k(b+k^R)] \\
& - [2b+k][b+k^R] \\
& \cdot [b(b+k)(b+k_N)+(2b+k)(k_N[k_A+k^R]+k_Ak^R)+bk_A(3b+2k)-b^2(b+k)] \\
= & b[b+k][k_A-b][b(b+k_N)+b(b+2k^R)+k(b+k^R)] \\
& - [2b+k][b+k^R][b(b+k)k_N+(2b+k)(k_N[k_A+k^R]+k_Ak^R)+bk_A(3b+2k)] \\
= & -b[b+k]b[b(b+k_N)+b(b+2k^R)+k(b+k^R)] \\
& + b[b+k]k_A[b(b+k_N)+b(b+2k^R)+k(b+k^R)] \\
& - [2b+k][b+k^R][b(b+k)k_N+(2b+k)(k_N[k_A+k^R]+k_Ak^R)] \\
& - k_A[2b+k][b+k^R]b[3b+2k] \\
= & -b[b+k]b[b(b+k_N)+b(b+2k^R)+k(b+k^R)] \\
& - [2b+k][b+k^R][b(b+k)k_N+(2b+k)k_Nk^R] \\
& + k_A\{b[b+k][b(b+k_N)+b(b+2k^R)+k(b+k^R)] \\
& - [2b+k][b+k^R][(2b+k)(k_N+k^R)+b(3b+2k)]\}. \tag{289}
\end{aligned}$$

The coefficient on k_A in (289) is:

$$\begin{aligned}
& b[b+k][b(b+k_N)+b(b+2k^R)+k(b+k^R)] \\
& - [2b+k][b+k^R][(2b+k)(k_N+k^R)+b(3b+2k)] \\
= & k_N[b(b+k)b-(2b+k)(b+k^R)(2b+k)] \\
& + b[b+k][b^2+b(b+2k^R)+k(b+k^R)] \\
& - [2b+k][b+k^R][(2b+k)k^R+b(3b+2k)] \\
= & k_N[b(b+k)b-(2b+k)(b+k^R)(2b+k)] \\
& + k^R[b(b+k)(2b+k)-(2b+k)(b+k^R)(2b+k)]
\end{aligned}$$

$$+ b[b+k][2b^2+kb] - [2b+k][b+k^R]b[3b+2k]. \quad (290)$$

The coefficient on k_N in (290) is:

$$b[b+k]b - [2b+k][b+k^R][2b+k] < 0. \quad (291)$$

The inequality in (291) holds because $b < b+k^R$, $b+k < 2b+k$, and $b < 2b+k$.

The coefficient on k^R in (290) is:

$$b[b+k][2b+k] - [2b+k][b+k^R][2b+k] < 0. \quad (292)$$

The inequality in (292) holds because $b < b+k^R$ and $b+k < 2b+k$.

The last line in (290) is:

$$\begin{aligned} & b[b+k][2b^2+kb] - [2b+k][b+k^R]b[3b+2k] \\ &= b^2[b+k][2b+k] - [2b+k][b+k^R]b[3b+2k] \\ &\stackrel{s}{=} b[b+k] - [b+k^R][3b+2k] < 0. \end{aligned} \quad (293)$$

The inequality in (293) holds because $b < b+k^R$ and $b+k < 3b+2k$.

(290) – (293) imply that the coefficient on k_A in (289) is negative. Therefore, (289) implies:

$$\frac{\partial^2 S_2(\bar{p})}{\partial \bar{p} \partial c_N} < 0. \quad (294)$$

$\bar{p}^*$ satisfies:

$$\frac{\partial S_2(\bar{p}^*)}{\partial \bar{p}} - r \frac{\partial V_2(\bar{p}^*)}{\partial \bar{p}} = 0. \quad (295)$$

Totally differentiating (295) with respect to c_N provides:

$$\begin{aligned} & \frac{\partial^2 S_2(\bar{p}^*)}{(\partial \bar{p})^2} \frac{\partial \bar{p}^*}{\partial c_N} + \frac{\partial^2 S_2(\bar{p}^*)}{\partial \bar{p} \partial c_N} - r \left[\frac{\partial^2 V_2(\bar{p}^*)}{(\partial \bar{p})^2} \frac{\partial \bar{p}^*}{\partial c_N} + \frac{\partial^2 V_2(\bar{p}^*)}{\partial \bar{p} \partial c_N} \right] = 0 \\ \Rightarrow \quad & \frac{\partial \bar{p}^*}{\partial c_N} = - \frac{\frac{\partial^2 S_2(\bar{p}^*)}{\partial \bar{p} \partial c_N} - r \frac{\partial^2 V_2(\bar{p}^*)}{\partial \bar{p} \partial c_N}}{\frac{\partial^2 S_2(\bar{p}^*)}{(\partial \bar{p})^2} - r \frac{\partial^2 V_2(\bar{p}^*)}{(\partial \bar{p})^2}} = - \frac{\frac{\partial^2 S_2(\bar{p}^*)}{\partial \bar{p} \partial c_N} - r \frac{\partial^2 V_2(\bar{p}^*)}{\partial \bar{p} \partial c_N}}{\frac{\partial^2 W_2(\bar{p}^*)}{(\partial \bar{p})^2}} < 0. \end{aligned}$$

The inequality follows from (284) and (294), because $\frac{\partial^2 W_2(\bar{p}^*)}{(\partial \bar{p})^2} < 0$ (from (173) and Lemmas 2 and 3).

To prove that $\frac{d\bar{p}^*}{dk_A} > 0$, observe that (2) implies:

$$\frac{\partial D}{\partial k_A} = [2b+k][k_N+k^R] + b[3b+2k] > 0. \quad (296)$$

(20) and (296) imply that for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\frac{dq_A}{dk_A} = -\frac{q_A}{D} \frac{\partial D}{\partial k_A} < 0. \quad (297)$$

(29) and (297) imply that for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\begin{aligned} \frac{\partial q_N}{\partial k_A} &= -\left[\frac{k_A + k^R}{b + k^R}\right] \frac{\partial q_A}{\partial k_A} - \left[\frac{1}{b + k^R}\right] q_A \\ &= \left[\frac{k_A + k^R}{b + k^R}\right] \frac{q_A}{D} \frac{\partial D}{\partial k_A} - \left[\frac{1}{b + k^R}\right] q_A \stackrel{s}{=} [k_A + k^R] \frac{1}{D} \frac{\partial D}{\partial k_A} - 1. \end{aligned} \quad (298)$$

(2), (296), and (298) imply that $\frac{\partial q_N}{\partial k_A} > 0$ because:

$$\begin{aligned} \frac{\partial q_N}{\partial k_A} > 0 &\Leftrightarrow [k_A + k^R] \frac{\partial D}{\partial k_A} > D \\ &\Leftrightarrow [2b + k] [k_N (k_A + k^R) + k_A k^R] + b k_A [3b + 2k] - b^2 [b + k] \\ &< [k_A + k^R] [(2b + k) (k_N + k^R) + b (3b + 2k)] \\ &\Leftrightarrow [2b + k] [k_N (k_A + k^R) + k_A k^R] + b k_A [3b + 2k] - b^2 [b + k] \\ &< [2b + k] [k_N + k^R] [k_A + k^R] + b [k_A + k^R] [3b + 2k] \\ &\Leftrightarrow [2b + k] [k_N (k_A + k^R) + k_A k^R] + b k_A [3b + 2k] - b^2 [b + k] \\ &< [2b + k] \left[k_N (k_A + k^R) + k^R k_A + (k^R)^2 \right] + b [k_A + k^R] [3b + 2k]. \end{aligned} \quad (299)$$

It is apparent that the inequality in (299) holds.

Because $Q(\bar{p})$ is linear in $\bar{p}$:

$$Q(\bar{p}) = Q(\bar{p}_0) + \frac{\partial Q}{\partial \bar{p}} [\bar{p} - \bar{p}_0] \quad \text{for } \bar{p} \in (\bar{p}_0, \bar{p}_d). \quad (300)$$

(24) implies that for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\frac{\partial Q}{\partial \bar{p}} = \frac{[b + k] [b + k_N]}{D}. \quad (301)$$

(296) and (301) imply:

$$\frac{\partial Q}{\partial \bar{p} \partial k_A} = -\frac{[b + k] [b + k_N]}{D^2} \frac{\partial D}{\partial k_A} < 0. \quad (302)$$

(6) implies that $\bar{p}_0$ does not vary with k_A . Lemma A1 implies that $Q(\bar{p}_0)$ does not vary with k_A . Therefore, (300) and (302) imply that for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\frac{\partial Q(\bar{p})}{\partial k_A} = \frac{\partial Q}{\partial \bar{p} \partial k_A} [\bar{p} - \bar{p}_0] < 0. \quad (303)$$

In summary, (297), (299), and (303) imply:

$$\frac{dq_A}{dk_A} < 0, \quad \frac{dq_N}{dk_A} > 0, \quad \text{and} \quad \frac{dQ}{dk_A} < 0 \quad \text{for all } \bar{p} \in (\bar{p}_0, \bar{p}_d). \quad (304)$$

(20) implies that for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$:

$$\begin{aligned} D q_A = & \left[3b^2 + 2b(k + k_N + k^R) + k(k_N + k^R) \right] [\bar{p} - c_A] \\ & + b[b + k^R][a - c] - [2b + k][b + k^R][a - c_N] \end{aligned}$$

which is not a function of k_A . Therefore, (268) implies:

$$\begin{aligned} G_{k_A} = & r b [b + k][b + k_N] \frac{\partial q_N}{\partial k_A} \\ & - b [2b^2 + b k + b k_N + 2b k^R + k k^R + b^2 r + 2b r k^R + b r k + r k k^R] \frac{\partial Q}{\partial k_A} \\ & - b [b + k][b + k_N] \frac{\partial q_A}{\partial k_A}. \end{aligned} \quad (305)$$

(304) and (305) imply:

$$G_{k_A} > 0. \quad (306)$$

(268) implies:

$$g_{k_A} = 0. \quad (307)$$

(268), (269), (274), (306), and (307) imply:

$$\frac{d\bar{p}^*}{dk_A} = \frac{G_{k_A}}{g - G_{\bar{p}}} > 0. \quad \blacksquare$$

B. Benchmark Setting where R is a Monopoly Supplier.

R 's problem, [M], when it is the sole supplier of the product is:

$$\begin{aligned} & \text{Maximize}_{q_A \geq 0, q_N \geq 0} P_A(q_A + q_N) q_A + [a - b(q_A + q_N)] q_N - C^R(q_A, q_N) \\ & \text{where } P_A(Q) = \begin{cases} \bar{p} & \text{if } P(Q) \geq \bar{p} \\ P(Q) & \text{if } P(Q) < \bar{p}. \end{cases} \end{aligned} \quad (308)$$

Definitions

$$\bar{p}_{0M} \equiv c_A + \frac{a - c_N}{2b + k_N + k^R} [b + k^R]. \quad (309)$$

$$\bar{p}_{dM} \equiv \frac{a[(b + k^R)(k_A + k_N) - b k_N + k_A k_N] + b c_N [k_A - b] + b c_A [b + k_N]}{[2b + k^R][k_A + k_N] - b k_N + k_N k_A}. \quad (310)$$

$$\bar{p}_{bM} \equiv \frac{a \left[(b + k^R) (k_A + k_N) + k_N k_A \right] + b \left[c_N k_A + c_A k_N \right]}{\left[2b + k^R \right] \left[k_A + k_N \right] + k_N k_A}. \quad (311)$$

Assumptions

1. $\left[k_A + k^R \right] \left[2b + k_N + k^R \right] > \left[b + k^R \right]^2$.
2. $0 \leq c_A \leq c_N < a$.
3. $0 \leq k_A \leq k_N$.

We characterize equilibrium outcomes when R is the monopoly supplier as follows.

Lemma 4 establishes that the price cap does not bind, and so has no impact on equilibrium outputs or prices, when $\bar{p} > \bar{p}_{bM}$.

Lemma 5 establishes that the market price of oil that is shipped without using the Alliance input is $\bar{p}$ when $\bar{p} \in [\bar{p}_{dM}, \bar{p}_{bM}]$, where $\bar{p}_{dM} < \bar{p}_{bM}$.

Lemma 6 establishes that the market price of oil exceeds $\bar{p}$ and R ships a positive amount of oil using the Alliance input (so $q_A > 0$) when $\bar{p} \in (\bar{p}_{0M}, \bar{p}_{dM})$.

Lemma 7 establishes that R does not ship any oil using the Alliance input and the market price exceeds $\bar{p}$ when $\bar{p} \leq \bar{p}_{0M}$.

Lemma 4. *If $\bar{p} > \bar{p}_{bM}$, then in equilibrium:*

$$\begin{aligned} q_A &= \frac{[a - c_A] [2b + k_N + k^R] - [a - c_N] [2b + k^R]}{[2b + k^R] [k_A + k_N] + k_A k_N}, \\ q_N &= \frac{[a - c_N] [2b + k_A + k^R] - [a - c_A] [2b + k^R]}{[2b + k^R] [k_A + k_N] + k_A k_N}, \text{ and} \\ Q^R &= \frac{[a - c_N] k_A + [a - c_A] k_N}{[2b + k^R] [k_A + k_N] + k_A k_N}. \end{aligned}$$

Proof. (308) implies that when the price cap does not bind, [M] is:

$$\begin{aligned} \text{Maximize}_{q_A, q_N} \quad & [a - b(q_A + q_N)] [q_A + q_N] - c_A q_A - \frac{k_A}{2} [q_A]^2 \\ & - c_N q_N - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_A + q_N]^2. \end{aligned} \quad (312)$$

Differentiating (312) with respect to q_A provides:³

$$\begin{aligned} a - b[q_A + q_N] - b[q_A + q_N] - c_A - k_A q_A - k^R[q_A + q_N] &= 0 \\ \Rightarrow a - [2b + k^R][q_A + q_N] - c_A - k_A q_A &= 0. \end{aligned} \quad (313)$$

Differentiating (312) with respect to q_N provides:

$$\begin{aligned} a - b[q_A + q_N] - b[q_A + q_N] - c_N - k_N q_N - k^R[q_A + q_N] &= 0 \\ \Rightarrow a - [2b + k^R]q_A - [2b + k_N + k^R]q_N - c_N &= 0 \\ \Rightarrow q_N &= \frac{a - [2b + k^R]q_A - c_N}{2b + k_N + k^R}. \end{aligned} \quad (314)$$

(313) and (314) imply:

$$\begin{aligned} a - [2b + k^R] \left[q_A + \frac{a - (2b + k^R)q_A - c_N}{2b + k_N + k^R} \right] - c_A - k_A q_A &= 0 \\ \Leftrightarrow a[2b + k_N + k^R] - [2b + k^R][q_A(2b + k_N + k^R) + a - (2b + k^R)q_A - c_N] \\ &\quad - c_A[2b + k_N + k^R] - k_A q_A[2b + k_N + k^R] = 0 \\ \Leftrightarrow a[2b + k_N + k^R] - [2b + k^R][q_A k_N + a - c_N] \\ &\quad - c_A[2b + k_N + k^R] - k_A q_A[2b + k_N + k^R] = 0 \\ \Leftrightarrow [a - c_A][2b + k_N + k^R] - [2b + k^R][q_A k_N + a - c_N] \\ &\quad - k_A q_A[2b + k_N + k^R] = 0 \\ \Leftrightarrow [a - c_A][2b + k_N + k^R] - [(2b + k^R)(k_A + k_N) + k_A k_N]q_A \\ &\quad - [a - c_N][2b + k^R] = 0 \\ \Leftrightarrow q_A &= \frac{[a - c_A][2b + k_N + k^R] - [a - c_N][2b + k^R]}{[2b + k^R][k_A + k_N] + k_A k_N}. \end{aligned} \quad (315)$$

By symmetry:

$$q_N = \frac{[a - c_N][2b + k_A + k^R] - [a - c_A][2b + k^R]}{[2b + k^R][k_A + k_N] + k_A k_N}. \quad (316)$$

(315) and (316) imply:

$$Q^R = q_A + q_N$$

³It is readily verified that the determinant of the Hessian associated with [M] is $[2b + k^R]k_N + [2b + k^R]k_A > 0$.

$$\begin{aligned}
&= \frac{1}{[2b + k^R][k_A + k_N] + k_A k_N} \{ [a - c_A][2b + k_N + k^R] - [a - c_N][2b + k^R] \\
&\quad + [a - c_N][2b + k_A + k^R] - [a - c_A][2b + k^R] \} \\
&= \frac{[a - c_N]k_A + [a - c_A]k_N}{[2b + k^R][k_A + k_N] + k_A k_N}. \tag{317}
\end{aligned}$$

(317) implies:

$$\begin{aligned}
P(Q) &= a - bQ^R = a - b \frac{[a - c_N]k_A + [a - c_A]k_N}{[2b + k^R][k_A + k_N] + k_A k_N} \\
&= \frac{a[(2b + k^R)(k_A + k_N) + k_A k_N] - b[a - c_N]k_A - b[a - c_A]k_N}{[2b + k^R][k_A + k_N] + k_A k_N} \\
&= \frac{a[(b + k^R)(k_A + k_N) + k_N k_A] + ab[k_A + k_N] - ab[k_A + k_N] + b[c_N k_A + c_A k_N]}{[2b + k^R][k_A + k_N] + k_N k_A} \\
&= \frac{a[(b + k^R)(k_A + k_N) + k_N k_A] + b[c_N k_A + c_A k_N]}{[2b + k^R][k_A + k_N] + k_N k_A} = \bar{p}_{bM}. \quad \blacksquare
\end{aligned}$$

Lemma 5. Suppose $\bar{p} \in (\bar{p}_{dM}, \bar{p}_{bM}]$, where $\bar{p}_{dM} < \bar{p}_{bM}$. Then in equilibrium:

$$\begin{aligned}
q_A &= \frac{b[c_N - c_A] + k_N[a - \bar{p}]}{b[k_A + k_N]}; \quad q_N = \frac{[a - \bar{p}]k_A - b[c_N - c_A]}{b[k_A + k_N]}; \\
P(Q) &= \bar{p}; \quad \text{and} \quad Q^R = \frac{a - \bar{p}}{b}. \tag{318}
\end{aligned}$$

Proof. To prove that $\bar{p}_{dM} < \bar{p}_{bM}$, observe that (310) and (311) imply:

$$\begin{aligned}
\bar{p}_{bM} > \bar{p}_{dM} &\Leftrightarrow \frac{a[(b + k^R)(k_A + k_N) + k_N k_A] + b[c_N k_A + c_A k_N]}{[2b + k^R][k_A + k_N] + k_N k_A} \\
&> \frac{a[(b + k^R)(k_A + k_N) - b k_N + k_A k_N] + b c_N[k_A - b] + b c_A[b + k_N]}{[2b + k^R][k_A + k_N] - b k_N + k_N k_A} \\
&\Leftrightarrow \frac{a[(b + k^R)(k_A + k_N) + k_N k_A] + b[c_N k_A + c_A k_N]}{[2b + k^R][k_A + k_N] + k_N k_A} \\
&> \frac{a[(b + k^R)(k_A + k_N) + k_A k_N] - ab k_N + b[c_N k_A + c_A k_N] - b^2[c_N - c_A]}{[2b + k^R][k_A + k_N] + k_N k_A - b k_N}. \tag{319}
\end{aligned}$$

Definitions. $N_1 \equiv a[(b + k^R)(k_A + k_N) + k_N k_A] + b[c_N k_A + c_A k_N] > 0$.

$$D_1 \equiv [2b + k^R][k_A + k_N] + k_N k_A > 0. \tag{320}$$

(319) and (320) imply:

$$\bar{p}_{bM} > \bar{p}_{dM} \Leftrightarrow \frac{N_1}{D_1} > \frac{N_1 - a b k_N - b^2 [c_N - c_A]}{D_1 - b k_N}. \quad (321)$$

(320) implies:

$$D_1 - b k_N = [2b + k^R] k_A + [b + k^R] k_N + k_N k_A > 0. \quad (322)$$

(320) and (322) imply that (321) holds if $N_1 - a b k_N - b^2 [c_N - c_A] \leq 0$.

Suppose $N_1 - a b k_N - b^2 [c_N - c_A] > 0$. Then (321) implies:

$$\begin{aligned} \bar{p}_{bM} > \bar{p}_{dM} &\Leftrightarrow N_1 D_1 - b k_N N_1 > D_1 N_1 - a b k_N D_1 - b^2 [c_N - c_A] D_1 \\ &\Leftrightarrow a b k_N D_1 + b^2 [c_N - c_A] D_1 > b k_N N_1 \\ &\Leftrightarrow a k_N D_1 + b [c_N - c_A] D_1 > k_N N_1 \\ &\Leftrightarrow [a k_N + b (c_N - c_A)] D_1 > k_N N_1. \end{aligned} \quad (323)$$

(320) and (323) imply:

$$\begin{aligned} \bar{p}_{bM} > \bar{p}_{dM} &\Leftrightarrow [a k_N + b (c_N - c_A)] \{ [2b + k^R] [k_A + k_N] + k_N k_A \} \\ &> a k_N [(b + k^R) (k_A + k_N) + k_N k_A] + b k_N [c_N k_A + c_A k_N] \\ \Leftrightarrow a k_N \{ [2b + k^R] [k_A + k_N] + k_N k_A \} &+ b [c_N - c_A] \{ [2b + k^R] [k_A + k_N] + k_N k_A \} \\ &> a k_N [b + k^R] [k_A + k_N] + a (k_N)^2 k_A + b k_N [c_N k_A + c_A k_N] \\ \Leftrightarrow a k_N [2b + k^R] [k_A + k_N] + a (k_N)^2 k_A &+ b [c_N - c_A] [2b + k^R] [k_A + k_N] \\ &+ b k_N k_A [c_N - c_A] \\ &> a k_N [b + k^R] [k_A + k_N] + a (k_N)^2 k_A + b c_N k_N k_A + b c_A (k_N)^2 \\ \Leftrightarrow a k_N b [k_A + k_N] + b [c_N - c_A] [2b + k^R] [k_A + k_N] &+ b k_N k_A [c_N - c_A] \\ &> b c_N k_N k_A + b c_A (k_N)^2 \\ \Leftrightarrow a k_N b [k_A + k_N] + b [c_N - c_A] [2b + k^R] [k_A + k_N] &- b k_N k_A c_A > b c_A (k_N)^2 \\ \Leftrightarrow a k_N b [k_A + k_N] + b [c_N - c_A] [2b + k^R] [k_A + k_N] &> b c_A k_N [k_A + k_N] \\ \Leftrightarrow [a - c_A] k_N b [k_A + k_N] + b [c_N - c_A] [2b + k^R] [k_A + k_N] &> 0. \end{aligned} \quad (324)$$

The inequality in (324) holds because $a > c_N \geq c_A$, by assumption.

Next observe that [M] can be stated as:

$$\begin{aligned} \text{Maximize}_{q_A, Q^R} \Pi^R &= [P_A(Q^R) - c_A] q_A + [P(Q^R) - c_N] [Q^R - q_A] \\ &\quad - \frac{k_A}{2} [q_A]^2 - \frac{k_N}{2} [Q^R - q_A]^2 - \frac{k^R}{2} [Q^R]^2 \\ \text{where } P_A(Q^R) &= \begin{cases} \bar{p} & \text{if } P(Q^R) \geq \bar{p} \\ P(Q^R) & \text{if } \bar{p} > P(Q^R). \end{cases} \end{aligned} \quad (325)$$

Then when $q_A > 0$ and there exists a range of $\bar{p}$ for which $P(Q) = \bar{p}$, the necessary conditions for a solution to [M] are:

$$\frac{\partial \Pi^R}{\partial q_A} = P_A(Q^R) - c_A - k_A q_A - [P(Q^R) - c_N] + k_N [Q^R - q_A] = 0 \quad \text{and} \quad (326)$$

$$\frac{\partial^+ \Pi^R}{\partial Q^R} \leq 0 \quad \text{and} \quad \frac{\partial^- \Pi^R}{\partial Q^R} \geq 0 \quad \text{for all } \bar{p} \in [\bar{p}_{dM}, \bar{p}_{bM}]. \quad (327)$$

Recall that $\frac{\partial^- \Pi^R}{\partial Q^R}$ denotes the left-sided derivative of Π^R with respect to Q^R , which is relevant when $P_A(\cdot) = \bar{p}$, and $\frac{\partial^+ \Pi^R}{\partial Q^R}$ denotes the right-sided derivative of Π^R with respect to Q^R , which is relevant when $P_A(\cdot) = P(Q^R)$.

If $P_A(Q^R) = \bar{p}$, then (326) implies:

$$\begin{aligned} \bar{p} - c_A - k_A q_A - [\bar{p} - c_N] + k_N [Q^R - q_A] &= 0 \\ \Leftrightarrow c_N - c_A - k_A q_A + k_N Q^R - k_N q_A &= 0. \end{aligned} \quad (328)$$

If $P_A(Q^R) = \bar{p}$, then $Q^R = \frac{a - \bar{p}}{b}$, so (328) implies:

$$\begin{aligned} c_N - c_A - k_A q_A + k_N \left[\frac{a - \bar{p}}{b} \right] - k_N q_A &= 0 \\ \Leftrightarrow c_N - c_A + k_N \left[\frac{a - \bar{p}}{b} \right] &= q_A [k_A + k_N] \\ \Leftrightarrow q_A &= \frac{c_N - c_A + k_N \left[\frac{a - \bar{p}}{b} \right]}{k_A + k_N} = \frac{b [c_N - c_A] + k_N [a - \bar{p}]}{b [k_A + k_N]}. \end{aligned} \quad (329)$$

(329) implies that when $Q^R = \frac{a - \bar{p}}{b}$:

$$\begin{aligned} q_N = Q^R - q_A &= \frac{a - \bar{p}}{b} - \frac{b [c_N - c_A] + k_N [a - \bar{p}]}{b [k_A + k_N]} \\ &= \frac{a [k_A + k_N] - \bar{p} [k_A + k_N] - b [c_N - c_A] - k_N [a - \bar{p}]}{b [k_A + k_N]} \end{aligned}$$

$$= \frac{a k_A - \bar{p} k_A - b [c_N - c_A]}{b [k_A + k_N]} = \frac{[a - \bar{p}] k_A - b [c_N - c_A]}{b [k_A + k_N]}. \quad (330)$$

It remains to prove that the inequalities in (327) hold. (325) implies:

$$\begin{aligned} \frac{\partial^+ \Pi^R}{\partial Q^R} \leq 0 &\Leftrightarrow -b q_A - b [Q^R - q_A] + a - b Q^R - c_N - k_N [Q^R - q_A] - k^R Q^R \leq 0 \\ &\Leftrightarrow a - 2b Q^R - c_N - k_N Q^R + k_N q_A - k^R Q^R \leq 0 \\ &\Leftrightarrow a - b Q^R - b Q^R - c_N - k_N [Q^R - q_A] - k^R Q^R \leq 0 \\ &\Leftrightarrow \bar{p} - [b + k^R] Q^R - c_N - k_N q_N \leq 0 \\ &\Leftrightarrow \bar{p} \leq [b + k^R] Q^R + c_N + k_N q_N. \end{aligned} \quad (331)$$

(330) and (331) imply:

$$\begin{aligned} \frac{\partial^+ \Pi^R}{\partial Q^R} \leq 0 &\Leftrightarrow \bar{p} \leq [b + k^R] \frac{a - \bar{p}}{b} + c_N + k_N \frac{[a - \bar{p}] k_A - b [c_N - c_A]}{b [k_A + k_N]} \\ &\Leftrightarrow \bar{p} b [k_A + k_N] \leq [b + k^R] [a - \bar{p}] [k_A + k_N] + c_N b [k_A + k_N] \\ &\quad + k_N [(a - \bar{p}) k_A - b (c_N - c_A)] \\ &\Leftrightarrow \bar{p} b [k_A + k_N] \leq [a - \bar{p}] [(b + k^R) (k_A + k_N) + k_N k_A] \\ &\quad + c_N b [k_A + k_N] - k_N b [c_N - c_A] \\ &\Leftrightarrow \bar{p} [b (k_A + k_N) + (b + k^R) (k_A + k_N) + k_N k_A] \\ &\quad \leq a [(b + k^R) (k_A + k_N) + k_N k_A] + c_N b [k_A + k_N] - k_N b [c_N - c_A] \\ &\Leftrightarrow \bar{p} [(2b + k^R) (k_A + k_N) + k_N k_A] \\ &\quad \leq a [(b + k^R) (k_A + k_N) + k_N k_A] + b c_N [k_A + k_N - k_N] + b k_N c_A \\ &\Leftrightarrow \bar{p} \leq \frac{a [(b + k^R) (k_A + k_N) + k_N k_A] + b c_N k_A + b c_A k_N}{[2b + k^R] [k_A + k_N] + k_N k_A} = \bar{p}_{bM}. \end{aligned} \quad (332)$$

(325) implies:

$$\begin{aligned} \frac{\partial^- \Pi^R}{\partial Q^R} \geq 0 &\Leftrightarrow a - 2b Q^R - c_N + b q_A - k_N [Q^R - q_A] - k^R Q^R \geq 0 \\ &\Leftrightarrow a - 2b Q^R - c_N + b q_A - k_N q_N - k^R Q^R \geq 0 \\ &\Leftrightarrow \bar{p} - b Q^R - c_N + b q_A - k_N q_N - k^R Q^R \geq 0 \\ &\Leftrightarrow \bar{p} \leq [b + k^R] Q^R + c_N - b q_A + k_N q_N \geq 0 \end{aligned}$$

$$\Leftrightarrow \bar{p} - [b + k^R] Q^R - c_N + b q_A - k_N q_N \geq 0. \quad (333)$$

(329), (330), and (333) imply:

$$\begin{aligned} \frac{\partial \Pi^R}{\partial Q^R} \geq 0 &\Leftrightarrow \bar{p} \geq [b + k^R] \frac{a - \bar{p}}{b} + c_N - b \frac{b[c_N - c_A] + k_N[a - \bar{p}]}{b[k_A + k_N]} \\ &\quad + k_N \frac{[a - \bar{p}]k_A - b[c_N - c_A]}{b[k_A + k_N]} \\ \Leftrightarrow \bar{p} b[k_A + k_N] &\geq [b + k^R][a - \bar{p}][k_A + k_N] + c_N b[k_A + k_N] \\ &\quad - b[b(c_N - c_A) + k_N(a - \bar{p})] + k_N[(a - \bar{p})k_A - b(c_N - c_A)] \\ \Leftrightarrow \bar{p} b[k_A + k_N] &\geq [a - \bar{p}][(b + k^R)(k_A + k_N) - b k_N + k_N k_A] + c_N b[k_A + k_N] \\ &\quad - b^2[c_N - c_A] - k_N b[c_N - c_A] \\ \Leftrightarrow \bar{p} [b(k_A + k_N) + (b + k^R)(k_A + k_N) - b k_N + k_N k_A] \\ &\geq a[(b + k^R)(k_A + k_N) - b k_N + k_N k_A] + c_N b[k_A + k_N] \\ &\quad - b^2[c_N - c_A] - k_N b[c_N - c_A] \\ \Leftrightarrow \bar{p} [(2b + k^R)(k_A + k_N) - b k_N + k_N k_A] \\ &\geq a[(b + k^R)(k_A + k_N) - b k_N + k_N k_A] + c_N [b(k_A + k_N) - b^2 - k_N b] \\ &\quad + [b^2 + b k_N] c_A \\ \Leftrightarrow \bar{p} &\geq \frac{a[(b + k^R)(k_A + k_N) - b k_N + k_N k_A] + b c_N [k_A + k_N - b - k_N] + [b + k_N] b c_A}{[2b + k^R][k_A + k_N] - b k_N + k_N k_A} \\ \Leftrightarrow \bar{p} &\geq \frac{a[(b + k^R)(k_A + k_N) - b k_N + k_N k_A] + b c_N [k_A - b] + b c_A [b + k_N]}{[2b + k^R][k_A + k_N] - b k_N + k_N k_A} \\ \Leftrightarrow \bar{p} &\geq \bar{p}_{dM}. \quad \blacksquare \end{aligned}$$

Lemma 6. *If $\bar{p} \in (\bar{p}_{0M}, \bar{p}_{dM}]$, then in equilibrium:*

$$\begin{aligned} q_A &= \frac{[2b + k_N + k^R][\bar{p} - c_A] - [b + k^R][a - c_N]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2}, \\ q_N &= \frac{[k_A + k^R][a - c_N] - [b + k^R][\bar{p} - c_A]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2}, \text{ and} \end{aligned}$$

$$Q^R \equiv q_A + q_N = \frac{[k_A - b][a - c_N] + [b + k_N][\bar{p} - c_A]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2}.$$

Proof. (308) implies that if $q_A > 0$ and $\bar{p} < P(Q)$, then [M] is:

$$\begin{aligned} \text{Maximize}_{q_A, q_N} \quad & \bar{p} q_A + [a - b(q_A + q_N)] q_N - c_A q_A - \frac{k_A}{2} [q_A]^2 \\ & - c_N q_N - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_A + q_N]^2. \end{aligned}$$

The necessary conditions for an interior solution to [M] when $\bar{p} < P(Q)$ are:⁴

$$q_A : \quad \bar{p} - b q_N - c_A - k_A q_A - k^R [q_A + q_N] = 0; \quad (334)$$

$$q_N : \quad a - b [q_A + q_N] - b q_N - c_N - k_N q_N - k^R [q_A + q_N] = 0. \quad (335)$$

(334) implies:

$$\bar{p} - b q_N - c_A - k^R q_N = [k_A + k^R] q_A \Rightarrow q_A = \frac{\bar{p} - c_A}{k_A + k^R} - \left[\frac{b + k^R}{k_A + k^R} \right] q_N. \quad (336)$$

(335) implies:

$$\begin{aligned} a - b q_A - c_N - k^R q_A &= [2b + k_N + k^R] q_N \\ \Rightarrow q_N &= \frac{a - c_N}{2b + k_N + k^R} - \left[\frac{b + k^R}{2b + k_N + k^R} \right] q_A. \end{aligned} \quad (337)$$

(334) also implies:

$$\bar{p} - c_A - k_A q_A - k^R q_A = [b + k^R] q_N \Rightarrow q_N = \frac{\bar{p} - c_A}{b + k^R} - \left[\frac{k_A + k^R}{b + k^R} \right] q_A. \quad (338)$$

(337) and (338) imply:

$$\begin{aligned} \frac{a - c_N}{2b + k_N + k^R} - \left[\frac{b + k^R}{2b + k_N + k^R} \right] q_A &= \frac{\bar{p} - c_A}{b + k^R} - \left[\frac{k_A + k^R}{b + k^R} \right] q_A \\ \Rightarrow \left[\frac{b + k^R}{2b + k_N + k^R} - \frac{k_A + k^R}{b + k^R} \right] q_A &= \frac{a - c_N}{2b + k_N + k^R} - \frac{\bar{p} - c_A}{b + k^R} \\ \Rightarrow \left\{ [b + k^R]^2 - [k_A + k^R][2b + k_N + k^R] \right\} q_A \\ &= [b + k^R][a - c_N] - [2b + k_N + k^R][\bar{p} - c_A] \\ \Rightarrow q_A &= \frac{[b + k^R][a - c_N] - [2b + k_N + k^R][\bar{p} - c_A]}{[b + k^R]^2 - [k_A + k^R][2b + k_N + k^R]} \end{aligned}$$

⁴It is readily verified that the Hessian associated with this problem is $[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2 > 0$. This inequality holds, by assumption.

$$\Rightarrow q_A = \frac{[2b + k_N + k^R][\bar{p} - c_A] - [b + k^R][a - c_N]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2}. \quad (339)$$

(338) and (339) imply:

$$\begin{aligned} q_N &= \frac{\bar{p} - c_A}{b + k^R} - \left[\frac{k_A + k^R}{b + k^R} \right] \left[\frac{(2b + k_N + k^R)(\bar{p} - c_A) - (b + k^R)(a - c_N)}{(k_A + k^R)(2b + k_N + k^R) - (b + k^R)^2} \right] \\ &= \frac{[\bar{p} - c_A] \left[(k_A + k^R)(2b + k_N + k^R) - (b + k^R)^2 \right]}{[b + k^R] \left[(k_A + k^R)(2b + k_N + k^R) - (b + k^R)^2 \right]} \\ &\quad - \left[\frac{k_A + k^R}{b + k^R} \right] \left[\frac{(2b + k_N + k^R)(\bar{p} - c_A) - (b + k^R)(a - c_N)}{(k_A + k^R)(2b + k_N + k^R) - (b + k^R)^2} \right] \\ &= \frac{[k_A + k^R][b + k^R][a - c_N] - [b + k^R]^2[\bar{p} - c_A]}{[b + k^R] \left[(k_A + k^R)(2b + k_N + k^R) - (b + k^R)^2 \right]} \\ &= \frac{[k_A + k^R][a - c_N] - [b + k^R][\bar{p} - c_A]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2}. \end{aligned} \quad (340)$$

(339) and (340) imply:

$$Q = Q^R = q_A + q_N = \frac{[k_A - b][a - c_N] + [b + k_N][\bar{p} - c_A]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2}. \quad (341)$$

It remains to show that $q_A > 0$ and $\bar{p} \leq P(Q)$ when $\bar{p} \in (\bar{p}_{0M}, \bar{p}_{dM}]$. (339) implies that $q_A > 0$ if:

$$\begin{aligned} &[2b + k_N + k^R][\bar{p} - c_A] - [b + k^R][a - c_N] > 0 \\ \Leftrightarrow \bar{p} &> c_A + \frac{[b + k^R][a - c_N]}{2b + k_N + k^R} = \bar{p}_{0M}. \end{aligned}$$

(341) implies that $P(Q^R) \geq \bar{p}$ if:

$$\begin{aligned} a - b \frac{[k_A - b][a - c_N] + [b + k_N][\bar{p} - c_A]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} &\geq \bar{p} \\ \Leftrightarrow a - b \frac{[k_A - b][a - c_N] - [b + k_N]c_A}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} \\ &\geq \bar{p} + \bar{p} \frac{b[b + k_N]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} \end{aligned}$$

$$\begin{aligned}
&\Leftrightarrow a + \frac{-b[k_A - b][a - c_N] + b[b + k_N]c_A}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} \\
&\quad \geq \bar{p} \frac{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2 + b[b + k_N]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} \\
&\Leftrightarrow a + \frac{-b[k_A - b]a + b[k_A - b]c_N + b[b + k_N]c_A}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} \\
&\quad \geq \bar{p} \frac{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2 + b[b + k_N]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} \\
&\Leftrightarrow \frac{a \left[(k_A + k^R)(2b + k_N + k^R) - (b + k^R)^2 \right] - b[k_A - b]a + b[k_A - b]c_N + b[b + k_N]c_A}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} \\
&\quad \geq \bar{p} \frac{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2 + b[b + k_N]}{[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2} \\
&\Leftrightarrow a \left[(k_A + k^R)(2b + k_N + k^R) - (b + k^R)^2 \right] - b[k_A - b]a + b[k_A - b]c_N + b[b + k_N]c_A \\
&\quad \geq \bar{p} \left\{ [k_A + k^R][2b + k_N + k^R] - [b + k^R]^2 + b[b + k_N] \right\} \\
&\Leftrightarrow a \left[(k_A + k^R)(2b + k_N + k^R) - (b + k^R)^2 - b(k_A - b) \right] + b[k_A - b]c_N + b[b + k_N]c_A \\
&\quad \geq \bar{p} \left\{ [k_A + k^R][2b + k_N + k^R] - [b + k^R]^2 + b[b + k_N] \right\}. \tag{342}
\end{aligned}$$

Observe that:

$$\begin{aligned}
&[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2 - b[k_A - b] \\
&= 2bk_A + k_A k_N + k_A k^R + 2bk^R + k^R k_N + [k^R]^2 - b^2 - 2bk^R - [k^R]^2 - bk_A + b^2 \\
&= bk_A + k_A k_N + k_A k^R + k^R k_N + bk_N - bk_N = [b + k^R][k_A + k_N] - bk_N + k_A k_N. \tag{343}
\end{aligned}$$

Further observe that:

$$\begin{aligned}
&[k_A + k^R][2b + k_N + k^R] - [b + k^R]^2 + b[b + k_N] \\
&= 2bk_A + k_A k_N + k_A k^R + 2bk^R + k^R k_N + [k^R]^2 - b^2 - 2bk^R - [k^R]^2 + b^2 + bk_N \\
&= 2bk_A + k_A k_N + k_A k^R + k^R k_N + bk_N \\
&= 2bk_A + k_A k_N + k_A k^R + k^R k_N - bk_N + 2bk_N \\
&= 2bk_A + 2bk_N + k_A k^R + k^R k_N - bk_N + k_A k_N
\end{aligned}$$

$$= [2b + k^R] [k_A + k_N] - b k_N + k_N k_A. \quad (344)$$

(342) – (344) imply that $P(Q^R) \geq \bar{p}$ if:

$$\begin{aligned} & a \{ [b + k^R] [k_A + k_N] - b k_N + k_A k_N \} + b [k_A - b] c_N + b [b + k_N] c_A \\ & \geq \bar{p} \{ [2b + k^R] [k_A + k_N] - b k_N + k_N k_A \} \\ \Leftrightarrow \bar{p} & \leq \frac{a [(b + k^R) (k_A + k_N) - b k_N + k_A k_N] + b [k_A - b] c_N + b [b + k_N] c_A}{[2b + k^R] [k_A + k_N] - b k_N + k_N k_A} \\ \Leftrightarrow \bar{p} & \leq \bar{p}_{dM}. \end{aligned}$$

Thus, $q_A > 0$ and $\bar{p} \leq P(Q)$ when $\bar{p} \in (\bar{p}_{0M}, \bar{p}_{dM}]$ ■

Lemma 7. *If $\bar{p} \leq \bar{p}_{0M}$, then in equilibrium:*

$$q_A = 0 \quad \text{and} \quad q_N = \frac{a - c_N}{2b + k_N + k^R}.$$

Proof. (308) implies that when $q_A = 0$, R 's problem is:

$$\text{Maximize}_{q_N \geq 0} [a - b q_N - c_N] q_N - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_N]^2. \quad (345)$$

(345) implies that R 's profit-maximizing choice of $q_N > 0$ is determined by:

$$a - 2b q_N - c_N - k_N q_N - k^R q_N = 0 \Rightarrow q_N = \frac{a - c_N}{2b + k_N + k^R}. \quad (346)$$

(346) implies:

$$\begin{aligned} P(Q) &= a - b \left[\frac{a - c_N}{2b + k_N + k^R} \right] = \frac{a [2b + k_N + k^R] - b [a - c_N]}{2b + k_N + k^R} \\ &= \frac{a [b + k_N + k^R] + b c_N}{2b + k_N + k^R}. \end{aligned} \quad (347)$$

(309) and (347) imply:

$$\begin{aligned} \bar{p}_{0M} < P(Q) &\Leftrightarrow c_A + \frac{[b + k^R] [a - c_N]}{2b + k_N + k^R} < \frac{a [2b + k_N + k^R] - b [a - c_N]}{2b + k_N + k^R} \\ \Leftrightarrow \frac{[b + k^R] [a - c_N]}{2b + k_N + k^R} &< \frac{[a - c_A] [2b + k_N + k^R] - b [a - c_N]}{2b + k_N + k^R} \\ \Leftrightarrow \frac{[2b + k^R] [a - c_N]}{2b + k_N + k^R} &< \frac{[a - c_A] [2b + k_N + k^R]}{2b + k_N + k^R} \\ \Leftrightarrow [2b + k^R] [a - c_N] &< [a - c_A] [2b + k_N + k^R]. \end{aligned}$$

The last inequality here holds because $c_A \leq c_N < a$ and $k_N > 0$. Therefore, $\bar{p} < P(Q)$ when $\bar{p} \leq \bar{p}_{0M}$.

It remains to show that $q_A = 0$ when $\bar{p} \leq \bar{p}_{0M}$. Because $\bar{p} < P(Q)$ when $\bar{p} \leq \bar{p}_{0M}$, $q_A = 0$ if R 's profit declines as q_A increases above 0, i.e., if:

$$\begin{aligned} & \frac{\partial}{\partial q_A} \left\{ [\bar{p} - c_A] q_A + [a - b(q_A + q_N) - c_N] q_N \right. \\ & \quad \left. - \frac{k_A}{2} [q_A]^2 - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_N + q_A]^2 \right\} \Big|_{q_A=0} \leq 0 \\ & \Leftrightarrow \bar{p} - c_A - b q_N - k^R q_N \leq 0 \Leftrightarrow \bar{p} \leq c_A + [b + k^R] q_N \\ & \Leftrightarrow \bar{p} \leq c_A + \frac{a - c_N}{2b + k_N + k^R} [b + k^R] = \bar{p}_{0M}. \end{aligned} \quad (348)$$

The last inequality in (348) reflects (314). ■

To compare the effects of a price cap on R 's revenue under monopoly and duopoly, suppose the maintained assumptions in both the monopoly and duopoly settings hold. Also let $Q_{dbM}^R(\bar{p})$ denote R 's total output when $\bar{p} \in (\bar{p}_{dM}, \bar{p}_{bM}]$. Then (318) implies:

$$Q_{dbM}^R(\bar{p}) = \frac{a - \bar{p}}{b} \Rightarrow \frac{\partial Q_{dbM}^R(\bar{p})}{\partial \bar{p}} = -\frac{1}{b}. \quad (349)$$

Let $Q_{db}^R(\bar{p})$ denote R 's total output when $\bar{p} \in (\bar{p}_d, \bar{p}_b]$ and R faces a rival. Recall from Lemma A3 that:

$$Q_{db}^R(\bar{p}) = \frac{a[b + k] + bc - \bar{p}[2b + k]}{b[b + k]} \Rightarrow \frac{\partial Q_{db}^R(\bar{p})}{\partial \bar{p}} = -\frac{2b + k}{b[b + k]}. \quad (350)$$

Conclusion 1. $\frac{\partial Q_{db}^R(\bar{p})}{\partial \bar{p}} < \frac{\partial Q_{dbM}^R(\bar{p}_M)}{\partial \bar{p}} < 0$ when $\bar{p}_M \in (\bar{p}_{dM}, \bar{p}_{bM}]$ and $\bar{p} \in (\bar{p}_d, \bar{p}_b]$.

Proof. (349) and (350) imply:

$$\frac{\partial Q_{dbM}^R(\bar{p})}{\partial \bar{p}} > \frac{\partial Q_{db}^R(\bar{p})}{\partial \bar{p}} \Leftrightarrow -\frac{1}{b} > -\frac{2b + k}{b[b + k]} \Leftrightarrow b + k < 2b + k \Leftrightarrow b > 0. \quad \blacksquare$$

Define $V_{dmM}(\bar{p})$ to be R 's revenue when $\bar{p} \in (\bar{p}_{dM}, \bar{p}_{bM}]$. Then (349) implies:

$$V_{dmM}(\bar{p}) = \bar{p} Q_{dbM}^R(\bar{p}) = \bar{p} \left[\frac{a - \bar{p}}{b} \right]. \quad (351)$$

(350) implies that when R faces a rival, R 's revenue when $\bar{p} \in (\bar{p}_d, \bar{p}_b]$ is:

$$V_{dm}(\bar{p}) = \bar{p} \left[\frac{a(b + k) + bc - \bar{p}(2b + k)}{b(b + k)} \right]. \quad (352)$$

Conclusion 2. $\frac{\partial V_{dm}(\bar{p})}{\partial \bar{p}} < \frac{\partial V_{dmM}(\bar{p})}{\partial \bar{p}}$ when $\bar{p} \in (\bar{p}_{dM}, \bar{p}_{bM}]$ and $\bar{p} \in (\bar{p}_d, \bar{p}_b]$.

Proof. (351) implies:

$$\frac{\partial V_{dbM}(\bar{p})}{\partial \bar{p}} = \frac{a - 2\bar{p}}{b}. \quad (353)$$

(352) implies:

$$\frac{\partial V_{db}(\bar{p})}{\partial \bar{p}} = \frac{a[b+k] + bc - 2\bar{p}[2b+k]}{b[b+k]}. \quad (354)$$

(353) and (354) imply:

$$\begin{aligned} \frac{\partial V_{db}(\bar{p})}{\partial \bar{p}} < \frac{\partial V_{dbM}(\bar{p})}{\partial \bar{p}} &\Leftrightarrow \frac{a[b+k] + bc - 2\bar{p}[2b+k]}{b[b+k]} < \frac{a - 2\bar{p}}{b} \\ &\Leftrightarrow a[b+k] + bc - 2\bar{p}[2b+k] < [b+k][a - 2\bar{p}] \\ &\Leftrightarrow bc + \bar{p}[2(b+k) - 2(2b+k)] < 0 \Leftrightarrow bc < 2b\bar{p} \Leftrightarrow \bar{p} > \frac{c}{2}. \end{aligned}$$

The last inequality here holds because, by assumption, $\bar{p} \geq \bar{p}_d > c$. ■

C. Welfare Analysis Involving Profit Rather than Revenue.

Now consider the duopoly setting where R 's profit replaces R 's revenue in the welfare function. Call this the $W - \Pi$ setting. Welfare in the $W - \Pi$ setting is:

$$W_{\Pi}(\bar{p}) \equiv S(\bar{p}) - r \Pi^R(\bar{p}) \quad (355)$$

where $r > 0$ is a parameter and $S(\cdot)$, which denotes consumer surplus, is:

$$S(\bar{p}) = aQ - \frac{b}{2} Q^2 - \bar{p} q_A - P(Q)[q_N + q]. \quad (356)$$

Definitions. $S_{0d}(\bar{p})$ is consumer surplus when $\bar{p} \in [\bar{p}_0, \bar{p}_d]$.

$S_{db}(\bar{p})$ is consumer surplus when $\bar{p} \in (\bar{p}_d, \bar{p}_b)$.

$\Pi_{0d}^R(\bar{p})$ is R 's profit when $\bar{p} \in [\bar{p}_0, \bar{p}_d]$.

$\Pi_{db}^R(\bar{p})$ is R 's profit when $\bar{p} \in (\bar{p}_d, \bar{p}_b)$. $\Pi_b^R(\bar{p}_b)$ is R 's profit when $\bar{p} = \bar{p}_b$.

$W_{\Pi 0d}(\bar{p})$ is welfare when $\bar{p} \in [\bar{p}_0, \bar{p}_d]$.

$W_{\Pi db}(\bar{p})$ is welfare when $\bar{p} \in (\bar{p}_d, \bar{p}_b)$.

Lemma 8. In the $W - \Pi$ setting:

$$\frac{\partial^2 \Pi_{db}^R(\bar{p})}{\partial (\bar{p})^2} < 0, \quad \frac{\partial^2 S_{db}(\bar{p})}{\partial (\bar{p})^2} > 0, \quad \text{and} \quad \frac{\partial^2 W_{\Pi db}(\bar{p})}{\partial (\bar{p})^2} > 0 \quad \text{for } \bar{p} \in (\bar{p}_d, \bar{p}_b). \quad (357)$$

Proof. Lemma A3 implies that when $\bar{p} \in (\bar{p}_d, \bar{p}_b)$, so $P(Q) = \bar{p}$:

$$Q = \frac{a - \bar{p}}{b} \Rightarrow \frac{\partial Q}{\partial \bar{p}} = -\frac{1}{b}. \quad (358)$$

(356) and (358) imply that for $\bar{p} \in (\bar{p}_d, \bar{p}_b)$, where $P(Q) = \bar{p}$:

$$\begin{aligned} \frac{\partial S_{db}(\bar{p})}{\partial \bar{p}} &= a \frac{\partial Q}{\partial \bar{p}} - b Q \frac{\partial Q}{\partial \bar{p}} - Q - \bar{p} \frac{\partial Q}{\partial \bar{p}} = -\frac{a}{b} + Q - Q + \frac{\bar{p}}{b} \\ &= -\frac{a - \bar{p}}{b} \Rightarrow \frac{\partial^2 S_{db}(\bar{p})}{\partial (\bar{p})^2} = \frac{1}{b} > 0. \end{aligned} \quad (359)$$

(110) implies that R 's revenue in the $W - \Pi$ setting when $\bar{p} \in (\bar{p}_d, \bar{p}_b)$ is:

$$\frac{\partial V_{dp}(\bar{p})}{\partial \bar{p}} = \frac{a[b + k] + bc - 2\bar{p}[2b + k]}{b[b + k]} \Rightarrow \frac{\partial^2 V_{dp}(\bar{p})}{\partial (\bar{p})^2} = -\frac{2[2b + k]}{b[b + k]} < 0. \quad (360)$$

Let $C_{db}^R(\bar{p})$ denote R 's total cost as a function of the price cap $\bar{p}$ when $\bar{p} \in (\bar{p}_d, \bar{p}_b)$. Then:

$$\frac{\partial C_{db}^R(\bar{p})}{\partial \bar{p}} = c_A \frac{\partial q_A}{\partial \bar{p}} + c_N \frac{\partial q_N}{\partial \bar{p}} + k_A q_A \frac{\partial q_A}{\partial \bar{p}} + k_N q_N \frac{\partial q_N}{\partial \bar{p}} + k^R [q_A + q_N] \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right].$$

Lemma A3 implies that $\frac{\partial q_N}{\partial \bar{p}}$ and $\frac{\partial q_A}{\partial \bar{p}}$ do not vary with $\bar{p}$ when $\bar{p} \in (\bar{p}_d, \bar{p}_b)$. Therefore:

$$\frac{\partial^2 C_{db}^R(\bar{p})}{\partial (\bar{p})^2} = k_A \left[\frac{\partial q_A}{\partial \bar{p}} \right]^2 + k_N \left[\frac{\partial q_N}{\partial \bar{p}} \right]^2 + k^R \left[\frac{\partial q_A}{\partial \bar{p}} + \frac{\partial q_N}{\partial \bar{p}} \right]^2 \geq 0. \quad (361)$$

(360) and (361) imply:

$$\frac{\partial^2 \Pi_{db}^R(\bar{p})}{\partial (\bar{p})^2} = \frac{\partial^2 V_{dp}(\bar{p})}{\partial (\bar{p})^2} - \frac{\partial^2 C_{db}^R(\bar{p})}{\partial (\bar{p})^2} < 0. \quad (362)$$

(355), (359), and (362) imply:

$$\frac{\partial^2 W_{\Pi db}(\bar{p})}{\partial (\bar{p})^2} = \frac{\partial^2 S_{db}(\bar{p})}{\partial (\bar{p})^2} - r \frac{\partial^2 \Pi_{db}^R(\bar{p})}{\partial (\bar{p})^2} > 0. \quad \blacksquare$$

Lemma 9. $\Pi_{0d}^R(\bar{p}_0) < \Pi_b^R(\bar{p}_b)$ in the $W - \Pi$ setting.

Proof. Lemma A1 implies that $Q^R(\bar{p}_0) = q_N(\bar{p}_0)$ because $q_A(\bar{p}_0) = 0$.

$$\text{Definition. } D_N \equiv [2b + k_N + k^R][2b + k] - b^2. \quad (363)$$

(363) and Lemmas A1 and A3 imply:

$$Q^R(\bar{p}_0) = \frac{[a - c_N][2b + k] - b[a - c]}{D_N} < \frac{[b + k][a - \bar{p}_b] - b[\bar{p}_b - c]}{b[b + k]} = Q^R(\bar{p}_b)$$

$$\begin{aligned}
&\Leftrightarrow \frac{a[b+k] + ab - c_N[2b+k] - ba + bc}{D_N} < \frac{[b+k]a - [b+k]\bar{p}_b - b\bar{p}_b + bc}{b[b+k]} \\
&\Leftrightarrow \frac{a[b+k] + bc - c_N[2b+k]}{D_N} < \frac{[b+k]a + bc - [2b+k]\bar{p}_b}{b[b+k]} \\
&\Leftrightarrow \frac{a[b+k] + bc - c_N[2b+k]}{D_N} b[b+k] - [b+k]a - bc < -[2b+k]\bar{p}_b \\
&\Leftrightarrow \frac{a[b+k] + bc}{2b+k} - \frac{a[b+k] + bc - c_N[2b+k]}{[2b+k]D_N} b[b+k] > \bar{p}_b \\
&\Leftrightarrow \frac{a[b+k] + bc}{2b+k} - \frac{[a(b+k) + bc]b[b+k] - c_N[2b+k]b[b+k]}{[2b+k]D_N} > \bar{p}_b \\
&\Leftrightarrow \frac{1}{[2b+k]D_N} \{ [a(b+k) + bc] [(2b+k_N + k^R)(2b+k) - b^2 - b(b+k)] \\
&\quad + c_N[2b+k]b[b+k] \} > \bar{p}_b \\
&\Leftrightarrow \frac{1}{[2b+k]D_N} \{ [a(b+k) + bc] [(2b+k_N + k^R)(2b+k) - b(2b+k)] \\
&\quad + c_N[2b+k]b[b+k] \} > \bar{p}_b \\
&\Leftrightarrow \frac{[a(b+k) + bc] [2b+k_N + k^R - b] + c_N b[b+k]}{D_N} > \bar{p}_b \\
&\Leftrightarrow \frac{[a(b+k) + bc] [b+k_N + k^R] + c_N b[b+k]}{D_N} > \bar{p}_b. \tag{364}
\end{aligned}$$

As established in the proof of Proposition 4, just below (149), $\bar{p}_b$ is increasing in k_A . Therefore, (8) implies that when $k_A \leq k_N$:

$$\begin{aligned}
\bar{p}_b &\leq \frac{[a(b+k) + bc] [2k_N(b+k^R) + (k_N)^2] + 2bc_N[b+k]k_N}{2b[b+k]k_N + (k_N)^2[2b+k] + 2k_N[2b+k][b+k^R]} \\
&= \frac{[a(b+k) + bc] [2(b+k^R) + k_N] + 2bc_N[b+k]}{2b[b+k] + k_N[2b+k] + 2[2b+k][b+k^R]} \\
&= \frac{[a(b+k) + bc] [b+k^R + \frac{k_N}{2}] + bc_N[b+k]}{b[b+k] + \frac{k_N}{2}[2b+k] + [2b+k][b+k^R]} \\
&= \frac{[a(b+k) + bc] [b+k^R + \frac{k_N}{2}] + bc_N[b+k]}{[2b+k][b+k^R + \frac{k_N}{2}] + b[b+k]} \\
&= \frac{[a(b+k) + bc] [b+k^R + \frac{k_N}{2}] + bc_N[b+k]}{[2b+k][2b+k^R + \frac{k_N}{2}] - b^2}. \tag{365}
\end{aligned}$$

The last equality in (365) holds because:

$$\begin{aligned} [2b+k] \left[b + k^R + \frac{k_N}{2} \right] + b[b+k] &= [2b+k] \left[2b + k^R + \frac{k_N}{2} \right] - b[2b+k] + b[b+k] \\ &= [2b+k] \left[2b + k^R + \frac{k_N}{2} \right] - 2b^2 - bk + b^2 + bk = [2b+k] \left[2b + k^R + \frac{k_N}{2} \right] - b^2. \end{aligned}$$

(363), (364), and (365) imply that $Q^R(\bar{p}_0) < Q^R(\bar{p}_b)$ if:

$$\begin{aligned} &\frac{[a(b+k) + bc] \left[b + k^R + \frac{k_N}{2} \right] + bc_N [b+k]}{[2b+k] \left[2b + k^R + \frac{k_N}{2} \right] - b^2} \\ &< \frac{[a(b+k) + bc] \left[b + k^R + k_N \right] + bc_N [b+k]}{[2b+k] \left[2b + k^R + k_N \right] - b^2}. \end{aligned} \quad (366)$$

The proof of Lemma A7 establishes that this inequality holds, so $Q^R(\bar{p}_0) < Q^R(\bar{p}_b)$.

(12) implies that the rival's output q is determined by:

$$a - b[Q^R(\bar{p}) + q(\bar{p})] - c - bq(\bar{p}) - kq(\bar{p}) = 0 \Rightarrow q(\bar{p}) = \frac{a - bQ^R(\bar{p}) - c}{2b+k}. \quad (367)$$

Because $Q^R(\bar{p}_0) < Q^R(\bar{p}_b)$, (367) implies that $q(\bar{p}_0) > q(\bar{p}_b)$. R 's profit when $\bar{p} = \bar{p}_0$ is:

$$\begin{aligned} \Pi_{0d}^R(\bar{p}_0) &= q_N(\bar{p}_0) [P(q_N(\bar{p}_0) + q(\bar{p}_0)) - c_N] - \frac{k_N}{2} [q_N(\bar{p}_0)]^2 - \frac{k^R}{2} [q_N(\bar{p}_0)]^2 \\ &< q_N(\bar{p}_0) [P(q_N(\bar{p}_0) + q(\bar{p}_b)) - c_N] - \frac{k_N}{2} [q_N(\bar{p}_0)]^2 - \frac{k^R}{2} [q_N(\bar{p}_0)]^2 \\ &\leq \max_{q_N \geq 0, q_A \geq 0} \{ q_N [P(q_N + q + q_A) - c_N] + q_A [P(q_N + q + q_A) - c_A] \\ &\quad - \frac{k_A}{2} [q_A]^2 - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_A + q_N]^2 \} = \Pi_b^R(\bar{p}_b). \quad \blacksquare \end{aligned}$$

Definition. $\bar{p}_\Pi^* \equiv \arg \max W_\Pi(\bar{p})$.

Conclusion 3. $\bar{p}_\Pi^* \in [\bar{p}_0, \bar{p}_d]$.

Proof. Lemma 8 implies that $\bar{p}_\Pi^* \notin (\bar{p}_d, \bar{p}_b)$. Equilibrium outcomes do not vary with $\bar{p}$ when $\bar{p} < \bar{p}_0$ (because $q_A = 0$ for all such $\bar{p}$). Similarly, equilibrium outcomes do not vary with $\bar{p}$ when $\bar{p} > \bar{p}_b$ (because the price cap does not bind for all such $\bar{p}$). Therefore: (i) welfare does not vary with $\bar{p}$ when $\bar{p} \leq \bar{p}_0$; and (ii) welfare does not vary with $\bar{p}$ when $\bar{p} \geq \bar{p}_b$. Consequently, $\bar{p}_\Pi^* \in [\bar{p}_0, \bar{p}_d] \cup \bar{p}_b$.

It remains to show that $\bar{p}_\Pi^* \neq \bar{p}_b$. The proof of Lemma 9 establishes that:

$$Q^R(\bar{p}_0) < Q^R(\bar{p}_b) \quad (368)$$

where $Q^R(\bar{p})$ is R 's total output when the price cap is $\bar{p}$. Proposition 2 implies:

$$Q^R(\bar{p}_b) < Q^R(\bar{p}_d). \quad (369)$$

(368) and (369) imply that $Q^R(\bar{p}_0) < Q^R(\bar{p}_b) < Q^R(\bar{p}_d)$. Lemma A2 implies that $Q^R(\bar{p})$ is continuous and monotonically increasing in $\bar{p}$ for $\bar{p} \in (\bar{p}_0, \bar{p}_d)$. Therefore, the Intermediate Value Theorem implies that there exists a $\bar{p}_E \in (\bar{p}_0, \bar{p}_d)$ such that:

$$Q^R(\bar{p}_E) = Q^R(\bar{p}_b). \quad (370)$$

(12) implies that the rival's output q is determined by:

$$a - b [Q^R(\bar{p}) + q(\bar{p})] - c - b q(\bar{p}) - k q(\bar{p}) = 0. \quad (371)$$

(370) and (371) imply:

$$q(\bar{p}_E) = q(\bar{p}_b). \quad (372)$$

(370) and (372) imply:

$$Q(\bar{p}_E) = Q(\bar{p}_b) \text{ and } P(Q(\bar{p}_E)) = P(Q(\bar{p}_b)). \quad (373)$$

Observe that:

$$\begin{aligned} \Pi_{0d}^R(\bar{p}_E) &= \bar{p}_E q_A(\bar{p}_E) + P(Q(\bar{p}_E)) q_N(\bar{p}_E) - C^R(q_A(\bar{p}_E), q_N(\bar{p}_E)) \\ &< P(Q(\bar{p}_E)) q_A(\bar{p}_E) + P(Q(\bar{p}_E)) q_N(\bar{p}_E) - C^R(q_A(\bar{p}_E), q_N(\bar{p}_E)) \\ &= P(Q(\bar{p}_b)) q_A(\bar{p}_b) + P(Q(\bar{p}_b)) q_N(\bar{p}_b) - C^R(q_A(\bar{p}_b), q_N(\bar{p}_b)) = \Pi_b^R(\bar{p}_b). \end{aligned} \quad (374)$$

The inequality in (374) holds because $\bar{p}_E < P(Q(\bar{p}_E))$, since $\bar{p}_E \in (\bar{p}_0, \bar{p}_d)$. The penultimate equality in (374) reflects (373). The last equality in (374) holds because $P(Q(\bar{p}_b)) = \bar{p}_b$.

(356) and (373) imply:

$$\begin{aligned} S_{0d}(\bar{p}_E) &= a Q(\bar{p}_E) - \frac{b}{2} Q(\bar{p}_E)^2 - P(Q(\bar{p}_E)) [q(\bar{p}_E) + q_N(\bar{p}_E)] - \bar{p}_E q_A(\bar{p}_E) \\ &> a Q(\bar{p}_E) - \frac{b}{2} Q(\bar{p}_E)^2 - P(Q(\bar{p}_E)) [q(\bar{p}_E) + q_N(\bar{p}_E) + q_A(\bar{p}_E)] \\ &= a Q(\bar{p}_E) - \frac{b}{2} Q(\bar{p}_b)^2 - P(Q(\bar{p}_E)) Q(\bar{p}_E) \\ &= a Q(\bar{p}_E) - \frac{b}{2} Q(\bar{p}_b)^2 - P(Q(\bar{p}_b)) Q(\bar{p}_b) = S(\bar{p}_b). \end{aligned} \quad (375)$$

The inequality in (375) holds because $\bar{p}_E < P(Q(\bar{p}_E))$, since $\bar{p}_E \in (\bar{p}_0, \bar{p}_d)$. (374) and (375) imply that consumer surplus is higher and R 's profit is lower in the $W - \Pi$ setting when $\bar{p} = \bar{p}_E$ than when $\bar{p} = \bar{p}_b$. Therefore, welfare is strictly greater when $\bar{p} = \bar{p}_E$ than when $\bar{p} = \bar{p}_b$, so $\bar{p}_{\Pi}^* \neq \bar{p}_b$. ■

D. Benchmark Setting with Exogenous Prices.

Finally, consider the benchmark *exogenous price setting* in which the price of output supplied using A 's input is set (exogenously) at $\bar{p}_A$ and the price of output supplied without

using A 's input is set (exogenously) at $\bar{p}_N$.

Conclusion 4. R reduces both q_A and $Q^R = q_A + q_N$ as $\bar{p}_A$ declines in the exogenous price setting.

Proof. In the exogenous price setting, R chooses q_A and q_N to:

$$\text{Maximize } \bar{p}_A q_A + \bar{p}_N q_N - c_A q_A - \frac{k_A}{2} [q_A]^2 - c_N q_N - \frac{k_N}{2} [q_N]^2 - \frac{k^R}{2} [q_A + q_N]^2.$$

The necessary conditions for a solution to this problem, [P-E], are:

$$\begin{aligned} \bar{p}_A - c_A - k_A q_A - k^R [q_A + q_N] &\leq 0 & q_A [\cdot] &= 0; \\ \bar{p}_N - c_N - k_N q_N - k^R [q_A + q_N] &\leq 0 & q_N [\cdot] &= 0. \end{aligned} \quad (376)$$

(376) implies that if $q_A = 0$ and $q_N > 0$ at the solution to [P-E]:

$$\begin{aligned} \bar{p}_N - c_N - k_N q_N - k^R q_N &= 0 \Rightarrow q_N = \frac{\bar{p}_N - c_N}{k_N + k^R} \\ \Rightarrow \frac{\partial q_N}{\partial \bar{p}_N} &= \frac{1}{k_N + k^R} > 0 \quad \text{and} \quad \frac{\partial q_N}{\partial \bar{p}_A} = 0. \end{aligned}$$

(376) also implies that if $q_N = 0$ and $q_A > 0$ at the solution to [P-E]:

$$\begin{aligned} \bar{p}_A - c_A - k_A q_A - k^R q_A &= 0 \Rightarrow q_A = \frac{\bar{p}_A - c_A}{k_A + k^R} \\ \Rightarrow \frac{\partial q_A}{\partial \bar{p}_A} &= \frac{1}{k_A + k^R} > 0 \quad \text{and} \quad \frac{\partial q_A}{\partial \bar{p}_N} = 0. \end{aligned}$$

(376) further implies that if $q_A > 0$ and $q_N > 0$ at the solution to [P-E]:

$$\begin{aligned} \bar{p}_A - c_A - k_A q_A &= k^R [q_A + q_N] \Rightarrow q_A [k_A + k^R] = \bar{p}_A - c_A - k^R q_N \\ \Rightarrow q_A &= \frac{\bar{p}_A - c_A - k^R q_N}{k_A + k^R}; \text{ and} \end{aligned} \quad (377)$$

$$\begin{aligned} \bar{p}_N - c_N - k_N q_N &= k^R [q_A + q_N] \Rightarrow q_N [k_N + k^R] = \bar{p}_N - c_N - k^R q_A \\ \Rightarrow q_N &= \frac{\bar{p}_N - c_N - k^R q_A}{k_N + k^R}. \end{aligned} \quad (378)$$

(377) and (378) imply:

$$q_N = \frac{\bar{p}_N - c_N}{k_N + k^R} - \frac{k^R}{k_N + k^R} \left[\frac{\bar{p}_A - c_A - k^R q_N}{k_A + k^R} \right]$$

$$\begin{aligned}
\Rightarrow q_N & \left[1 - \frac{(k^R)^2}{(k_A + k^R)(k_N + k^R)} \right] = \frac{\bar{p}_N - c_N}{k_N + k^R} - \frac{k^R [\bar{p}_A - c_A]}{[k_A + k^R][k_N + k^R]} \\
\Rightarrow q_N & \left[(k_A + k^R)(k_N + k^R) - (k^R)^2 \right] = [\bar{p}_N - c_N][k_A + k^R] - k^R[\bar{p}_A - c_A] \\
\Rightarrow q_N & = \frac{[\bar{p}_N - c_N][k_A + k^R] - k^R[\bar{p}_A - c_A]}{[k_A + k^R][k_N + k^R] - (k^R)^2}. \tag{379}
\end{aligned}$$

(377) and (379) imply:

$$\begin{aligned}
q_A & = \frac{\bar{p}_A - c_A}{k_A + k^R} - \frac{k^R}{k_A + k^R} \left\{ \frac{[\bar{p}_N - c_N][k_A + k^R] - k^R[\bar{p}_A - c_A]}{[k_A + k^R][k_N + k^R] - (k^R)^2} \right\} \\
& = \frac{1}{[k_A + k^R] \{ [k_A + k^R][k_N + k^R] - (k^R)^2 \}} \\
& \quad \cdot \left\{ [\bar{p}_A - c_A] \left\{ [k_A + k^R][k_N + k^R] - (k^R)^2 \right\} - k^R[\bar{p}_N - c_N][k_A + k^R] \right. \\
& \quad \left. + (k^R)^2[\bar{p}_A - c_A] \right\} \\
& = \frac{[\bar{p}_A - c_A][k_A + k^R][k_N + k^R] - k^R[\bar{p}_N - c_N][k_A + k^R]}{[k_A + k^R] \{ [k_A + k^R][k_N + k^R] - (k^R)^2 \}} \\
& = \frac{[\bar{p}_A - c_A][k_N + k^R] - k^R[\bar{p}_N - c_N]}{[k_A + k^R][k_N + k^R] - (k^R)^2}. \tag{380}
\end{aligned}$$

(379) and (380) imply:

$$\begin{aligned}
\frac{\partial q_A}{\partial \bar{p}_A} & = \frac{k_N + k^R}{[k_A + k^R][k_N + k^R] - (k^R)^2} > 0, \text{ and} \\
\frac{\partial (q_A + q_N)}{\partial \bar{p}_A} & = \frac{k_N + k^R - k^R}{[k_A + k^R][k_N + k^R] - (k^R)^2} \\
& = \frac{k_N}{[k_A + k^R][k_N + k^R] - (k^R)^2} > 0. \blacksquare \tag{381}
\end{aligned}$$

E. The *Simplified Cost Setting*.

Definitions.

$$\bar{p}_2 \equiv \frac{b c_A [b + k] + a [b + k] k^R + b c k^R}{[2b + k][b + k^R] - b^2}.$$

$$\bar{p}_3 \equiv \frac{a [b + k^R] [b + k] + b c_A [b + k] + b c [b + k^R]}{[2b + k^R] [2b + k] - b^2}. \quad (382)$$

Assumptions

1. $C^R(q_A, q_N) = c_A q_A + c_N q_N + \frac{k^R}{2} [q_A + q_N]^2$, where $k^R > 0$.
2. $c_N > c_A > 0$.
3. $a > \max\{c, c_N\}$.
4. $[a - c_A] [2b + k] > b [a - c]$.
5. $\bar{p}_2 > c$.
6. All other model features are as specified in the main analysis.

Lemma 10. $\bar{p}_3 > \bar{p}_2$.

Proof. (382) implies that the lemma holds if:

$$\begin{aligned}
& \{ a [b + k] [b + k^R] + b c_A [b + k] + b c [b + k^R] \} \{ [2b + k] [b + k^R] - b^2 \} \\
& > \{ a [b + k] k^R + b c_A [b + k] + b c k^R \} \{ [2b + k^R] [2b + k] - b^2 \} \\
\Leftrightarrow & a [b + k] [2b + k] [b + k^R]^2 - a b^2 [b + k] [b + k^R] \\
& + b c_A [b + k] [2b + k] [b + k^R] - b^3 c_A [b + k] \\
& + b c [2b + k] [b + k^R]^2 - b^3 c [b + k^R] \\
& > a [b + k] [2b + k] k^R [2b + k^R] - a b^2 [b + k] k^R \\
& + b c_A [b + k] [2b + k] [2b + k^R] - b^3 c_A [b + k] \\
& + b c [2b + k] k^R [2b + k^R] - b^3 c k^R \\
\Leftrightarrow & a [b + k] [2b + k] \left[b^2 + 2b k^R + (k^R)^2 - 2b k^R - (k^R)^2 \right] - a b^3 [b + k] \\
& - b^2 c_A [b + k] [2b + k] \\
& + b c [2b + k] \left[b^2 + 2b k^R + (k^R)^2 - 2b k^R - (k^R)^2 \right] - b^4 c > 0 \\
\Leftrightarrow & a b^2 [b + k] [2b + k] - a b^3 [b + k] - b^2 c_A [b + k] [2b + k] \\
& + b^3 c [2b + k] - b^4 c > 0
\end{aligned}$$

$$\begin{aligned}
&\Leftrightarrow ab^2[b+k][2b+k-b] + b^2c_A[2b+k][b-(b+k)] - b^4c > 0 \\
&\Leftrightarrow ab^2[b+k]^2 - b^2kc_A[2b+k] - b^4c > 0 \\
&\Leftrightarrow ab^2[b^2+2bk+k^2] - 2b^3kc_A - b^2k^2c_A - b^4c > 0 \\
&\Leftrightarrow ab^4 + 2ab^3k + ab^2k^2 - 2b^3kc_A - b^2k^2c_A - b^4c > 0 \\
&\Leftrightarrow [a-c]b^4 + 2b^3k[a-c_A] + b^2k^2[a-c_A] > 0.
\end{aligned}$$

The last inequality here holds because $a > c$ and $a > c_A$, by assumption. ■

Conclusion 5. $q_A = 0$ or $q_A = 0$ in equilibrium in the simplified cost setting.

Proof. Let $\Pi^R(q_A, q_N)$ denote R 's profit when it supplies q_A units using A 's input and q_N units without using this input. We will show in three distinct cases that if R is initially supplying strictly positive amounts of output both using A 's input and not using A 's input (so $q_A > 0$ and $q_N > 0$), then R can strictly increase its profit by setting $q_A = 0$ or $q_N = 0$.

Case 1. $P_A(Q) - c_A > P(Q) - c_N$. If $q_A > 0$ and $q_N > 0$ in this case, then:

$$\Pi^R(q_A, q_N) = [P_A(Q) - c_A]q_A + [P(Q) - c_N]q_N - \frac{k^R}{2}[q_A + q_N]^2 \quad (383)$$

$$< [P_A(Q) - c_A][q_A + q_N] - \frac{k^R}{2}[q_A + q_N]^2 = \Pi^R(q_A + q_N, 0). \quad (384)$$

(384) implies that R could increase its profit by selling its entire output, $Q^R = q_A + q_N$, using A 's input, and supplying no output without using the input.

Case 2. $P_A(Q) - c_A < P(Q) - c_N$. If $q_A > 0$ and $q_N > 0$ in this case, then:

$$\begin{aligned}
\Pi^R(q_A, q_N) &= [P_A(Q) - c_A]q_A + [P(Q) - c_N]q_N - \frac{k^R}{2}[q_A + q_N]^2 \\
&< [P(Q) - c_N][q_A + q_N] - \frac{k^R}{2}[q_A + q_N]^2 = \Pi^R(0, q_A + q_N).
\end{aligned} \quad (385)$$

(385) implies that R could increase its profit by selling its entire output, $Q^R = q_A + q_N$, without using A 's input, and supplying no output using the input.

Case 3. $P_A(Q) - c_A = P(Q) - c_N$. In this case:

$$P(Q) - P_A(Q) = c_N - c_A > 0 \Rightarrow P_A(Q) < P(Q) \Rightarrow P_A(Q) = \bar{p}. \quad (386)$$

(386) implies that if $q_A > 0$ and $q_N > 0$ in this case, then:

$$\begin{aligned}
\frac{\partial \Pi^R(q_A, q_N)}{\partial q_A} &= -bq_N + \bar{p} - c_A - k^R[q_A + q_N] = 0 \\
&\Rightarrow \bar{p} - c_A - k^R[q_A + q_N] = bq_N > 0.
\end{aligned} \quad (387)$$

(386) implies that in this case:

$$\begin{aligned}
\Pi^R(q_A, q_N) &= [P_A(Q) - c_A] q_A + [P(Q) - c_N] q_N - \frac{k^R}{2} [q_A + q_N]^2 \\
&= [P_A(Q) - c_A] [q_A + q_N] - \frac{k^R}{2} [q_A + q_N]^2 \\
&= [\bar{p} - c_A] [q_A + q_N] - \frac{k^R}{2} [q_A + q_N]^2 = \Pi^R(q_A + q_N, 0). \tag{388}
\end{aligned}$$

(388) implies that R would secure the same profit if it sold its entire output, $Q^R = q_A + q_N$, using A 's input. (388) also implies:

$$\frac{\partial \Pi^R(q_A + q_N, 0)}{\partial q_A} = \bar{p} - c_A - k^R [q_A + q_N] > 0. \tag{389}$$

The inequality in (389) reflects (387). (389) implies that R could increase its profit by marginally increasing its total output and selling the entire output using A 's input. ■

Lemma 11. *Suppose $\bar{p} \geq \bar{p}_3$. Then in equilibrium in the simplified cost setting:*

$$\begin{aligned}
\bar{p} \geq P(Q), \quad q^* &= \frac{[a - c] [2b + k^R] - b[a - c_A]}{[2b + k^R] [2b + k] - b^2}, \quad q_N^* = 0, \text{ and} \\
q_A^* &= \frac{[a - c_A] [2b + k] - b[a - c]}{[2b + k^R] [2b + k] - b^2}. \tag{390}
\end{aligned}$$

Proof. Consider a putative equilibrium in which outputs are as specified in (390). Then in this equilibrium, for $Q^* = q^* + q_A^* + q_N^*$:

$$\begin{aligned}
P(Q^*) &= a - b[q_A^* + q_N^* + q^*] = a - b q_A^* - b q^* \\
&= a - b \left[\frac{(a - c_A)(2b + k) - b(a - c)}{(2b + k^R)(2b + k) - b^2} \right] - b \left[\frac{(a - c)(2b + k^R) - b(a - c_A)}{(2b + k^R)(2b + k) - b^2} \right] \\
&= a - b \left[\frac{(a - c_A)(2b + k) - b(a - c) + (a - c)(2b + k^R) - b(a - c_A)}{(2b + k^R)(2b + k) - b^2} \right] \\
&= a - b \left[\frac{(a - c_A)(b + k) + (a - c)(b + k^R)}{(2b + k^R)(2b + k) - b^2} \right] \\
&= \frac{a[(2b + k^R)(2b + k) - b^2] - b[a - c_A][b + k] - b[a - c][b + k^R]}{[2b + k^R][2b + k] - b^2} \\
&= \frac{a[(2b + k^R)(2b + k) - b^2 - b(b + k) - b(b + k^R)] + b c_A [b + k] + b c [b + k^R]}{[2b + k^R][2b + k] - b^2}
\end{aligned}$$

$$\begin{aligned}
&= \frac{a \left[(2b + k^R)(2b + k) - b(2b + k) - b(b + k^R) \right] + b c_A [b + k] + b c [b + k^R]}{[2b + k^R][2b + k] - b^2} \\
&= \frac{a \left[(b + k^R)(2b + k) - b(b + k^R) \right] + b c_A [b + k] + b c [b + k^R]}{[2b + k^R][2b + k] - b^2} \\
&= \frac{a [b + k^R] [b + k] + b c_A [b + k] + b c [b + k^R]}{[2b + k^R][2b + k] - b^2} = \bar{p}_3.
\end{aligned} \tag{391}$$

The last inequality in (391) reflects (382). (391) implies that $\bar{p} \geq P(Q^*)$ in the putative equilibrium when $\bar{p} \geq \bar{p}_3$.

We now establish that R cannot increase its profit by deviating unilaterally to supply q_A and q_N that differ from their values in (390). (383) and (391) imply that R 's profit in the simplified cost setting when $\bar{p} \geq \bar{p}_3$ is:

$$\begin{aligned}
\Pi^R(q_A, q_N) &= [P_A(Q) - c_A] q_A + [P(Q) - c_N] q_N - \frac{k^R}{2} [q_A + q_N]^2 \\
&= [a - bQ - c_A] q_A + [a - bQ - c_N] q_N - \frac{k^R}{2} [q_A + q_N]^2 \\
\Rightarrow \frac{\partial \Pi^R(q_A, q_N)}{\partial q_A} &= a - b[q + q_A + q_N] - c_A - b q_A - b q_N - k^R [q_A + q_N] \\
&= a - b q - 2b q_A - 2b q_N - c_A - k^R [q_A + q_N]
\end{aligned} \tag{392}$$

$$\Rightarrow \frac{\partial^2 \Pi^R(q_A, q_N)}{\partial (q_A)^2} = -2b - k^R < 0. \tag{393}$$

(393) implies that $\Pi^R(q_A, q_N)$ is strictly concave in q_A .

(390) and (392) imply that in the putative equilibrium:

$$\begin{aligned}
\left. \frac{\partial \Pi^R(q_A, q_N)}{\partial q_A} \right|_{q_A = q_A^*, q_N = q_N^*} &= a - b q^* - 2b q_A^* - c_A - k^R q_A^* = 0 \\
\Leftrightarrow a - c_A - b q^* &= [2b + k^R] q_A^* \\
\Leftrightarrow [a - c_A] \left[(2b + k^R)(2b + k) - b^2 \right] - b \left[(a - c)(2b + k^R) - b(a - c_A) \right] \\
&= [2b + k^R] \left[(a - c_A)(2b + k) - b(a - c) \right] \\
\Leftrightarrow [a - c_A] \left[(2b + k^R)(2b + k) - b^2 - (2b + k^R)(2b + k) + b^2 \right] \\
&\quad - b[a - c] \left[2b + k^R - (2b + k^R) \right] = 0 \\
\Leftrightarrow [a - c_A] 0 - b[a - c] 0 &= 0.
\end{aligned} \tag{394}$$

The last equality in (394) holds. (393) and (394) imply:

$$[P(Q) - c_A] q_A - \frac{k^R}{2} [q_A]^2 \leq \Pi^R(q_A^*, 0) \text{ for all } q_A. \quad (395)$$

(395) implies that R cannot increase its profit by deviating from the putative equilibrium by producing distinct q_A and q_N because (383) implies:

$$\begin{aligned} \Pi^R(q_A, q_N) &= [P(Q) - c_N] q_N + [P_A(Q) - c_A] q_A - \frac{k^R}{2} [q_A + q_N]^2 \\ &\leq [P(Q) - c_N] q_N + [P(Q) - c_A] q_A - \frac{k^R}{2} [q_A + q_N]^2 \\ &< [P(Q) - c_A] [q_A + q_N] - \frac{k^R}{2} [q_A + q_N]^2 \leq \Pi^R(q_A^*, 0). \end{aligned} \quad (396)$$

The first inequality in (396) holds because $P_A(Q) \leq P(Q)$ for all Q . The second inequality in (396) holds because $c_A < c_N$, by assumption. The last inequality in (396) reflects (395). (396) implies that R cannot increase its profit by undertaking such a deviation.

Finally, we establish that the rival cannot increase its profit by unilaterally deviating to supply some output other than the q^* specified in (390). The rival's profit is:

$$\begin{aligned} \Pi(q) &= [P(Q) - c] q - \frac{k}{2} q^2 = [a - b(q + q_A + q_N) - c] q - \frac{k}{2} q^2 \\ \Rightarrow \frac{\partial \Pi(q)}{\partial q} &= a - bQ - c - bq - kq \end{aligned} \quad (397)$$

$$\Rightarrow \frac{\partial^2 \Pi(q)}{\partial q^2} = -2b - k < 0. \quad (398)$$

(390) and (397) imply that at the putative equilibrium:

$$\begin{aligned} \left. \frac{\partial \Pi(q)}{\partial q} \right|_{q=q^*} &= a - bq_A^* - 2bq^* - c - kq^* = a - [2b + k]q^* - bq_A^* \\ &= a - [2b + k] \left[\frac{(a - c)(2b + k^R) - b(a - c_A)}{(2b + k^R)(2b + k) - b^2} \right] \\ &\quad - b \left[\frac{(a - c_A)(2b + k) - b(a - c)}{(2b + k^R)(2b + k) - b^2} \right] - c \\ &= [a - c] [(2b + k^R)(2b + k) - b^2] - [2b + k] [(a - c)(2b + k^R) - b(a - c_A)] \\ &\quad - b[(a - c_A)(2b + k) - b(a - c)] = 0 \\ &= [a - c] [(2b + k^R)(2b + k) - b^2 - (2b + k)(2b + k^R) + b^2] \\ &\quad - b[a - c_A][2b + k - (2b + k)] = 0 \end{aligned} \quad (399)$$

(398) and (399) imply that the rival cannot increase its profit by unilaterally deviating from the putative equilibrium. Therefore, the outputs in (390) constitute an equilibrium because neither firm can increase its profit by undertaking a unilateral deviation. ■

Lemma 12. *In the simplified cost setting, there exists a $\tilde{p} \in [\bar{p}_2, \bar{p}_3)$ such that in equilibrium, for all $\bar{p} \in (\tilde{p}, \bar{p}_3]$:*

$$P(Q^*) = \bar{p}, \quad q_A^* = \frac{a[b+k] + bc - \bar{p}[2b+k]}{b[b+k]}, \quad q_N^* = 0, \quad \text{and} \quad q^* = \frac{\bar{p} - c}{b+k}. \quad (400)$$

Proof. Suppose an equilibrium exists in which prices and outputs are as specified in (400). We will show that neither R nor the rival can increase its profit by deviating unilaterally from the putative equilibrium. We first consider three possible deviations by R .

Deviation 1. R deviates unilaterally to set $q_A \neq q_A^*$ and $q_N = 0$.

(383) implies:

$$\Pi^R(q_A, 0) = [P_A(Q) - c_A] q_A - \frac{k^R}{2} [q_A]^2 \quad (401)$$

$$\Rightarrow \frac{\partial \Pi^R(q_A, 0)}{\partial q_A} = P_A(Q) - c_A + q_A \frac{\partial P_A(Q)}{\partial q_A} - k^R q_A \quad (402)$$

$$\Rightarrow \frac{\partial^2 \Pi^R(q_A, 0)}{\partial (q_A)^2} = 2 \frac{\partial P_A(Q)}{\partial q_A} - k^R < 0. \quad (403)$$

The inequality in (403) holds because $\frac{\partial P_A(Q)}{\partial q_A} \in \{0, -b\}$. (403) implies that the identified deviation will not increase R 's profit if:

$$\frac{\partial^+ \Pi^R(q_A^*, 0)}{\partial q_A} \leq 0 \leq \frac{\partial^- \Pi^R(q_A^*, 0)}{\partial q_A}. \quad (404)$$

If (404) holds, then R 's profit declines as q_A increases above q_A^* or as q_A declines below q_A^* , given q^* and $q_N^* = 0$.

(400) and (402) imply that at the putative equilibrium:

$$\begin{aligned} \left. \frac{\partial^+ \Pi^R(q_A, 0)}{\partial q_A} \right|_{q_A=q_A^*} &= P_A(Q^*) - c_A + q_A^* \frac{\partial P_A(Q^*)}{\partial q_A} - k^R q_A^* \\ &= \bar{p} - c_A - [b + k^R] q_A^* = \bar{p} - c_A - [b + k^R] \left[\frac{a(b+k) + bc - \bar{p}(2b+k)}{b(b+k)} \right] \\ &= \frac{\bar{p}b[b+k] - bc_A[b+k] - a[b+k][b+k^R] - bc[b+k^R] + \bar{p}[2b+k][b+k^R]}{b[b+k]} \\ &= \frac{\bar{p}[b(b+k) + (2b+k)(b+k^R)] - bc_A[b+k] - a[b+k][b+k^R] - bc[b+k^R]}{b[b+k]} \leq 0 \end{aligned} \quad (405)$$

$$\begin{aligned}
\Leftrightarrow \bar{p} &\leq \frac{a[b+k][b+k^R] + bc_A[b+k] + bc[b+k^R]}{b[b+k] + [2b+k][b+k^R]} \\
\Leftrightarrow \bar{p} &\leq \frac{a[b+k][b+k^R] + bc_A[b+k] + bc[b+k^R]}{b[b+k] + [2b+k][2b+k^R] - b[2b+k]} \\
\Leftrightarrow \bar{p} &\leq \frac{a[b+k][b+k^R] + bc_A[b+k] + bc[b+k^R]}{[2b+k][2b+k^R] - b^2} = \bar{p}_3.
\end{aligned} \tag{406}$$

The first equality in (405) holds because $P_A(Q) = P(Q) = a - bQ$ as q_A increases above q_A^* (thereby reducing $P(Q)$ below $\bar{p}$). The last equality in (406) reflects (382).

(400) and (402) also imply that at the putative equilibrium:

$$\begin{aligned}
\left. \frac{\partial \Pi^R(q_A, 0)}{\partial q_A} \right|_{q_A=q_A^*} &= P_A(Q^*) - c_A + q_A^* \frac{\partial P_A(Q^*)}{\partial q_A} - k^R q_A^* = \bar{p} - c_A - k^R q_A^* \geq 0 \tag{407} \\
\Leftrightarrow \bar{p} - c_A - k^R \left[\frac{a(b+k) + bc - \bar{p}(2b+k)}{b(b+k)} \right] &\geq 0 \\
\Leftrightarrow \bar{p} - c_A - \left[\frac{a(b+k)k^R + bc k^R - \bar{p}(2b+k)k^R}{b(b+k)} \right] &\geq 0 \\
\Leftrightarrow \frac{[\bar{p} - c_A]b[b+k] - a[b+k]k^R - bc k^R + \bar{p}[2b+k]k^R}{b[b+k]} &\geq 0 \\
\Leftrightarrow \frac{\bar{p}[b(b+k) + (2b+k)k^R] - bc_A[b+k] - a[b+k]k^R - bc k^R}{b[b+k]} &\geq 0 \\
\Leftrightarrow \bar{p} \geq \frac{bc_A[b+k] + a[b+k]k^R + bc k^R}{b[b+k] + [2b+k]k^R} \\
\Leftrightarrow \bar{p} \geq \frac{bc_A[b+k] + a[b+k]k^R + bc k^R}{b[2b+k] + [2b+k]k^R - b^2} \\
\Leftrightarrow \bar{p} \geq \frac{bc_A[b+k] + a[b+k]k^R + bc k^R}{[2b+k][b+k^R] - b^2} = \bar{p}_2.
\end{aligned} \tag{408}$$

The second equality in (407) holds because $P_A(Q) = \bar{p}$ as q_A declines below q_A^* (thereby increasing $P(Q)$ above $\bar{p}$). The last equality in (408) reflects (382). (404), (406), and (408) imply that R cannot increase its profit by undertaking a deviation of this kind when $\bar{p} \in [\bar{p}_2, \bar{p}_3]$.

Deviation 2. R deviates unilaterally to set $q_A = 0$ and $q_N > 0$.

Let $q_A^*(\bar{p})$ denote q_A^* as defined in (400) as a function of $\bar{p}$. (382) and (400) imply:

$$q_A^*(\bar{p}_3) = \frac{a[b+k] + bc - \bar{p}_3[2b+k]}{b[b+k]}$$

$$\begin{aligned}
&= \frac{a[b+k] + bc - \frac{a[b+k][b+k^R] + bc_A[b+k] + bc[b+k^R]}{[2b+k][2b+k^R] - b^2} [2b+k]}{b[b+k]} \\
&= \frac{[(2b+k)(2b+k^R) - b^2][a(b+k) + bc]}{b[b+k][(2b+k)(2b+k^R) - b^2]} \\
&\quad - \frac{[a(b+k)(b+k^R) + bc_A(b+k) + bc(b+k^R)][2b+k]}{b[b+k][(2b+k)(2b+k^R) - b^2]} \\
&= \frac{[(2b+k)(2b+k^R) - b^2][a(b+k) + bc]}{b[b+k][(2b+k)(2b+k^R) - b^2]} \\
&\quad - \frac{\{[a(b+k) + bc][b+k^R] + bc_A[b+k]\}[2b+k]}{b[b+k][(2b+k)(2b+k^R) - b^2]} \\
&= \frac{[(2b+k)b - b^2][a(b+k) + bc] - bc_A[b+k][2b+k]}{b[b+k][(2b+k)(2b+k^R) - b^2]} \\
&= \frac{[2b+k-b][a(b+k) + bc] - c_A[b+k][2b+k]}{[b+k][(2b+k)(2b+k^R) - b^2]} \\
&= \frac{[b+k][a(b+k) + bc] - c_A[b+k][2b+k]}{[b+k][(2b+k)(2b+k^R) - b^2]} \\
&= \frac{a[b+k] + bc - c_A[2b+k]}{[2b+k][2b+k^R] - b^2} = \frac{[a - c_A][2b+k] - ab + bc}{[2b+k][2b+k^R] - b^2} \\
&= \frac{[a - c_A][2b+k] - b[a - c]}{[2b+k^R][2b+k] - b^2} = q_{A4}^* \tag{409}
\end{aligned}$$

where q_{A4}^* reflects the value of q_A^* identified in (390). Let q_N^{max} denote the value of q_N that maximizes R 's profit when $q_A = 0$, given q . Then, given q , for all q_N :

$$\Pi^R(0, q_N) = [P(q + q_N) - c_N]q_N - \frac{k^R}{2} [q_N]^2 \leq \Pi^R(0, q_N^{max}). \tag{410}$$

q_N^{max} exists because it is apparent from (410) that $\Pi^R(0, q_N)$ is a strictly concave function of q_N . Observe that, given q^* as specified in (390):

$$\begin{aligned}
\Pi^R(0, q_N^{max}) &< [P(q^* + q_N^{max}) - c_A] q_N^{max} - \frac{k^R}{2} [q_N^{max}]^2 \\
&\leq \Pi^R(q_{A4}^*, 0) = \Pi^R(q_A^*(\bar{p}_3), 0). \tag{411}
\end{aligned}$$

The first inequality in (411) holds because $c_A < c_N$, by assumption. The second inequality in (411) holds because, from Lemma 11, $q_{A4}^* = q_A^*(\bar{p}_3)$ is the value of q_A that maximizes R 's profit when $q_N = 0$ and $q = q^*$. The equality in (411) reflects (409).

R 's profit when $q = q^*$, $q_N = 0$, and $q_A = q_A^*(\bar{p})$ is:

$$\Pi^R(q_A^*(\bar{p}), 0) = [P_A(q_A^*(\bar{p}) + q^*(\bar{p})) - c_A] q_A^*(\bar{p}) - \frac{k^R}{2} [q_A^*(\bar{p})]^2. \quad (412)$$

(400) and (412) imply that $\Pi^R(q_A^*(\bar{p}), 0)$ is a continuous function of $\bar{p}$. Therefore, (411) implies that there exists a $\bar{p}' \leq \bar{p}_3$ such that, for all $\bar{p} \in (\bar{p}', \bar{p}_3]$:

$$\Pi^R(q_A^*(\bar{p}), 0) > \Pi^R(0, q_N^{max}) \geq \Pi^R(0, q_N) \text{ for all } q_N.$$

Consequently, when $\bar{p} \in (\bar{p}', \bar{p}_3]$, R cannot increase its profit by undertaking a deviation of the specified type.

Deviation 3. R deviates unilaterally to set $q_A > 0$ and $q_N > 0$.

Conclusion 5 establishes that such a deviation generates less profit for R than R secures by setting $q_A = 0$ or $q_N = 0$. Therefore, the foregoing findings regarding Deviations 1 and 2 imply that R cannot increase its profit by undertaking a unilateral deviation in which $q_A > 0$ and $q_N > 0$.

In summary, R cannot increase its profit by undertaking a unilateral deviation when $\bar{p} \in [\tilde{\bar{p}}, \bar{p}_3]$ where $\tilde{\bar{p}} = \max\{\bar{p}', \bar{p}_2\}$. It remains to establish that the rival cannot increase its profit by changing q unilaterally. The rival's profit is:

$$\Pi(q) = [a - bQ - c]q - \frac{k}{2}q^2 \Rightarrow \frac{\partial \Pi(q)}{\partial q} = a - bQ - c - bq - kq \quad (413)$$

$$\Rightarrow \frac{\partial^2 \Pi(q)}{\partial q^2} = -2b - k < 0. \quad (414)$$

(413) implies that when $P(Q) = \bar{p}$:

$$\begin{aligned} \left. \frac{\partial \Pi(q)}{\partial q} \right|_{q=q^*} &= a - bQ - c - bq^* - kq^* \\ &= \bar{p} - c - bq^* - kq^* = 0 \Leftrightarrow q^* = \frac{\bar{p} - c}{b + k}. \end{aligned} \quad (415)$$

(400) implies that the last equality in (415) holds at the putative equilibrium. (414) and (415) imply that the rival cannot increase its profit by deviating from the proposed equilibrium.

We have proved that neither firm can increase its profit by deviating from the outputs specified in (400). Therefore, these outputs constitute an equilibrium. ■

Conclusion 6. *In the simplified cost setting, there exists a $\tilde{\bar{p}} \in [\bar{p}_2, \bar{p}_3)$ such that R 's revenue, $V(\bar{p})$, is strictly decreasing in $\bar{p}$ for $\bar{p} \in (\tilde{\bar{p}}, \bar{p}_3]$.*

Proof. Let $\tilde{\bar{p}}$ denote the smallest $\bar{p}$ for which the outcomes identified in (400) prevail in equilibrium. (400) implies that when $\bar{p} \in (\tilde{\bar{p}}, \bar{p}_3]$:

$$V(\bar{p}) = \bar{p} q_A = \bar{p} \left[\frac{a(b+k) + bc - \bar{p}(2b+k)}{b(b+k)} \right]$$

$$\begin{aligned}
\Rightarrow \quad \frac{\partial V(\bar{p})}{\partial \bar{p}} &= \frac{a[b+k] + bc - 2\bar{p}[2b+k]}{b[b+k]} \\
\Rightarrow \quad \frac{\partial V(\bar{p})}{\partial \bar{p}} < 0 &\Leftrightarrow a[b+k] + bc - 2\bar{p}[2b+k] < 0 \\
&\Leftrightarrow \bar{p} > \frac{a[b+k] + bc}{2[2b+k]} \equiv \bar{p}_{V_{3M}}. \tag{416}
\end{aligned}$$

(382) and (416) imply:

$$\begin{aligned}
\bar{p}_{V_{3M}} < \bar{p}_3 &\Leftrightarrow \frac{a[b+k] + bc}{2[2b+k]} < \frac{a[b+k^R][b+k] + bc_A[b+k] + bc[b+k^R]}{[2b+k^R][2b+k] - b^2} \\
&\Leftrightarrow \frac{a[b+k] + bc}{2[2b+k]} < \frac{[a(b+k) + bc][b+k^R] + bc_A[b+k]}{[2b+k^R][2b+k] - b^2}. \tag{417}
\end{aligned}$$

The inequality in (417) holds if:

$$\begin{aligned}
&\frac{a[b+k] + bc}{2[2b+k]} < \frac{[a(b+k) + bc][b+k^R]}{[2b+k^R][2b+k] - b^2} \\
&\Leftrightarrow \frac{1}{2[2b+k]} < \frac{b+k^R}{[2b+k^R][2b+k] - b^2} \\
&\Leftrightarrow [2b+k^R][2b+k] - b^2 < 2[2b+k][b+k^R] \\
&\Leftrightarrow [b+k^R][2b+k] + b[2b+k] - b^2 < 2[2b+k][b+k^R] \\
&\Leftrightarrow b[2b+k] - b^2 < [2b+k][b+k^R] \Leftrightarrow -b^2 < [2b+k]k^R.
\end{aligned}$$

The last inequality here holds. Therefore, (416) and (417) imply that $\frac{\partial V(\bar{p})}{\partial \bar{p}} < 0$ when $\bar{p} \in (\max\{\tilde{\bar{p}}, \bar{p}_{VM}\}, \bar{p}_3]$. ■

F. The Modified Baseline Setting.

Table TA1 reports how equilibrium outcomes change as parameter values change in the modified baseline setting. Recall that $P(Q) = mQ^{-\frac{1}{\varepsilon}}$ where $m = 10^6$ and $\varepsilon = 2$ in this setting.

Parameter Variation	$\frac{\bar{p}_b - \bar{p}_d}{\bar{p}_b}$	$\frac{V(\bar{p}_d) - V(\bar{p}_b)}{V(\bar{p}_b)}$	$\bar{p}^*$	$\frac{\bar{p}^*}{\bar{p}_b}$	$\frac{W(\bar{p}^*) - W(\bar{p}_b)}{W(\bar{p}_b)}$
1.50 m	0.05	0.14	88.06	0.95	0.01
0.50 m	0.06	0.15	43.01	0.95	0.01
1.10 ε	0.04	0.13	121.56	0.96	0.01
0.90 ε	0.07	0.16	34.82	0.93	0.02
1.50 c_A	0.05	0.14	67.59	0.95	0.01
0.50 c_A	0.05	0.14	67.12	0.95	0.01
1.50 k_A	0.04	0.12	70.64	0.96	0.01
0.5 k_A	0.07	0.19	61.51	0.93	0.03
1.50 k^R	0.05	0.14	69.00	0.95	0.01
0.50 k^R	0.05	0.15	65.72	0.95	0.02
1.50 c_N	0.05	0.15	67.59	0.95	0.02
0.50 c_N	0.05	0.14	67.36	0.95	0.01
1.50 k_N	0.06	0.16	68.29	0.94	0.02
0.50 k_N	0.04	0.12	65.48	0.96	0.01
1.50 c	0.05	0.14	67.59	0.95	0.02
0.50 c	0.05	0.14	67.36	0.95	0.01
1.50 k	0.07	0.16	70.64	0.94	0.02
0.50 k	0.03	0.12	61.27	0.97	0.01

Table TA1. The Effects of Changing Baseline Parameters.

The first column in Table A4 identifies the single parameter that is changed in the modified baseline setting and the amount by which it is changed. All other parameters remain at their levels in the baseline setting.⁵ The remaining columns in Table TA1 identify outcomes that arise in equilibrium, corresponding to the entries in Table A1 in the Appendix of the paper. The welfare calculations in the last column assume that $r = \frac{1}{2}$.

⁵For example, the first row of data in Table TA1 records the outcomes that arise in equilibrium when m is increased by 50% above its level in the modified baseline setting with $P(Q) = mQ^{-\frac{1}{\varepsilon}}$, holding all other parameters at their values in this setting. Table TA1 considers relatively limited variation in ε because the second order condition for R 's problem is violated if ε becomes too small.

F. Supplemental Figures.

Figure A1 below complements Figure 2 in the paper by illustrating how consumer surplus ($S(\cdot)$) and R 's revenue ($V(\cdot)$) vary with the price cap ($\bar{p}$) in settings where $\bar{p}_{V_m} \in (\bar{p}_0, \bar{p}_d)$ and $\bar{p}_{SM} \in (\bar{p}_0, \bar{p}_d)$. Recall that $\bar{p}_{V_m}$ is the value of the price cap at which R 's equilibrium revenue is minimized and $\bar{p}_{SM}$ is the value of the price cap at which equilibrium consumer surplus is maximized.

Figure A2 below complements Figures 2 and 4 in the paper by depicting how consumer surplus ($S(\cdot)$), R 's revenue ($V(\cdot)$), and welfare ($W(\cdot) = S(\cdot) - \frac{1}{2} V(\cdot)$) vary with $\bar{p}$ in the baseline setting.